SPICELY: AI-Driven Platform to Facilitate Direct Access for Small-Scale Spice Farmers to International Buyers.

Mobile Application

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Abstract

Small-scale spice farmers in Sri Lanka are unable to reach international markets easily, points to the reliance on middlemen, the expense of intermediation, and restricted access to up-to-date accessing systems. This study presents SPICELY, an Android application that aims at linking small holder farmers in the spice sector directly to international buyers without using an intermediary which is likely to increase the market accessibility. The application and Java were used as the backend logic programming language and XML was used to design the user interface in Android Studio. Real-time database management through Firebase services, authentication, and safe data handling were provided with the help of AI algorithms created based on Python to analyse prices predictions. These were developed through the Agile Software Development Model which allowed testing iteratively, rapid prototyping, and constant user feedback. The combined approach of mixed methods was used, and the research covered both qualitative data collection in terms of interviews and focus groups and quantitative (testing of the system and user satisfaction surveys). The findings show that the system has effectively helped farmers gain access to the markets, mitigated the reliance of their businesses on intermediaries, and increased their profitability through the power of predictive analytics based on artificial intelligence and direct links to buyers.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
UI	User Interface
XML	eXtensible Markup Language
DBMS	Database Management System
DSR	Design Science Research
MVP	Minimum Viable Product
SDLC	Software Development Life Cycle
ERD	Entity Relationship Diagram
UML	Unified Modeling Language
IDE	Integrated Development Environment
IoT	Internet of Things
CRUD	Create, Read, Update, Delete
SQL	Structured Query Language
OTP	One-Time Password
M-Commerce	Mobile Commerce
GPS	Global Positioning System
JVM	Java Virtual Machine
NLP	Natural Language Processing

Table 1 : List of Abbreviation

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Spice farming is an important part of the economy of various developing nations, especially in the areas such as South Asia, where farming is the basis of livelihood and foreign exchange. In Sri Lanka, the production of spices, including cinnamon, pepper, cardamom and turmeric, not only provides local people with additional income, but also plays an important role in supporting export (Fernando, et al., 2021). Nonetheless, in spite of their significance, small-scale spice farmers are likely to be affected by substantial limitations in accessing the international markets in most cases. Such obstacles are the low bargaining power, the use of third parties, the absence of transparency in the market, and the risk of facing fluctuating prices (Jayasinghe & Weerahewa, 2020). This reliance on the middlemen tends to make the farmers earn very little of what will be the end price on the market hence their profitability and growth sources are restricted.

The modernity of digital technologies and agriculture, intensive development at the stage of Agricultural Technology (AgriTech), offers a new chance to solve these old problems. The utilization of Artificial Intelligence (AI) in agricultural value chains has been highlighted as an innovative process that has the potential to boost the efficiency of managing the supply chain, decision-making, and real-time market conditions available to farmers (Kamilaris, et al., 2017). The usage of AI-powered systems can help predict market prices, optimize production plans and make direct contacts between producers and buyers, which ensures shrinking the reliance on intermediaries (Ganguly et al., 2022). Using predictive analytics and automation of complex trade processes, AI can allow farmers to increase their competitiveness in the global market.

1.2 Problem Statement

The growing worldwide market interest for superior spices cannot reach small-scale producers because they face challenges gaining direct market connections with buyers. Traditional marketing systems create many intermediary organizations who decrease farmer profits while making supply chain management less efficient. The present situation demands a digital solution which allows farmers to establish connections with buyers and acquire contemporary pricing updates while developing superior sales procedures.

1.3 Research Questions

- What prevents local spices farmers who have small operations from accessing international markets?

Among the alleged obstacles of small-scale spice farmers to trade internationally, there are several barriers to entry. They are inability to access real time market information, reliance on middlemen, exposure to export standard, digital illiteracy and deficient infrastructure like internet connectivity and logistics support. In addition, they usually lack bargaining power and traceability systems, which are important factors of developing trust among the global buyers. All these constraints limit them to make a direct entry into the foreign markets and deliver their goods at competitive prices.

- Do AI-based solutions possess suitable methods to handle these problems effectively?

Indeed, AI-based solutions have a great potential to help overcome most of the problems smallholder spice farmers must deal with. Artificial intelligence will allow determining future market prices in the most accurate way, advising on the best choice of time to sell products and having ideas on whom to sell the products to, in addition to analyzing the quality of any product based on their recognitions. The tools will allow farmers to make evidence-based decisions to enable them to be less dependent on intermediaries and speculation. Using the scope of AI in a mobile application, the farmers obtain real-time analysis and intelligent automation improving their productivity and the ability to reach the buyers, therefore, closing the information and opportunity divide.

- What benefits and obstacles exist for an AI-based system that facilitates spice trade while its benefits and limitations must be identified.

The advantages of AI-based spice trading platform comprise better transparency in price, better trade practices, more efficient methods of accessing the market, and low cost of operation due to automation. It makes people trust more through traceability backed by blockchain, and farmers can compete in the world even with minimum technical skills. Nevertheless, they are also limited. These are the necessity of stable internet connection, resisting to implementing new technology, low level of digital literacy, as well as the complexity of AI (artificial intelligence) algorithms that can require an ongoing input of data and improvement. The issue of overcoming the difficulties necessitates strong training, user-friendly design as well as continuous technical assistance.

- What plans should be used to guarantee farmers and buyers fully implement the platform?

The multi-pronged implementation strategy is needed to make the adoption of the platform successful. These involve training farmers in digital literacy, localization of application using local languages, compatibility with low-end mobile devices and making users trust the application through good user help and transparent procedures. The government bodies, cooperatives, and NGOs may be a partner that helps to promote the platform and be able to give incentives to using it as an incentive at the very beginning. Also, ongoing user responses, frequent updates and offline

features can help make usage and the lifespan of the user friendly and quite engaging as it would meet user demands on the spot.

1.4 Research Objectives

I. Small-scale spice farmers encounter critical barriers when trying to connect to international marketplaces for their products.

This objective is aimed at describing and identifying the structural, technological and economic issues affecting the ability of the smallholder spice farmers to access the global markets. Such barriers could be the absence of digital infrastructure, the use of intermediaries, or scarcity of market intelligence, lack of supply chain trust, and inability to comply with international trade requirements. The analysis of these constraints is supposed to help uncover the causes of low market participation as well as establish the basis of a digital intervention that will focus on the practical characteristics of farmers.

II. A mobile application development using Artificial Intelligence will be created to enable direct market access.

The second objective will deal with the creation of a feasible digital solution as the mobile application. This platform, called Spicely, carries with it Artificial Intelligence (AI) to provide farmers with Artificial Intelligence-based in predictive pricing and buyer matching. The app is meant to alleviate a reliance on the middlemen and bring the farmers into direct contact with the global consumers. This entails the integration of invaluable features like product listing, AI-based price prediction, traceability enabled by blockchain, and safe payments by mobile device-trade that leads to an automated and smart marketplace in terms of spice trade.

III. An evaluation will determine how AI market intelligence platforms affect supply chain optimization along with price forecasting processes.

This goal assesses the efficiency of AI technologies as a change driver in providing traditional agricultural supply chains. In particular, the study examines how artificial intelligence-based insights could streamline and enhance the decision-making of farmers, going by the accuracy of pricing and the flow of goods along the supply chain. Values like Mean Absolute Percentage Error (MAPE) of price forecast, level of precision on the buyer-match and usability of the system are applied to gauge the real-life effects of the application on operational performance and financial consequences of farmers.

IV. A set of recommendations is proposed that will ensure both adoption success and sustainability of the platform.

The last objective is devoted to the development of actionable plans on how to achieve long-term adoption and scalability of the platform by the rural farmers and the international buyers. These are suggestions brought about by testing of the system, feedback and assessment by its users and market. The most attention is devoted to the development of language and accessibility options, offline capabilities to support low-connectivity regions, logistics services and training. These critical success factors take care of, by establishing that Spicely will be able to provide sustainable value where there exist variety and change in the agricultural situation.

1.5 Research Aims

1. A mobile application developed with AI capabilities functions as a platform linking international buyers directly with small-scale spice farmers.

This aim is pointing at the creation and implementation of an effective mobile app that would become a digital translation between the farmers of rural spices and international customers. The application allows the system to make intelligent matches, real-time price recommendations, and easy communications by having Artificial Intelligence (AI) as the essence of the application. The main aim is to remove the traditional access to markets by substituting the fragmented and intermediary-based model with a straightforward technologically powered and well-organized environment that allows direct interaction between farmers and buyers.

2. High transparency along with enhanced accessibility for smallholder farmers exists when data analytics based on AI powers the market.

The second aim takes care of the necessity of more visibility and trust in the agricultural transactions. Farmers will be able to know the trends on the market, consumer pattern, and performance of their products using AI-powered data analytics. Such openness enables the farmers to make informed decisions in terms of pricing and distribution. Also, the system improves the accessibility of smallholder farmers since most of them live or work in remote regions where they rarely have contact with any digital means.

3. The system would decrease the need for middlemen while minimizing expenses in spice market transactions.

The conventional spice markets usually include several levels of intermediaries that profit by sharing a part of the seller's revenues and adds to the overall expenses of the purchaser. The platform has the advantage of making direct transactions, so there is less reliance on middlemen

as account parameters like price negotiation and traceability are automated. This not only reduces the cost of transactions but also makes more profit go to the farmers, hence enhancing their financial positions.

4. Small-scale spice farmers can boost their market success and market competitiveness in international markets.

This aim considers long-term empowerment of the smallholder farmers. Farmers will be able to enhance the quality and positioning of their products in competitive international markets by using AI tools to price, predict the quality, and gain insights into the markets. The app allows them to achieve global standards, agility in responding to any changes in demand besides housing their operation in a more strategic manner. This, in its turn, increases their capability to compete with the bigger exporters and establish their stable positions in foreign markets.

5. Create advice for both policy improvements and digital literacy teaching programs so AI-driven agricultural technology solutions can more easily be adopted.

The last aim understands that ecosystem support requires an additional boost other than technology. System design is not the only factor that guarantees success in AI-driven platforms; user education and facilitating policy frameworks play important roles in the success of AI-driven platforms. This goal implies the production of policy recommendations that stimulate the growth of digital agriculture and the plan of training measures to develop the digital literacy of farmers. Such initiatives play a critical role in making technological improvement to be all-inclusive, sustainable and reach many beneficiaries as targeted.

1.6 Scope and Limitations

This study focuses on the spice farming sector, particularly small-scale farmers in developing regions. The research is limited to the design, development, and evaluation of the AI-driven platform, without extending into large-scale implementation. Additionally, factors such as digital literacy, infrastructural constraints, and policy implications are considered but not exhaustively analysed.

1.7 Thesis Structure

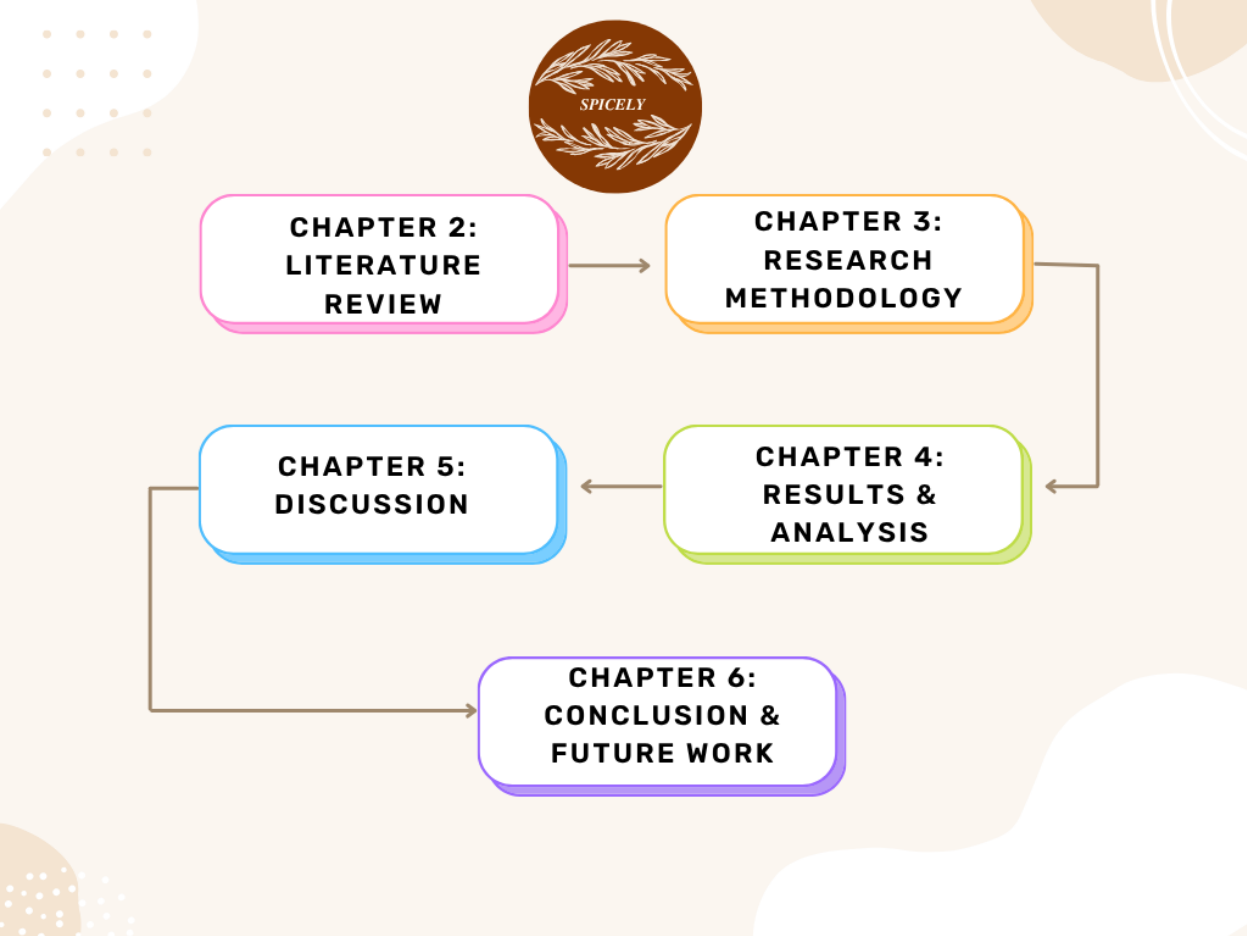


Figure 1 : Thesis Structure

Source: Author's own (2025)

2 CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The cultivation of crops stands as the fundamental economic driver within rural areas specifically for developing countries such as Sri Lanka where small-scale farmers maintain pivotal positions for national domestic food supply as well as export industries. The farming community consisting of spice growers produces leading agricultural value, but they stand among the most marginalized and least protected groups in global trade according to (FAO, 2020). The path spices take from their original farms until they reach international markets becomes complex because of broken supply chains and difficulties with market information access and limited power to negotiate among other barriers (Ruben, et al., 2019). Multiple barriers stop smallholder spice farmers from acquiring reasonable payments while blocking their ability to develop primary trading relationships with overseas market buyers.

Digital technology development during recent years introduced new solutions to solve long-standing difficulty barriers. Modern technologies including Artificial Intelligence (AI) as well as Blockchain and Mobile Commerce (m-commerce) have reintroduced efficiency improvements together with transparency and accessibility to various businesses (Wolfert, et al., 2017). Agriculture professionals identify these technologies as transformative instruments which modernize typical farming operations and reinforce market engagement and agricultural control for regional farmers (Kamilaris, et al., 2019). The literature presents minimal investigation regarding how these tools are being used by small-scale farmers within the spice industry.

Artificial intelligence solutions have established their ability to deliver exceptional performance in predictive analysis and crop monitoring and farm suggestion recommendations to farmers. These advanced systems can assist spice farmers by applying data analytics principles to their choices regarding growing methods and handling procedures and all aspects of market planning (Wamba & Queiroz, 2020). Blockchain technology serves as an innovative answer to develop trust and transparency through agri-food supply chain management using tracing elements and protected transactions alongside unchangeable documentation traveling from farm to eat (UNDP, 2021). Through mobile commerce platforms farmers currently can eliminate traditional market intermediaries while establishing direct sales relationships with customers which extends their market reach internationally (Aker & Mbiti, 2010).

High-end digital innovations continue to develop promising opportunities for spice farmers operating on a small scale, yet several access barriers prevent them from taking advantage. The barriers to digital adoption among small-scale spice farmers include insufficient digital literacy along with expensive technology acquisition costs and unsatisfactory rural infrastructure as well as difficulties in changing existing traditional agricultural methods (Ogunniyi & Ojebiyi, 2016). Present academic investigations center on substantial agricultural operations combined with rice and wheat and maize and rarely explore export spices like spices.

The objective of this literature review analyses the impact AI along with Blockchain technology and m-commerce has on agricultural value chains particularly regarding small-scale spice farmers. The research explores current agricultural technology applications worldwide to reveal research gaps while analysing opportunities and obstacles for creating an online system linking spice grower access to international buyers. Development of an AI-based mobile application solution specifically designed for Sri Lankan smallholder spice farmers starts from the groundwork presented within this chapter to achieve sustainable market integration and foster inclusive development.

2.2 The Role and Importance of Spice Farming in the Global Market

Historically speaking spices functioned as vital elements which defined worldwide commerce as well as financial systems together with cultural structures for extensive periods. Spice agriculture continues as a fundamental agricultural practice for developing nations because of its expensive value combined with expanding worldwide market needs and currency acquisition potential (Freedman, 2020). The global spice industry demonstrates sustained growth because consumers search for nutritious food and natural seasonings in addition to organic products (FAO, 2020).

The growing of spices holds special significance for tropical locations of Asia Africa and Latin America since these areas provide perfect climatic environments for peppers and cardamom and nutmeg and turmeric and cloves (Ravindran, 2017). India together with Sri Lanka and Indonesia and Vietnam maintain leadership positions among spice producers which drive international spice market exports (ITC, 2021). The global spice industry grew to exceed USD 10 billion in trading value in recent times according to the Food and Agriculture Organization (FAO, 2020).

Almost all small-scale farmers in Sri Lanka depend on spice cultivation as their primary livelihood choice for generating revenue. The market value of spices exceeds that of rice or maize fields on a per-hectare basis which draws small-scale farmers toward growing these profitable crops (Jayasinghe, et al., 2022). Spices such as cinnamon and cardamom store well in their dried state because they are non-perishable thus enabling better storage and export opportunities which minimize typical losses experienced during perishable crop storage (Kumar, et al., 2018).

The global spice market maintains high competition levels while existing primarily under control of international buyers together with processing companies. Small-scale spice cultivators operating in developing economies face challenges to integrate properly into worldwide value chains because they lack essential market insights and quality requirements as well as official certifications and overseas purchaser connections (Ruben, et al., 2019). International export processes together with middlemen control prevents farmers from securing the total potential value from their harvested goods thus restricting their financial returns even though international markets offer higher prices.

The global spice trade now prioritizes both excellent qualities along with complete traceability and ethical sourcing practices and international health and safety standards compliance (UNDP, 2021). The new trends apply additional stress to small farmers who need to fulfil these requirements even when they lack support or resources.

The world market heavily relies on spice farming which provides financial benefits through agricultural production and maintains export significance for various developing countries. The complete development of spice farming by smallholder farmers needs solutions for three main market access challenges and product identification requirements along with direct global buyer connections.

2.3 Challenges Faced by Small-Scale Spice Farmers

Farming operations of spices at a small scale are vital contributors to agricultural economies of developing nations. The remarkable potential for competitive market success of small spice farmers is reduced against numerous barriers that disrupt their capacity to succeed locally or internationally. Farmers encounter complex barriers of economic nature together with technical limitations and social issues and inadequate infrastructure which prevent them from reaching sustainability and productivity levels (Ruben, et al., 2019).

2.3.1 Limited Access to Market Information

Smallholder spice farmers face the main barrier of not having secure market data and accurate information in a timely manner. Farmer operations take place in rural isolation where poor communication channels prevent them from obtaining market price updates or industry demand analysis or buyer preference reports (Mittal & Mehar, 2016). Poor decisions in production planning and pricing alongside marketing strategies emerge because farmers lack reliable market information. As a result, farmers must settle for poor prices when selling their crops since they lack market alternatives and negotiation power (Bank, 2019).

2.3.2 Exploitation by Middlemen

The value chain of spices operates mainly through intermediaries who maintain control over transactions specifically in rural zones which farmers lack direct market access. Except for middleman businesses who use their power to acquire spices from farmers at costs much lower than international market value and keep most of the final price for themselves (Ruben, et al., 2019). Research reveals that middlemen gain control of transfer and warehousing operations as well as export relationships which results in farmer dependence and diminished profit potential (Jayasinghe, et al., 2022).

2.3.3 Inadequate Infrastructure and Logistics

If farmers work in rural areas, they face several critical issues because of poor infrastructure. Farmers frequently experience problems because they have no access to suitable storage infrastructure as well as essential transport networks and processing facilities to protect their agricultural output quality (Kumar, et al., 2018). International trade limitations arise from the lack of appropriate cold storage and drying facilities because they result in spice quality deterioration along with post-harvest losses (FAO, 2020).

2.3.4 Difficulty in Meeting International Quality Standards

International spice markets require strong standards which check both quality and food safety and traceability measures. International certification compliance through HACCP or ISO or organic standards demands substantial expenses on training personnel as well as infrastructure establishment and quality control activities (UNDP, 2021). The absence of technical expertise and financial strength among small-scale farmers prevents their ability to fulfil industry standards and consequently blocks market entry into lucrative exports (AgTech, 2025).

2.3.5 Lack of Access to Finance

The formal financial sector presents barriers to spice farmers when they try to obtain loans or credit facilities. Smallholder farmers find it hard to access financial services from institutions because lenders view agriculture as risky, and farmers lack collateral, and the production values are unstable (IFAD, 2020). Lack of proper financial resources prevents farmers from using modern farming practices and purchasing necessary inputs for upgrading their productivity levels.

2.3.6 Technological Barriers

Most rural spice farmers experience difficulties with digital technology access because of limited knowledge about digital tools and expensive smartphone prices and insufficient internet connectivity (Ogunniyi & Ojebiyi, 2016). Mobile applications along with AI tools and online marketplaces that aim to enhance agricultural trade remain out of reach for farmers because of these barriers (Wolfert, et al., 2017).

2.4 Digital Transformation in Agriculture (AgriTech)

Almost every aspect of farming underwent transformative change through digital technology because agriculture used to operate with manual practices and regional trade systems. AgriTech brings restructuring to the agricultural sector through its adoption of sophisticated technologies including AI alongside IoT and Blockchain together with m-commerce (Wolfert, et al., 2017). Modern agricultural sector depends heavily on AgriTech to advance its operations specifically through its positive effects on small-farm operations dealing with resource constraints and market accessibility issues and fragmented market systems.

The main goal of digital transformation within agricultural industries involves enhancing productivity levels together with operational efficiency and decision support functions while establishing better market connections (Kamilaris, et al., 2019). The global food and agricultural product demand rise has directed attention to market-oriented technology solutions which combine sustainability in farming practices.

2.4.1 The Role of AgriTech in Market Access

The main advantage of AgriTech technology is its ability to provide market opportunities to smallholder farmers. Today's market technology provides farms with the capability to find buyers directly thus bypassing the conventional practice of using middlemen who charge farmers uneven prices (Aker & Mbiti, 2010). Mobile applications and online marketplaces give farmers access to up-to-date market information which helps them determine better prices while broadening their customer reach out of local markets.

The capabilities of mobile technology to deliver empowerment to farmers are shown through M-Farm in Kenya and AgroStar in India by linking farmers with suppliers and buyers and advisory services (Bank, 2019). These platforms solve information problems between farmers and buyers which promotes fair agricultural trade transparency.

2.4.2 Impact of Mobile Commerce (M-Commerce)

Mobile commerce stands as an essential AgriTech element which enables small-scale farmers to execute buying and selling of agricultural products through mobile devices. The new technology presents market-integrated solutions to farmers by enabling flexible and convenient control of their sales operations (Mittal & Mehar, 2016). Through limited physical market access rural farmers obtain accessibility to national and international buyers through m-commerce platforms without physical travel requirements or dependence on intermediaries.

The platforms incorporate digital payment systems that allow fast secure financial transactions which create trust between farmers and buyers (Aker & Mbiti, 2010). The proper management of

payments together with documentation remains vital for spice farmers who need to trade internationally.

2.4.3 Integration of AI and Blockchain in AgriTech

Advanced technologies including AI and Blockchain systems strategically integrate with agricultural systems to enhance operational effectiveness and enhance transparency. The functions of AI technologies include predictive analytics together with supply chain optimization and intelligent decision support frameworks that allow farmers to handle their production risks in addition to market uncertainties (Wamba & Queiroz, 2020). AI solutions evaluate climate conditions to estimate infestation threats before offering farming guidance that suits environmental settings.

Blockchain technology enables supply chain businesses to establish complete tracking capabilities throughout the agricultural industry while maintaining complete visibility into operations. A secure digital ledger through Blockchain technology enables smallholder farmers to protect their recorded farming practices and harvest information and deals which strengthens international market trust (UNDP, 2021). The spice industry relies on detailed information about product origin alongside quality standards and sustainability practices because buyers make this requirement.

2.4.4 Challenges of Digital Transformation in Agriculture

Small-scale farmers encounter various obstacles when it comes to implementing the advantages of AgriTech technology. Digital solutions fail to spread widely due to farmers' inadequate digital skills as well as their limited internet connectivity which affects the accessibility of computing devices and rural region development gaps (Ogunniyi & Ojebiyi, 2016). The successful implementation of AgriTech systems require addressing both digital transaction security and privacy-related issues because farmers remain concerned about these factors (Wolfert, et al., 2017).

2.5 Artificial Intelligence (AI) Applications in Agriculture

Artificial Intelligence (AI) technology implementations in agricultural systems have brought major changes to how farmers cultivate and care for product crops and engage in product marketing activities. Smart farming technology results from the utilization of AI technologies as they provide both data-based decision assistance and resource management optimization and improved business network synchrony (Wamba & Queiroz, 2020). Smaller spice producers stand to benefit from AI through an improved efficiency and reduced expenses which upgrades their market position when they utilize mobile systems linked to digital supply networks.

2.5.1 AI for Precision Agriculture and Decision Support

Precise agricultural management relies on AI applications as its most valuable implementation in farming. Software technologies through AI let farmers enhance their crop management efficiency by acquiring instant information about soil characteristics and weather conditions and pest movements and plant health states (Wolfert, et al., 2017). Machine learning algorithms together with AI-directed drones and remote sensing imagery provide analyses to spot agricultural health conditions and soil deficiencies and irrigation requirements after capturing farmland survey pictures (Liakos, et al., 2018).

The utilisation of these systems allows farmers to apply fertilizers alongside pesticides and water with greater accuracy thus they minimize waste and lower environmental impact while achieving optimal yields (Sharma, et al., 2020). Agricultural manufacturers who work with cinnamon and pepper face advantages from AI-driven precision farming because it ensures quality consistency at decreased waste levels after harvests.

2.5.2 AI for Market Prediction and Pricing Strategies

AI offers farmers the ability to forecast market demands as well as track price trends in the industry. The combination of historical spice price data with market demand data combined with seasonal variations and buyer preferences enables machine learning algorithms to make future market condition forecasts (Wamba & Queiroz, 2020). Small-scale farmers gain better profit potential by utilizing this data to determine optimal times for harvest storage and market entry.

Sub-agricultural spice growers who usually work through middlemen to get marketplace information now face better opportunities to buy and sell directly to international trade parties through AI-based pricing technology.

2.5.3 AI Chatbots and Virtual Assistants for Farmers

Preventative measures at an agricultural level employ chatbots and virtual assistants to supply individualized agricultural assistance and technical help to farmers. The tools provide answers for farmers who have inquiries about cultivation methods as well as pest control techniques and market entry solutions while also supplying weather prediction forecasts in their native tongues (Tripathy, et al., 2021). The chatbot "Darli AI gives regenerative farming guidance to African farmers working with small properties so these practitioners can implement sustainable methods beyond isolated rural areas (TIME, 2024).

AI chatbots can provide vital information about harvesting practices alongside international quality standards and packaging needs to farmers involved in spice production thus making them ready for global commercial activities.

2.5.4 Challenges in AI Adoption

The vast advantages of AI do not eliminate significant obstacles especially when serving smallholder farmers. The main impediments to AI adoption are farmers' insufficient access to digital technology and their restricted internet bandwidth and their low digital skills combined with expensive AI system costs (Ogunniyi & Ojebiyi, 2016). The distrust in AI systems grows when rural communities show limited understanding of digital rights because they remain concerned about data privacy together with algorithm transparency issues (Wolfert, et al., 2017).

Implementing capacity development initiatives together with user-friendly designs and lightweight AI applications suitable for low-end mobile devices with offline capabilities represents the solution to encounter these challenges.

2.6 Blockchain Technology for Traceability & Trust

The farm and food industry has found revolutionary solutions in Blockchain technology that both protects supply chain visibility and tracks products while assuring faith between growers and customers. International buyers alongside consumers from across the world have become preoccupied about spice authenticity and quality standards and system of origin (Kamilaris, et al., 2019) in global agricultural value chains that focus on export markets. Through blockchain technology the agricultural supply chain becomes secure through its transparent digital records which prevent any form of tampering (Saberi, et al., 2019).

2.6.1 Role of Blockchain in Agricultural Supply Chains

The decentralized digital ledger of Blockchain functions across multiple computers to track information thus making any modification of data extremely difficult to achieve without general node agreement (Casino, et al., 2019). The blockchain system enables agriculture producers to verify and document every process starting from planting through harvesting and packaging until shipping (Tripoli & Schmidhuber, 2018).

Blockchains provide small-scale spice farmers an opportunity to show their agricultural practices and quality certifications directly to foreign buyers because this technology allows them to prove product authenticity and sustainable sourcing (UNDP, 2021). These export markets have become more transparent regarding sustainable and organic ethical products because consumer demands are rising.

2.6.2 Traceability for Food Safety & Quality Assurance

The spice industry faces traceability issues as its most important problem. Totally interfering quality control procedures coupled with adulteration issues as well as contamination incidents can drive international buyers to lose faith in Sariya International's products and harm the reputation of Sariya International. Since Blockchain assigns every spice batch its distinct digital code or QR code customers and final buyers gain access to complete tracking information about spice origins, and they can observe handling procedures as well as test product certifications.

Traceability systems in agriculture as outlined by the Food and Agriculture Organization (FAO, 2020) help prevent food fraud while improving product quality and fostering consumer trust in agricultural supplies. Through blockchain technology the recording process auto-initiates while remaining available for all parties who hold authorization.

2.6.3 Enhancing Trust and Reducing Middlemen Exploitation

BC technology establishes trust management by allowing parties to bypass intermediary systems which used to control data sharing and pricing structures. Through blockchain implementation spice producers at a small scale can establish direct trading connections with global buyers thus eliminating multiple middlemen who typically adjust prices for their gain (Kamilaris, et al., 2019).

Smart contracts that function as blockchain features provide users with this additional capability. The automatic payment system within self-executing contracts ensures scheduled payments to both farmers and buyers through pre-established conditions (Casino, et al., 2019).

2.6.4 Challenges in Blockchain Adoption for Small-Scale Farmers

Smallholder spice farmers encounter various hurdles during the blockchain technology implementation process. These include:

- High cost of technology infrastructure
- The rural area demands access to internet services
- Low digital literacy level
- The implementation of blockchain technology proves difficult when merged with conventional farming methods (Wamba & Queiroz, 2020).

The reliability of blockchain depends on the accuracy of the data that farmers and cooperatives submit at the beginning. Insufficient training and mistrust about the system allows incorrect or fraudulent data entry to persist (Tripoli & Schmidhuber, 2018).

Success of adopting blockchain for small-scale spice farmers requires government policies alongside educational programs for farmers assisted by simple mobile interfaces.

2.7 Mobile Commerce (M-Commerce) Platforms for Farmers

Mobile commerce (m-commerce) works as a modern and effective platform which permits farmers primarily small-scale producers to make direct business contacts through mobile-based systems. The utilization of mobile applications and smartphones as tools enables agricultural product trading together with agricultural information acquisition and financial operations for farm activity management (Mittal & Mehar, 2016).

m-commerce platforms enabled by mobile phone expansion make rural transitioning in developing nations possible by breaking geographic and informational and financial obstacles (Aker & Mbiti, 2010).

2.7.1 Benefits of M-Commerce for Small-Scale Farmers

- Direct Market Access

M-commerce platforms help small-scale farmers establish direct buying connections with consumers from their local market and across their respective regions and other national territories. Traditional middlemen are cut out of the deal through these platforms because producers can now find better prices directly (Bank, 2019). Through direct market access spice farmers gain special benefits to sell valuable crops including cinnamon and pepper and cardamom directly to international purchasers without broker or exporter dependence.

- Real-Time Market Information

Formattable applications enable farmers to access current market information regarding prices and consumer demands and buyer specifications and agricultural press updates simultaneously. Mobile applications deliver current market data which enables farmers to base their harvesting dates and sale moments and pricing approaches (Mittal & Mehar, 2016).

- Financial Inclusion and Digital Payments

M-commerce platforms connect farmers to mobile payment systems which enable them to get instant and secure payments by using their mobile wallets or bank accounts. The system helps eliminate the problem of delayed payments that usually occurs when using intermediaries to market products (Aker & Mbiti, 2010). Mobile money services provide farmers with banking options such as savings accounts and credit facilities and insurance policies even though they were previously denied access to traditional banking institutions.

- Increased Bargaining Power

Mobile commerce platforms help smallholder farmers gain better market position by minimizing information imbalances while creating buyer-farmer interaction through which they can participate better in market competition (Sharma, et al., 2020).

2.7.2 Global Examples of Successful M-Commerce in Agriculture

Multiple nations have applied m-commerce platforms which achieved notable achievements.

- M-Farm in Kenya operates a SMS messaging service that gives farmers access to market price details and buyer information together with climate condition updates (Bank, 2019).
- Through its mobile-based system Twiga Foods (Kenya) links farm-to-retail operations directly which gives farmers better prices while reducing food losses during harvest time (Project, 2024).
- Indian farmers benefit from e-Choupal by using digital kiosks and mobile systems to get market insights and conduct trade and obtain agricultural guidance (Mittal & Mehar, 2016).

M-commerce platforms prove effective by increasing rural farmer economic opportunities while making them accessible to formal market systems through these examples.

2.7.3 M-Commerce Potential for Spice Farmers

Mobile commerce acts as a suitable solution which helps small-scale spice farmers address current obstacles in the marketplace. Such a specialized mobile platform would allow farmers to:

- Showcase their spices to international buyers through digital product catalogs.
- Spice farmers should establish direct communications with buyers for discussions about quality, quantity and pricing terms.
- Users should be able to receive safe mobile payments through dedicated payment systems.
- Users gain access to providers who deliver packaging materials in addition to shipping solutions and certifications.
- The platform would increase market exposure for farmers who seek to enter international spice markets.

A specialized marketplace will boost international market competitiveness for smallholder spice farmers because customers desire organic and traceable premium quality spices (Jaiswal, et al., 2022).

2.7.4 Challenges in Adopting M-Commerce

Several obstacles prevent small-scale farmers from adopting m-commerce while still benefiting from its advantages.

- Limited digital literacy among rural farmers.
- Poor mobile network connectivity in remote areas.
- Small-scale farmers face difficulties in adopting m-Commerce because smartphones and internet services have high prices for them.
- Trust issues in online transactions.
- Traditional marketing systems face a widespread reluctance among farmers to move away from conventional methods of sales (Ogunniyi & Ojebiyi, 2016).

Rural users need affordable digital tools and supportive infrastructure together with user-friendly mobile interfaces to overcome the barriers that farmer education must address to transform their digital practices.

2.8 Research Gap Identification

Agriculture now observes major changes in growing methods and marketplace operations through the combination of Artificial Intelligence (AI), Blockchain and Mobile Commerce (m-commerce) digital technologies across different global regions (Kamilaris, et al., 2019). These technologies have demonstrated their ability to improve operational efficiency together with transparency levels and market connectivity specifically for large-scale farmers engaged in commercial agriculture and staple crops production (Wolfert, et al., 2017) (Wamba & Queiroz, 2020).

The existing research about AgriTech innovations does not sufficiently address their practical use by small-scale spice farmers across developing countries especially within Sri Lanka. Digital agriculture platforms primarily support major food crops including rice and maize and coffee but neglect the development of platforms for valuable export spices (Ruben, et al., 2019).

Blockchain technology has established numerous uses for food supply chain traceability according to (Jaiswal, et al., 2022) but lacks clear applications for the spice market. International buyers find spices suitable for blockchain implementation because they need stringent quality assurance coupled with certification and traceability. There exist minimal blockchain solutions available for smallholder spice farmers to meet their specific requirements.

Feature	Existing AgriTech Apps	Proposed AI-Driven App
Target Crop	Food crops / Fruits / Vegetables	Specifically focused on Spices
Target User	General farmers or exporters	Small-scale Spice Farmers only
Market Access	Mainly local / domestic	Direct International Buyers
Artificial Intelligence	Rarely used	AI-powered market prediction, pricing suggestions, buyer matching
Blockchain Traceability	Very limited or absent	Integrated Blockchain for supply chain transparency
Mobile Commerce Features	Mostly price info & local trade	End-to-end trading platform
Farmer Empowerment	Advisory based	AI-Decision support + Independent selling without middlemen
Challenges Addressed	Crop production challenges	Market access + Traceability + International trade barriers

Table 2 : Research Gap Identification

3 CHAPTER 3: METHODOLOGY

3.1 Introduction

The chapter introduces the research methodology implemented so far to develop and test Spicely- a mobile application based on the concepts of Artificial Intelligence (AI), Blockchain, and Mobile Commerce (m-commerce) that aimed at narrowing the gap between small-scale Sri Lankan spice farmers and international buyers. The methodology explains the theoretical orientation, the research design, the development model, data programming tactics, and procedures of validating the system movements in the project. Such a mixed method also guaranteed that the practical design was guided by the needs of the real world and tested practically and empirically.

The application was to solve fundamental problems in value chains in the agricultural industry, including price manipulation by other intermediaries, inability to track a product, and accessibility to a good market data. To accomplish this task, the research included AI price forecasting, buyer matching algorithms, and Blockchain Traceability of products transparent participants and m-commerce ensuring the protection of transactions. These technologies were combined into a mobile platform that is based on the Digital literacy and resource endowment of rural farmers.

The research is based on the Design Science Research (DSR) approach, which is an appropriate way to develop and evaluate the innovative IT solutions that can solve the real-life problems. According to (Peffers, et al., 2007), DSR entails defining a problem, coming up with the solution, constructing an artifact as well as assessing its success. This is an organised form of research which allows the researcher to come up with useful contexts and provides a theory. By this study, DSR assists in the iterative construction of the Spicely application and offers a framework within the context of verifying its usefulness and effectiveness.

To be responsive and user-driven development process the research assimilated Agile method of software development. Short iterative development is made possible by agile methods, which fits into short development called sprints where functional parts of the app are rolled out and tested gradually. (Beck, 2001) claim that Agile promotes a constant user feedback and changeability to evolving requirements, which is of particular importance when opening to non-technical customers including farmers and rural traders.

Mixed-method research methodology has been used to collect the data to measure the system. This has entailed qualitative methods that comprise semi structured interviews together with focus group discussion to know the actual challenges and expectations of both farmers and buyers. The survey of the performance, usability testing and the survey on user satisfaction were also applied as quantitative methods in validating the performance of the system. (Tashakkori & Teddlie, 2003) claim that qualitative and quantitative studies show more and more comprehensive and reliable results when they are combined.

Description of methodology in details does not only make the research more transparent and reproducible but also can enable future researchers and developers to replicate or expand the study. An effective model to be used in developing a reliable, user-friendly, and socially sound digital platform can be created through combining Design Science Research, Agile development, and mixed-methods evaluation.

3.2 Research Design

Research design is the comprehensive approach, which combines all the various elements of the research together logically and coherently to facilitate the solution of the research problem. As part of this project, the objective was to design and test a mobile application that utilizes AI usage, blockchain, and mobile commerce to bridge the gap between small-scale spice farmers in Sri Lanka and foreign consumers. The quantitative evaluation techniques were considered to create a functional IT artifact under a design science research (DSR) paradigm.

Design Science Research suits well in projects whose aim is to construct new solutions to real-life issues using pieces of information technology (Peffers, et al., 2007). This type of research methodology champions an intensive process which involves the identification of the problem, the definition of requirements, artifact building and testing, and artifact drawing implications on theory and practice. Through DSR, not only were the Spicely developed following the needs of the users, but the entire process was also tested and checked in terms of user feedback that guaranteed that the product will be relevant in academic and practically efficient courses.

To further guarantee that the system positively solved end-user difficulties, Agile principles of development were integrated in the design of the research. Agile works well in the development of applications in changes environments; making changes in user needs and technical limitations change (Beck, 2001). It promotes the approach of an iterative development in the form of small and manageable sprints, where constant feedback is possible and the system may be improved. Using Agile, the development team could subsequently release usable app features (e.g. product upload, AI price prediction, buyer messaging) in turns and subsequently determine their usefulness and make timely adjustments.

In addition, a mixed-methods approach was embraced in the research whereby the study integrated the qualitative and quantitative research types to provide a clear understanding. Qualitative data collection methods were used to interview smallholder farmers and buyers in semi-structured interviews and focus groups to find out about their areas of pain, expectations and usability issues. This droves the needs and graphical user interface. The quantitative measures were then determined by means of usability testing, survey answers (along with SUS scores) and the system performance measures, which provided an objective measurement of the quality of this application in connection with user's needs (Tashakkori & Teddlie, 2003).

The integration of design science and Agile methodology and mixed-methods research allowed getting a complete and versatile design approach that can resonate with the multidisciplinary focus of the project. It makes sure that the artifact is relevant both technologically and friendly to the socio-economic situation of agricultural context of spice farmers in Sri Lanka. Such research design will finally make it possible to deliver a scalable, inclusive and impactful digital solution.

3.3 Software Development Model

In the chosen research, the software development method will be the Agile Software Development Model, namely, the principles of Scrum will be included to work with an iterative approach and ongoing user interaction. The reason why it was used in this model was because it is flexible, quick in delivery and therefore a very powerful model when it comes to obtaining stakeholder feedback hence of great benefit in areas that evolve quite rapidly like agriculture technology and development of user-centred system (Beck, et al., 2001).

Although Agile development is very flexible since the software project is broken down into small and manageable parts referred to as sprints. At the end of every sprint, a working portion of the application is produced that may be subjected to testing and stakeholder review and enhanced accordingly. Each sprint, in the case of the project, took about 2 weeks and involved the creation and improvement of separate modules, including, but not limited to, user registration, product listing, AI-assisted price prediction, blockchain traceability, and buyer communicator. This strategy resulted in development that would be consistent with the changing user requirements and technical needs.

Among the many strengths of Agile, the focus on early and continuous delivery of valuable software seems the best appropriate to the overall vision of creating a minimum viable product (MVP) and testing it in the real-life conditions of an agricultural environment. They gathered feedback by test users (mostly smallholder farmers and buyers), after every sprint which enabled the development team to work on usability concerns, feature gaps and contextual challenges including language barrier and connectivity.

The Scrum regime also led to the efficient work team in the development team. Positions like the Product Owner (who has the priority on the features), Scrum Master (who controls the process), and Developers (who constructs the system) were identifiable. Using regular meetings such as a sprint planning, daily stand-ups, sprint reviews and retrospectives, development was done with efficiency and with the quality and user in mind.

Finally, the Agile model was advantageous in terms of supporting an adaptive, expandable, and user-friendly application. It helped in provision of a strong system architecture and to be flexible to provide iterative enhancements. Not only did this development model improve the quality of the software, but also made sure that the product Spicely, was not only practically relevant, but also technically good enough, to be used by the target users in the rural agricultural areas.

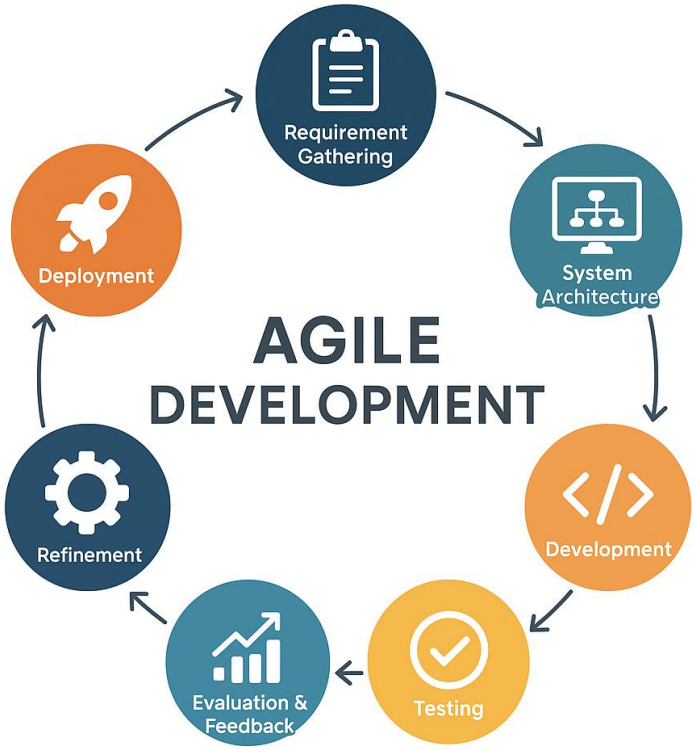


Figure 2 : Agile Software Development Model

Source: Author's own (2025)

3.4 Functional Requirements

- User Registration and Authentication

Users should access the system through secure registration processes by utilizing their mobile phone number as well as their email or social media accounts. When users sign up they will be identified as farmers or buyers. Role-based access control provides a system which grants users the ability to reach appropriate functions based on their assigned roles. A two-step authentication system involving secure one-time password verification or password-based encryption will manage user access.

- Product Upload and Management

The system must enable farmers to post specifications about their various spice merchandise. A complete listing requires the spice type, a product description, quality grade, available quantity, optional expected price and one or more images. Real-time inventory management demands that the system must enable users to make necessary changes to their listings as desired.

- AI-Based Price Prediction

The AI platform should generate automatic price advice for farmers through data monitoring historical price point changes, spice quality standards, seasonal market amounts and universal market shifts. The information system includes automatic price suggestion capabilities which enable farmers to receive appropriate market value for their products. This system will receive new information to enhance its precision automatically in the long run.

- Buyer Matching and Recommendation

A matching system should exist within the application to present recommended buyers for all spice listings. The recommendation engine needs to examine buyer preferences along with purchase records and product needs to present farmers with best-suited match possibilities. The designed system minimizes farmer search efforts while boosting selling efficiency through this feature.

- Product Browsing and Filtering for Buyers

A system must provide an interface that allows users to search and filter available spice listings. The system will enable viewers to sort spices according to their type, quantity, quality, location and price point. Buyers will make purchasing decisions with confidence because detailed product views show images, origin information, AI-recommended prices, along with traceability logs.

- Blockchain-Based Traceability

All spice products submitted in the application need to possess blockchain-enabled traceability records. The blockchain system contains a set of information which includes the farmer's anonymized identification, harvest timing and product information and transaction records. The cryptographic mechanism used in blockchain technology offers both unchangeable data and confirmable information for customers and regulatory groups to trust.

- Multilingual Support

The system must provide three language options which enable users to choose either English or Sinhala or Tamil. The application enables users to choose their language preference while signing up or through the settings menu. Users from all backgrounds can access the program through this language feature because it supports multiple languages.

3.5 Non-Functional Requirements

- Performance

The system needs to operate efficiently across various network conditions together with changing user loads. The performance specifications for essential operations like login should match product listing and AI price prediction functions for which system delays should reach three seconds maximum. Those users located in sparsely populated regions need fast-performing systems since their mobile networks provide constrained speeds. The infrastructure should receive optimization for handling active service requests simultaneously while avoiding system downtimes.

- Usability

The digital interface needs to simplify its operations because small-scale farmers use digital tools at different levels of competence. The platform's interface needs to provide a guided and intuitive design along with simple labels and a low text complexity for its icons. The application needs to adopt accessible design principles which incorporate readable fonts along with high-contrast visuals and large interactive elements for providing smooth use to new and experienced users alike.

- Scalability

The platform needs capabilities for growth through expansion of user base as well as increased product listings and new features. The system must provide capabilities to scale horizontally through different approaches that include additional servers or cloud instances deployment. The application maintains its scalability through these features because new user groups including cooperatives and exporters can join without requiring major changes to the design structure.

- Security

Safety remains vital for both protecting user-sensitive information and processing financial operations. The system should implement HTTPS encryption for secure data transmission with AES-256 encryption to safeguard all passwords and payment details and personal information. An enforced policy of role-based access control combined with secure session management must exist. The implementation of blockchain must include a security feature to create tamper-resistant records that track both transactions and traceability data.

- Reliability and Availability

The system requirements emphasize dependability because it needs to function inexactly while also ensuring small periods of breakdown and few mistakes. A cloud-based hosting system must exist with both high availability capabilities and redundant features enabled. The backend system needs built-in error handling combined with failover methods alongside data recovery protocols which guarantee uninterrupted service operations along with safeguarded user data against system failures and all network connectivity issues.

- Maintainability

Users need access to a framework which enables easy maintenance and debugging functions and room for updates. The maintenance approach requires modulo development while creating distinct program sections (authentication and AI component and blockchain implementation) alongside extensive documentation creation. The system design should allow developers to amend specific program blocks independently, so maintenance becomes more enduring.

- Compatibility

The application needs to support various Android devices especially older devices which rural farmers routinely use. A functional Android version starting from 6.0 Marshmallow must run on the application which needs only a small amount of device storage space to extend its availability to various users.

- Localization and Multilingual Support

The effective accessibility demands the app supports Sinhala Tamil and English languages for all user interface points. The application should support localization of content and every format of text alongside date displays and currency presentation and all region-specific phrases. Users should access the language switching option from the application settings menu for easy multilingual change at any time.

- Legal and Ethical Compliance

The application needs to meet all data protection standards and ethical requirements through safeguards for personal data together with informed consent requests during data collection. Special attention needs to be applied to blockchain records as well as AI profiling activities. The application needs to display terms of use information while users should grant permission before the collection or handling of sensitive data occurs.

3.6 Target Population and Sampling & Data Collection Methods

3.6.1 Target Population

The focus of this research includes Sri Lankan farmers who maintain small spice operations especially those growing valuable crops such as cinnamon pepper and cardamom. Sri Lankan farming communities struggle to gain position in worldwide supply chains because they depend on brokers and have minimal understanding of digital technology and limited access to market data. The study features four additional important groups as participants: international spice buyers, exporters, agriculture officers and technology experts who contribute to digital agricultural trade platforms with their expertise in supply chain management.

The research analysed the farming communities which maintain spice cultivation in rural districts of Matale, Kandy, and Galle districts. The study focused on these areas because they depended heavily on spice exports which made them essential for understanding the research objectives.

3.6.2 Sampling Strategy

The study adopted purposive sampling because this method targets specific individuals or groups according to the research problem (Creswell & Plano Clark, 2018). Further assurance was provided by this approach because it allowed participants to experience directly the problems the mobile application addressed before providing their feedback.

The sampling groups included:

- Thirty farmers who operate small-scale spice operations belong to selected districts.
- A total of 10 international businesses operating as export and sourcing entities work with Sri Lankan spice producers.
- A team of 5 agriculture officers together with 5 IT experts handle both trade facilitation and digital agriculture system operations.

A sample of 30 small-scale spice farmers along with 10 international buyers/exporters and 5 agriculture officers and IT experts was deemed suitable because it enabled diverse perspective collection within the given resources and time parameters for qualitative depth and quantitative user experience assessment.

3.6.3 Data Collection Methods

1. Semi-Structured Interviews (Qualitative)

A set of selected spice workers, exporters and agri-officers participated in one-to-one interview sessions. The interview method investigated various key aspects.

- The direct sale of agricultural products encounters difficulties from farmers aiming to sell their products.
- Digital readiness and familiarity with mobile apps.
- People anticipate how technology should assist their trade operations and their pricing activities.

The technique yielded valuable information about respondents' complex situations along with their attitudes and actual trade barriers they faced.

2. Focus Groups

Two meetings with six to eight participants took place in spice farms. The meetings with stakeholders served to create valuable group wisdom regarding essential application features as well as language preferences and expected traceability functions. Through group discussions participants enhanced each other's input to expose universal issues they faced.

3. Surveys and Questionnaires (Quantitative)

I distributed structured surveys both during and after the prototype tests to gauge three factors:

- Ease of use
- User satisfaction
- Users show trust in both the pricing algorithms running their transactions and blockchain-based verification systems.
- The subjects expressed willingness to use the application for concrete business transactions.

The evaluation process incorporated Likert-scale together with multiple-choice questionnaires for analysis purposes.

4. Field Observations

During training sessions within pilot areas, the researcher watched farmers interact with the prototype. The observation of farmers during testing sessions delivered feedback about how well the system worked for them and their device choices and preferred languages.

3.7 System Architecture

This architectural design framework provides support for developing and operating and extending the functionality of the proposed AI-powered mobile solution aimed at small-scale spice farmers. The designed platform unites three advanced technologies AI along with Blockchain together with m-commerce to fulfil application requirements both functionally and non-functionally.

The system operates using a Three-Tier Architecture Model with Presentation Layer and Application Layer and Data Layer as its components. The modular design enables better division between application sections while making the system easier to maintain and enables larger user bases.

3.8 Architecture Overview

Layer	Role	Key Technologies
Presentation Layer	User interaction through the mobile app interface	XML, Android (Java)
Application Layer	Business logic, AI processing, blockchain interaction	Node.js, Express.js, Python (TensorFlow, scikit-learn), Solidity

Table 3 : Architecture Overview

3.9 Presentation Layer (Frontend)

All user-related operations occur at the presentation layer. The application uses Android SDK (Kotlin/Java) development to support a wide spectrum of smartphones including low-end rural phones. Key features include:

- User registration/login interface
- Users can upload products through screens that allow them to add images together with description and quality criteria.
- AI-generated price display
- Buyer matching results
- Blockchain-verified traceability view
- Multilingual support (Sinhala, English)

The planning of user interfaces follows an optimal path by implementing basic navigation functions together with easy-to-use buttons as well as straightforward texts for farmers who lack digital literacy knowledge.

3.10 Application Layer

This system layer functions as the central processing unit since it handles essential decisions and main computations. Key components include:

- The system implements Firebase Authentication for user authentication and authorization through email accounts and phone numbers as well as social media logins.
- The AI Price Prediction Engine uses Python to create predictions regarding optimal spice market prices through analysis of historical sales figures and product qualities with consideration of worldwide market trends.
- The AI-based classification system uses Buyer Matching Algorithm to propose suitable buyers through combining information about product type and volume with buyer preferences.

RESTful APIs operate as the communication mechanism that enables back-end and front-end service interaction in this layer.

3.11 Data Layer

Secure reliable data storage occurs in the data layer which integrates two major components: Firebase Realtime Database and MongoDB Atlas databases.

- Firebase Realtime Database: Stores user profiles, product listings, price prediction and histories records.
- The product data consisting of large unstructured information such as images and farmer certificates gets stored within MongoDB Atlas.

3.12 Key Functional Flows

Farmer Workflow

- Account Registration

To make an account, the farmer puts in his/her personal information including the name, preferred email address, farm position, crops cultivated and his/her preferred language.

- Dashboard Access

After logging in, the farmer would be taken to a custom dashboard of the current listings, incoming orders and market trends.

- Add Spice Product

The farmer can post information on spices, e.g.:

  - Product name
  - Images
  - Harvest date
  - Quantity (kg)
  - Selected variety
  - Pricing per kg
  - IoT data (optional for traceability)

- AI Price prediction

The farmers are provided with the price forecasts on a month-by-month basis depending on such inputs as regional location, season, demand, supply, rainfall, temperature.

- Buyer Matching

AI in the system helps to automatically find interested buyers for the products of the farmer depending on the preferences of the buyers and the product categories.

- Chatbot Interaction

The farmer may get in touch with buyers via chatbot integrated on the interface to negotiate rates, provide additional information, and validate orders.

- Order Confirmation and processing

As soon as both sides come to an agreement, the order is accepted using the application, and information on the transaction is stored (optionally, on blockchain, which ensures its integrity).

Buyer Workflow

- Buyer Registration

The buyer registers themselves by writing business information, interested type of spices, location and contact details.

- Marketplace Browsing

The buyer uses the spice marketplace to search, making use of categories, spice name, product type, price, and origin, as well as location of supplier.

- Machine Learning Price Forecast

Consumers get to see price forecasts of the chosen spices based on real-time market trends and predictive AI analyses.

- Buyer-Farmer Matching

The system recommends best-compatible farmers and products depending on the search history of the buyer, his/her preferences, and market trends.

- Chatbot Communication

Chatbot allows buyers to start a conversation with the farmers to request some samples, negotiate or clarify the quantities of orders.

- Transaction Finalization

The buyer accepts purchase on mutual agreement basis after which logistics are delivered or integrated through the platform.

3.13 Technology Components

Component	Technology Used
Mobile Platform	Android (Java)
UI Design	XML Layouts
Backend Server	Node.js + Express.js
AI Module	Python
Blockchain	Ethereum, Solidity, MetaMask Integration
Database	Firebase Realtime DB
Hosting / Cloud	AWS EC2 / Google Cloud Functions
Security	JWT Authentication, AES Data Encryption

Table 4 : Technology Components

3.14 Use Case Diagram

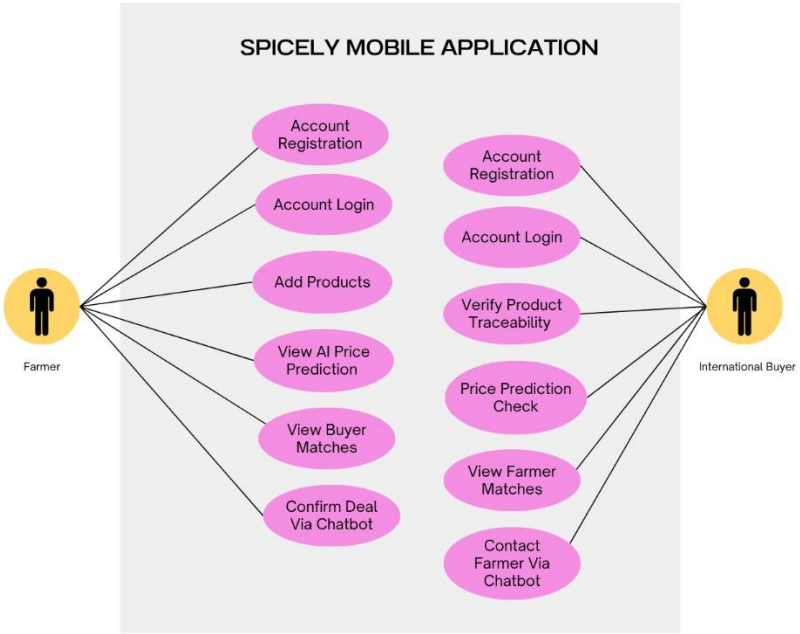


Figure 3 : Use Case Diagram

Source: Author's own (2025)

3.15 Class Diagram

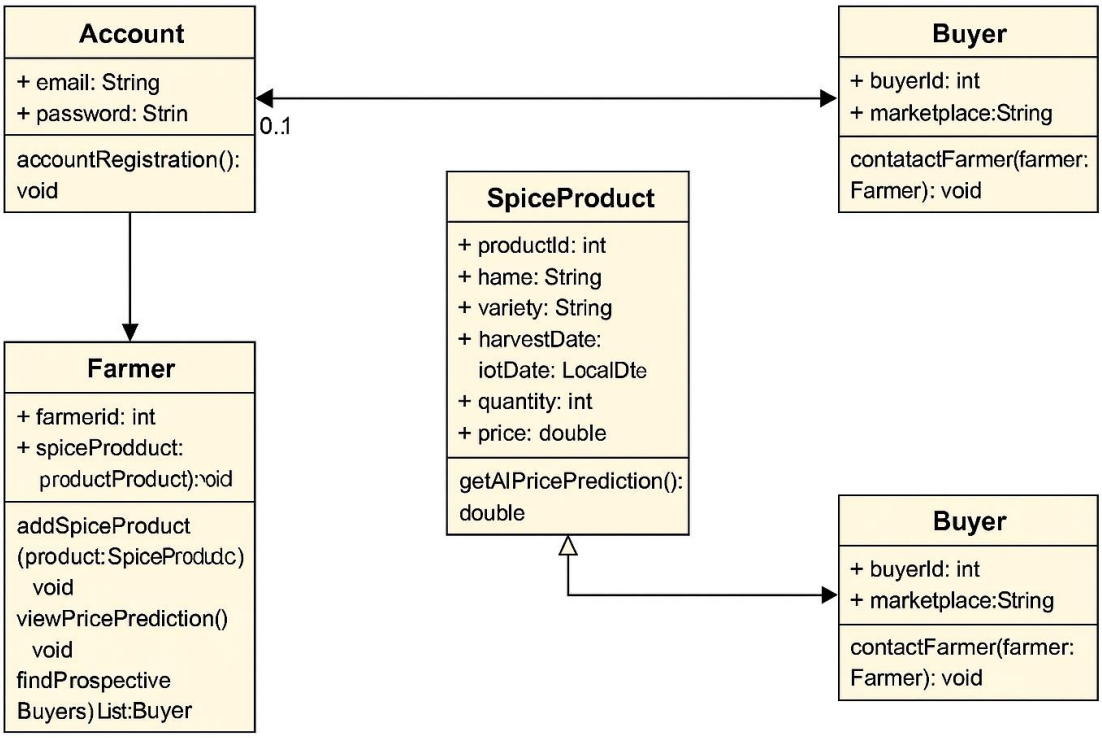


Figure 4: Class Diagram

Source: Author's own (2025)

3.16 Frontend XML Codes Explanation and UI Designs

- Splash Screen 1

This XML structure establishes the initial screen of onboarding with the use of a Relative layout with the essential elements which are image logo (ImageView), the title and description (TextViews), and two buttons Skip and Next arranged in a horizontal LinearLayout. It moves towards the clean, centralized design to convey the purpose of the app and orient the client in the process of onboarding.

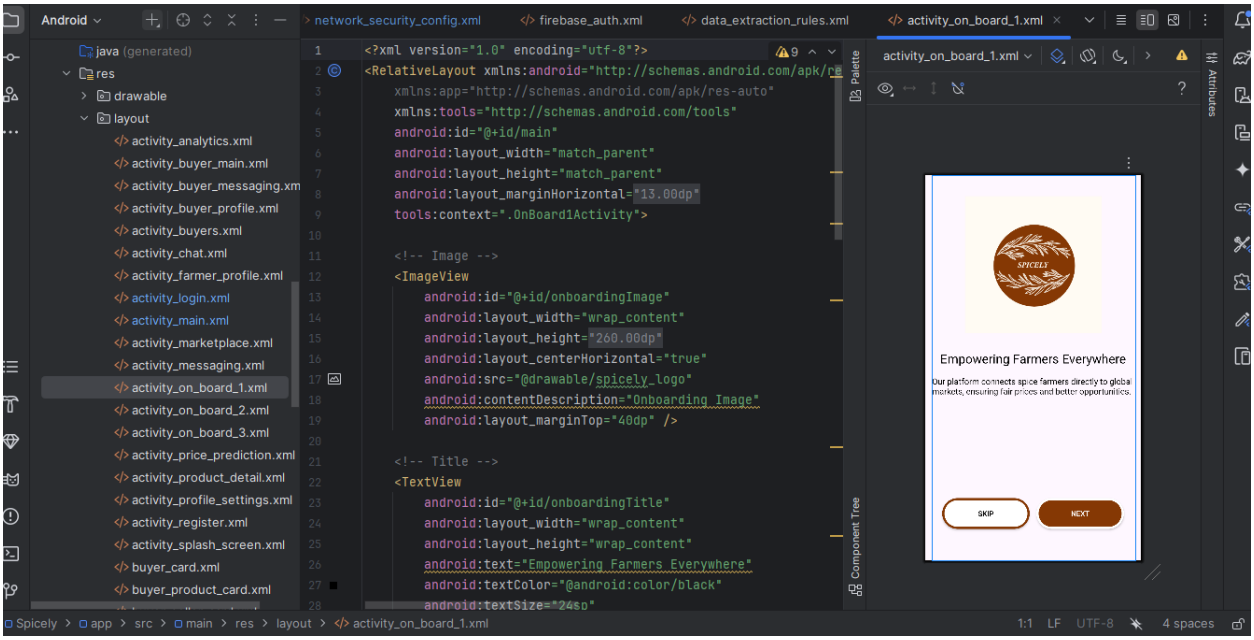


Figure 5 : Splash Screen 1

- Splash Screen 2

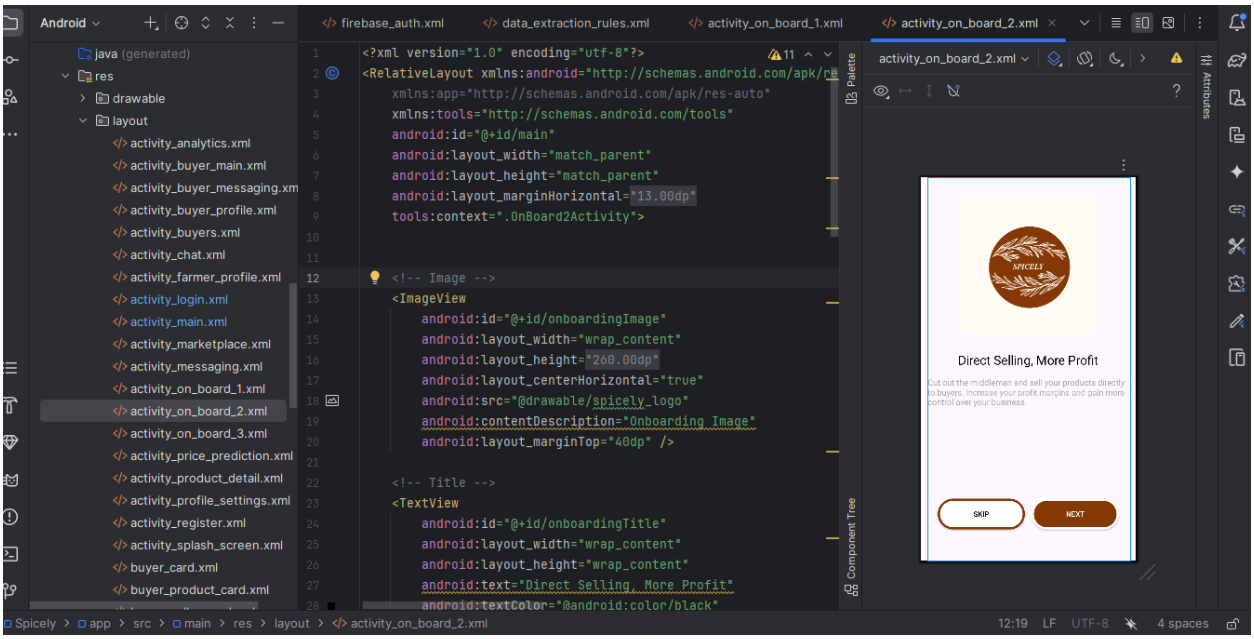


Figure 6 : Splash Screen 2

- Splash Screen 3

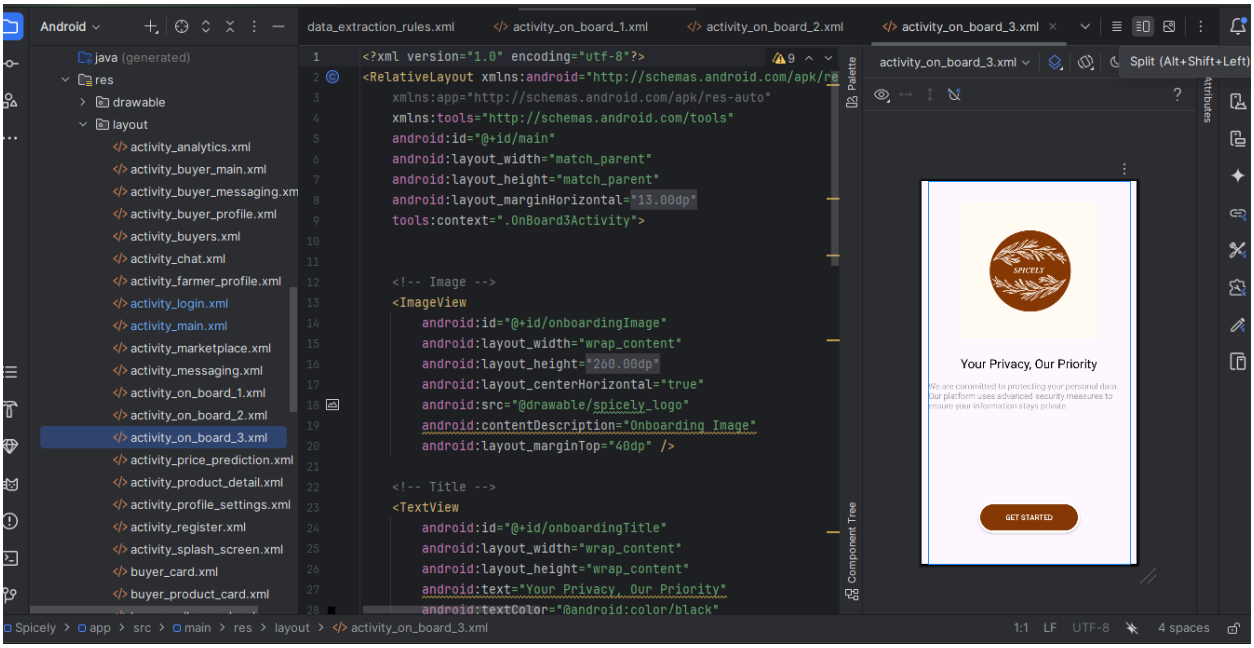


Figure 7 : Splash Screen 3

- Choose the Role Page

In this XML design, the registration screen of the Spicely mobile application will be set in which the user will be given the option to register as either a farmer or a buyer. A logo is shown at the top of the screen, and under it, there is a role selection area with a short description to show, and two buttons are located to select the type of the user.

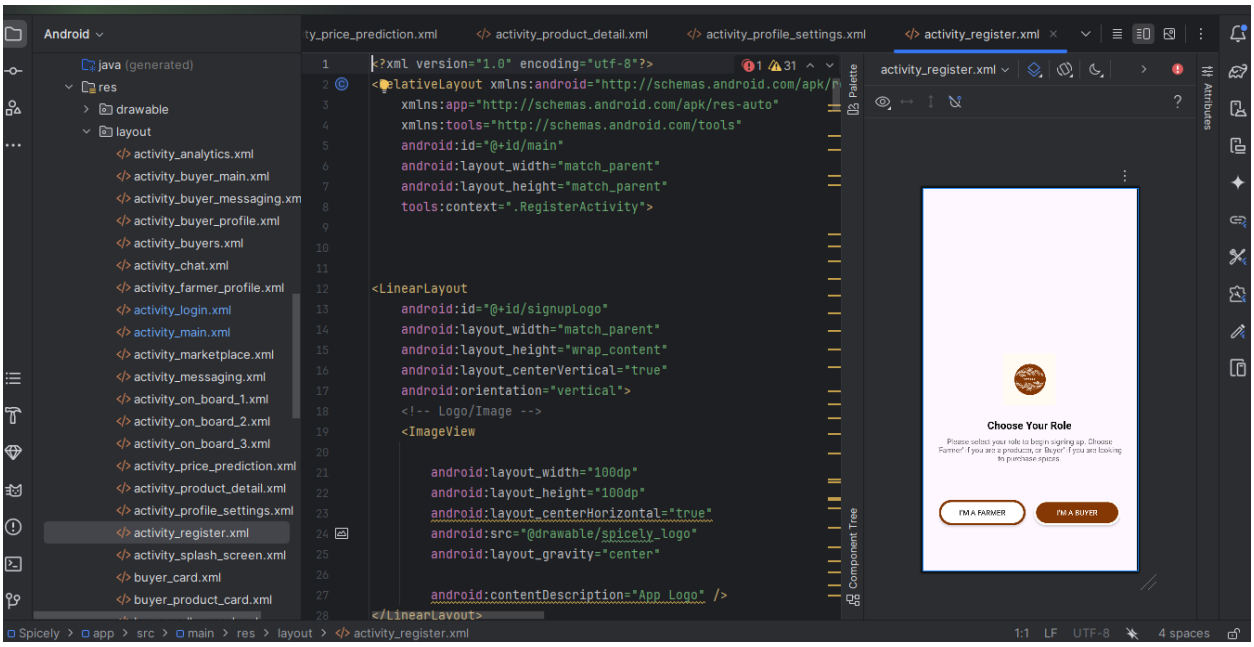


Figure 8 : Choose the Role Page

- Sign in Page

This XML layout is the login screen design for Spicely mobile application constructed with Linear Layout which is in vertical orientation. This includes centered logo image on top then followed by "Sign In" title. After that it is TextInputLayoutfor user's email and password. The password field has a visibility toggle icon, there's also a "Forgot Password?" Link is added under it.

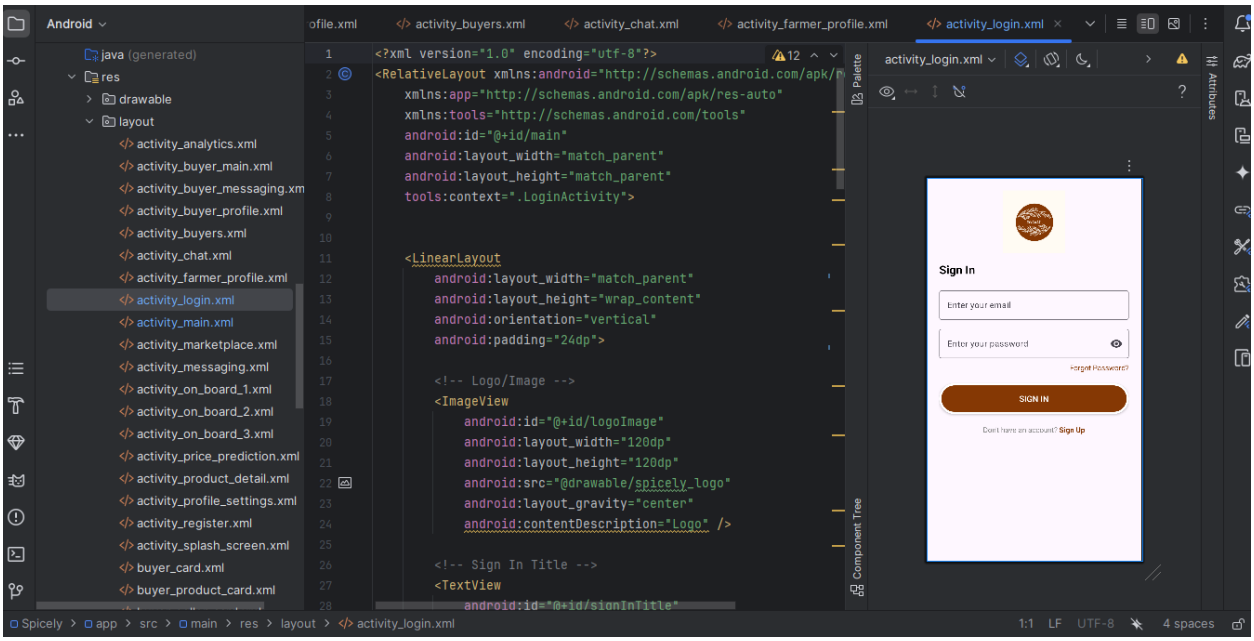


Figure 9 : Sign in Page

- Farmer’s Home Page

It is an XML layout that shapes the Market Trends (AnalyticsActivity) screen of the Spicely application whose implementation is comprised of a CoordinatorLayout to target the responsive scrolling experience with the aid of AppBarLayout and NestedScrollView. The UI will have three major sections in three CardView containers namely: Price Prediction, which will contain a line chart with a spice selection option; Best Time to Sell, which will contain insights into a recycler view; and Market Insights which gives the general trend such as an increase in demands, seasonality and premium pricing. They enable switching between core areas of an app by use of bottom navigation bar. The structure focuses on evidence-based determination among spice farmers.

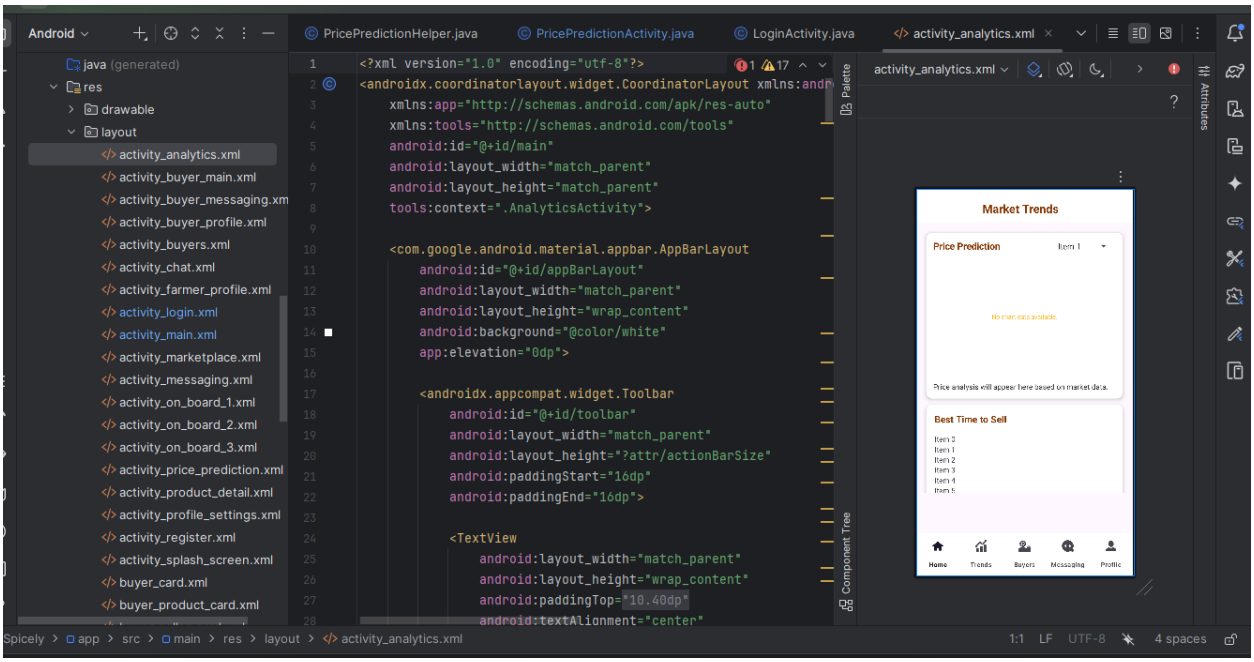


Figure 10 : Farmer’s Home Page

- Buyer's Home Page

This XML layout creates the Buyer Main Dashboard of the Spicely mobile application, which is created with the usage of CoordinatorLayout and NestedScrollView to make the interface easy to navigate and allow scrolling. The interface consists of a toolbar with the name of the app and a spices search bar and scrollable filter chips to the categories such as Pepper, Cardamom, and Cinnamon. The main content sections have the scrollable dynamic lists of New Offers, Popular Products and Favourite Sellers implemented through RecyclerView, to help buyers navigate to the listings of spices.

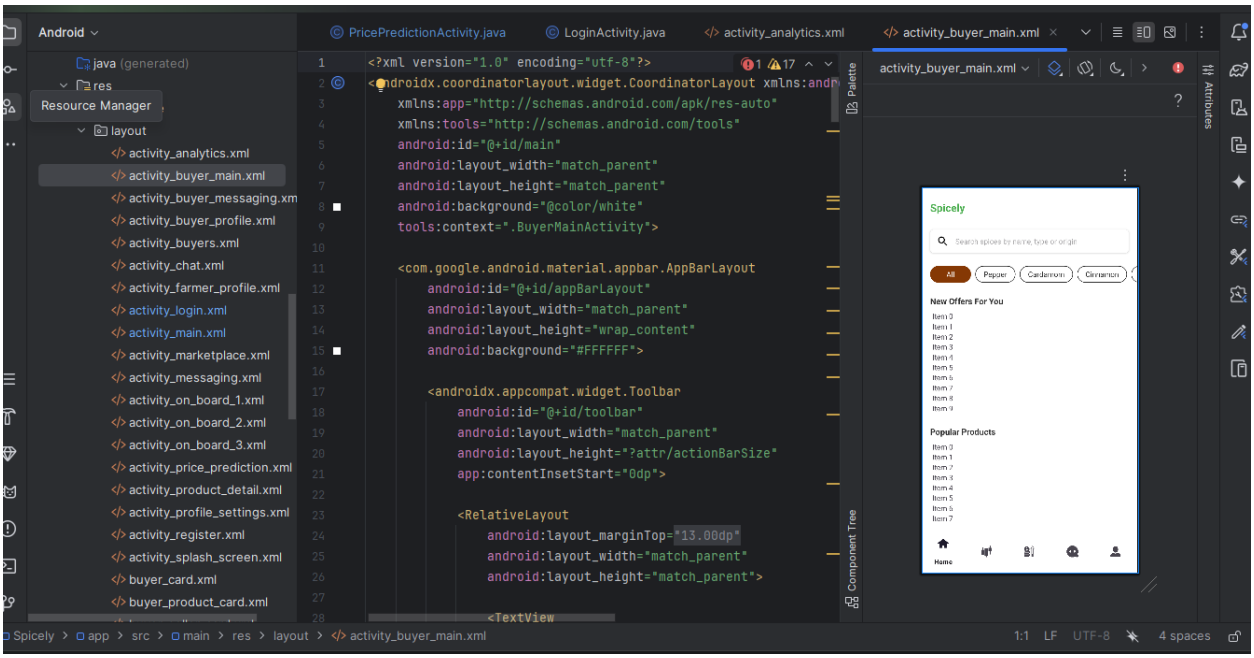


Figure 11 : Buyer's Home Page

- Buyer's Profile Page

This XML gives the design of the Buyer Profile Page of Spicely mobile app using a RelativeLayout and an adjustable Linear Layout so that it is responsive. It has a back button, a header and contains a series of fields to update the profile information including name, company, favorite spices and the region and the email field is read only. A section to upload a company image has been provided so that users can customize his/her profile. Some other features consist of a text link to change the password and the Update Info button that allows users to make changes. The fact that it uses TextInputLayout makes the UI of a text input prominent and clean based on Material Design.

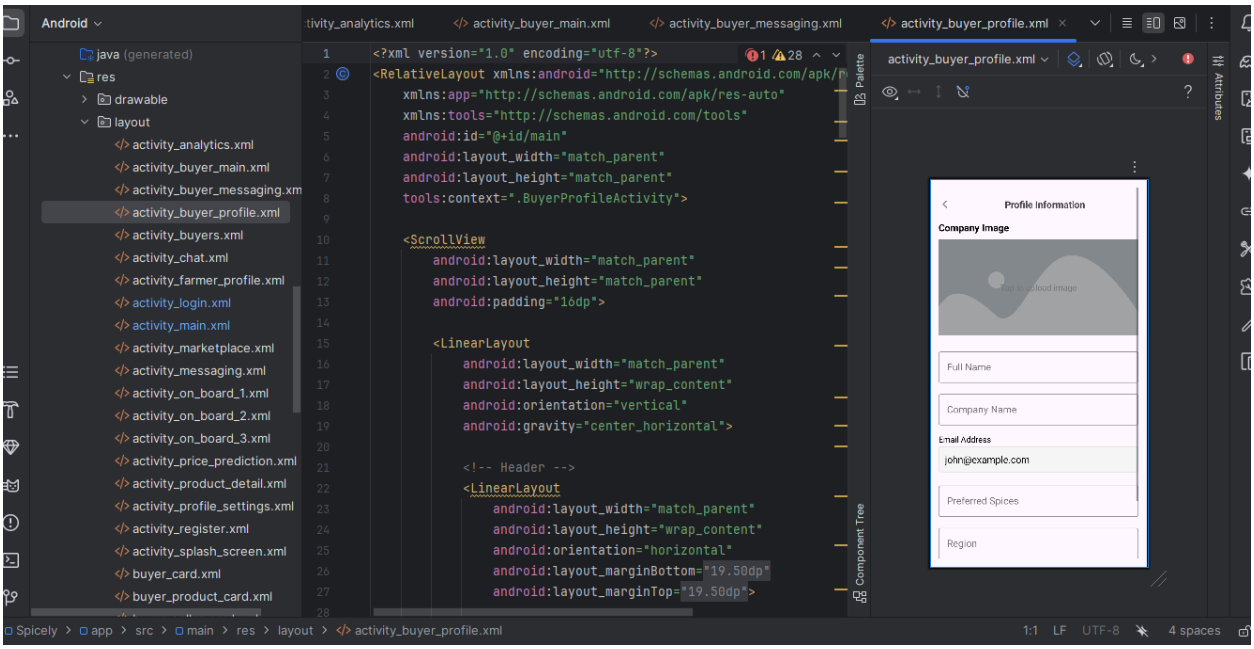


Figure 12 : Buyer's Profile Page

- Potential Buyer for Farmers Page

This XML structure is a definition of the screen, Potential Buyers in the Spicely farmer interface, where the design is created with CoordinatorLayout which is a container of scrollable content (NestedScrollView). According to the screen the feature of search enables the buyers to be filtered through the name, spice, or location. It also shows two RecyclerViews one of AI-generated recommended buyers and another one of all buyers. Its structure is based on CardView and LinearLayout to represent the structure of the content in a cool material design style, and it also allows the user navigation to have a bottom navigation bar that will make it easier to navigate between the sections of the app, making it far more intuitive and user-friendly.

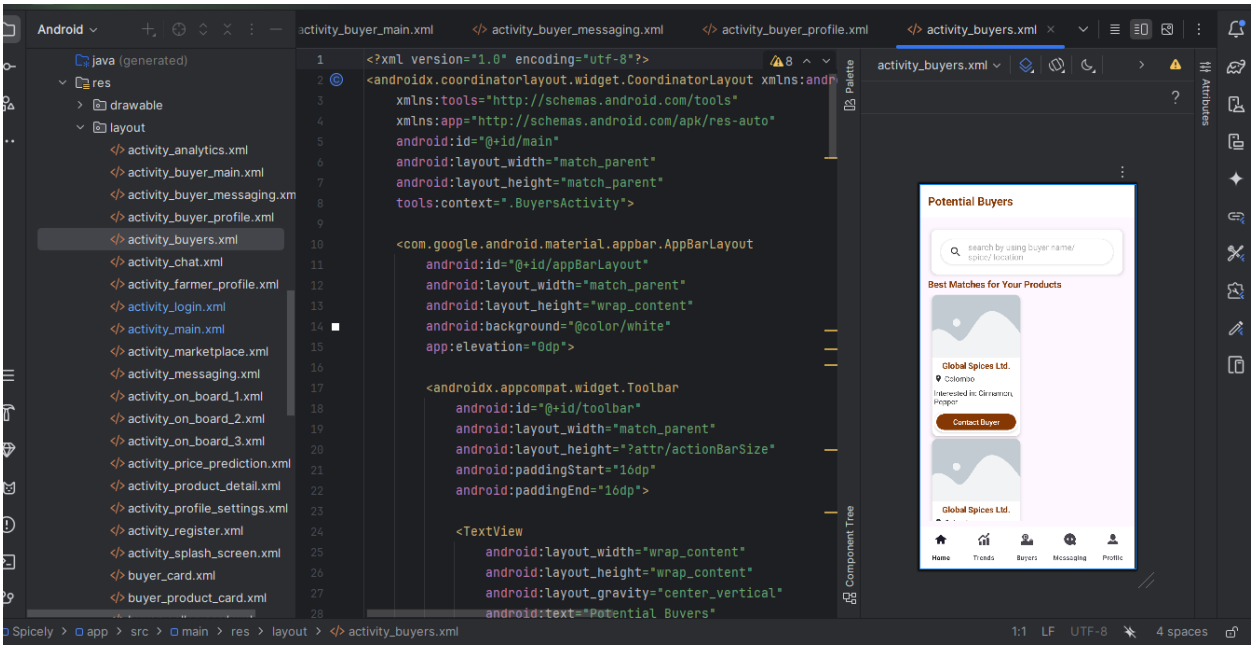


Figure 13 : Potential Buyer for Farmers Page

- Product Layout Page

This XML represents a custom card layout to display the individual product listings within the Spicely app of particular spice products. It displays an image of the product, a name of a spice, a variety, the date of the harvest, its quality, and a stock amount, the prices per kilogrammes, and a price in dollars in the vertically organized CardView. There is also the possibility of an additional IoT field value (e.g., soil moisture) and a delete icon to control the entries of the products. This interface empowers the user who, in this case, would be farmers to be able to present their spice inventory in a neat, neat, and easily understood manner in the application.

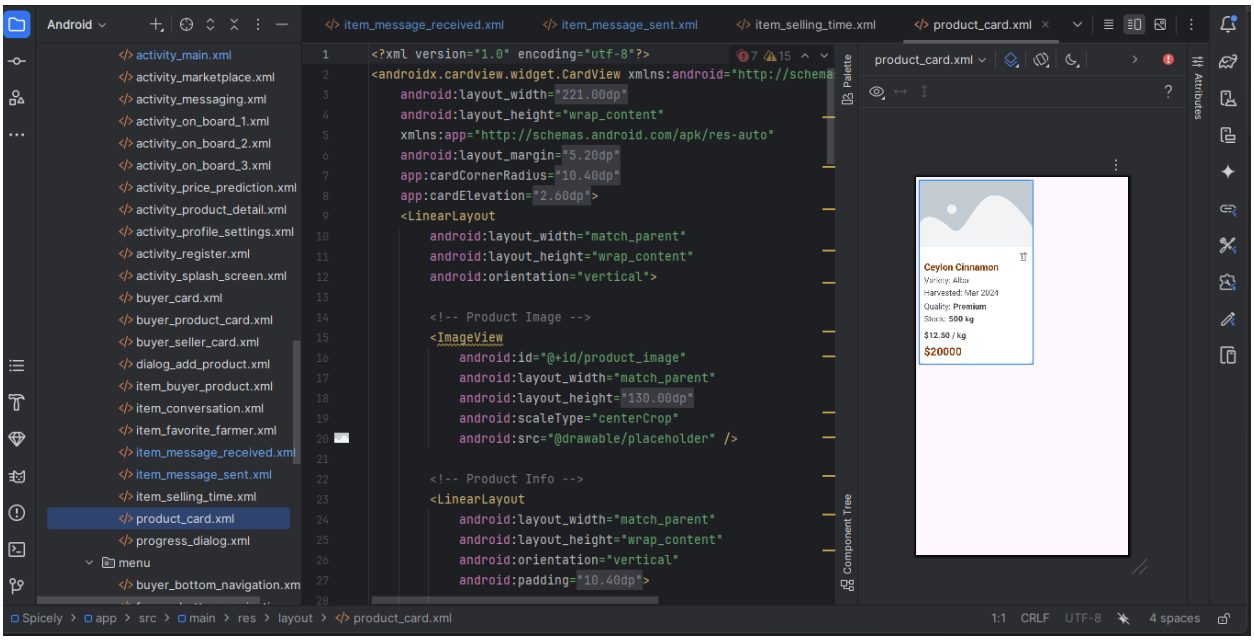


Figure 14 : Product Layout Page

- Buyer's Information Layout Page

This XML layout uses a custom card view to display details of buyers in a distilled yet attractive layout in Spicely app. There is a picture of the customer or the company, name and location of the company or the buyer as well as a small description of what kind of spice they are interested in. The users (e.g., farmers) can also message without any intermediaries the ease of contact through a clearly visible Contact Buyer button. The design enables fast identification and communication, which aids in simplifying the buyer-seller relationship process of the environment.

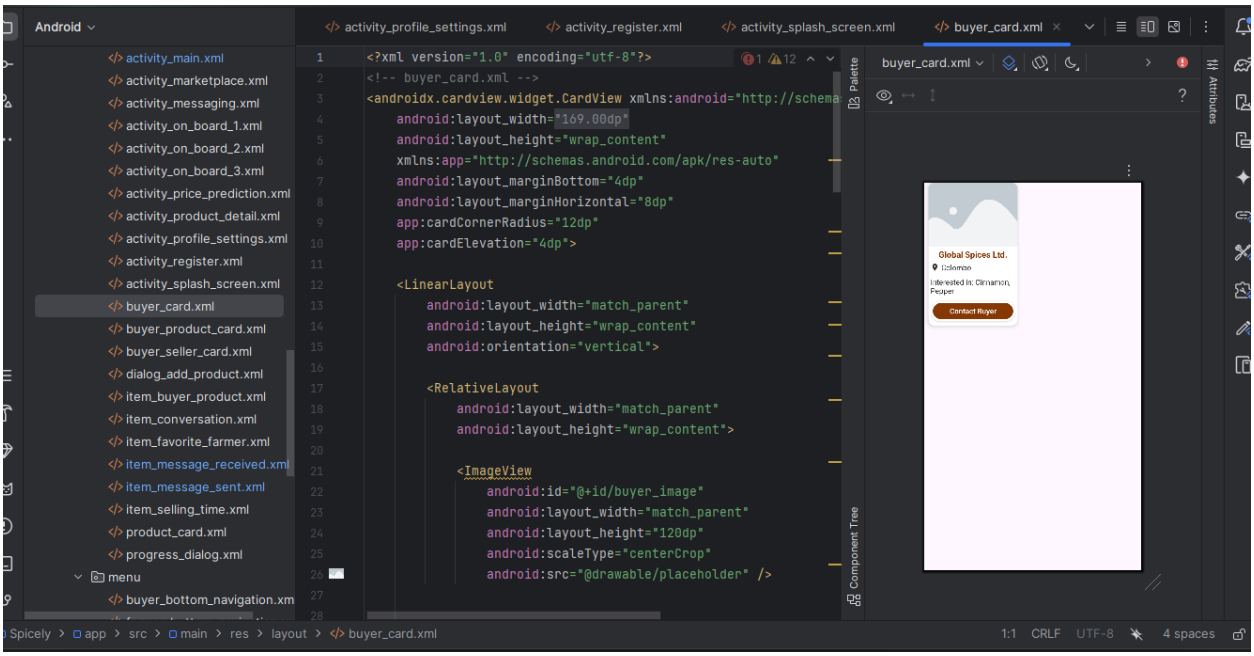


Figure 15 : Buyer's Information Layout Page

- Farmer's Information Layout Page

This XML uses a space saving seller card view that renders individual seller profiles in either a grid format or in a list. It contains a circular profile picture, a star icon to specify trusted sellers, the name of sellers, their address and a short sentence that indicates the number of products that a particular seller deals in (e.g., five products are made available). The placement is captured in the middle and is arranged in such a way that it aims at easing browsing and rapid recognition to provide a higher user experience when looking at several seller alternatives in the app.

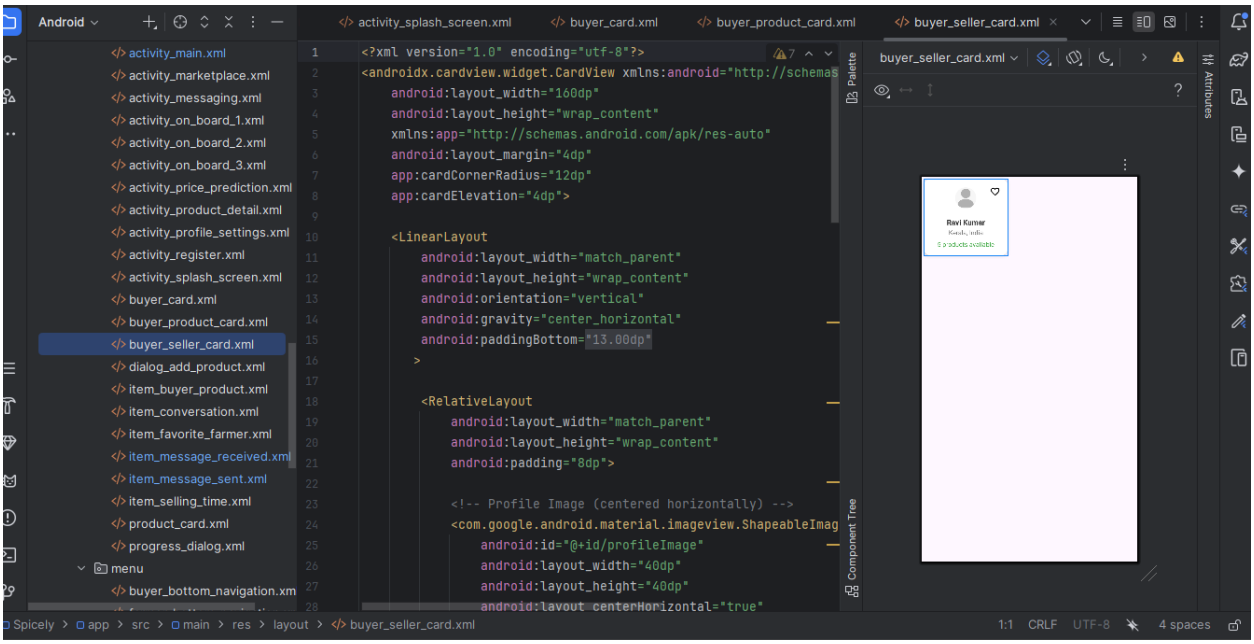


Figure 16 : Farmer's Information Layout Page

- Profile setting Page

It is an XML layout that obeys Profile & Settings screen in an Android app. It has a header containing a back button and title Profile & Settings and is followed by an interactive block which includes: Personal Information (redirects to user profile information page), Notifications switch (switches on or off notifications), Language (shows current language and gives user an opportunity to change it), and Help & Support (gives user help). All the sections are designed with the help of the LinearLayout and are appropriately padded, given elevation, and indicative of icons to achieve a lightweight and convenient presentation of the settings.

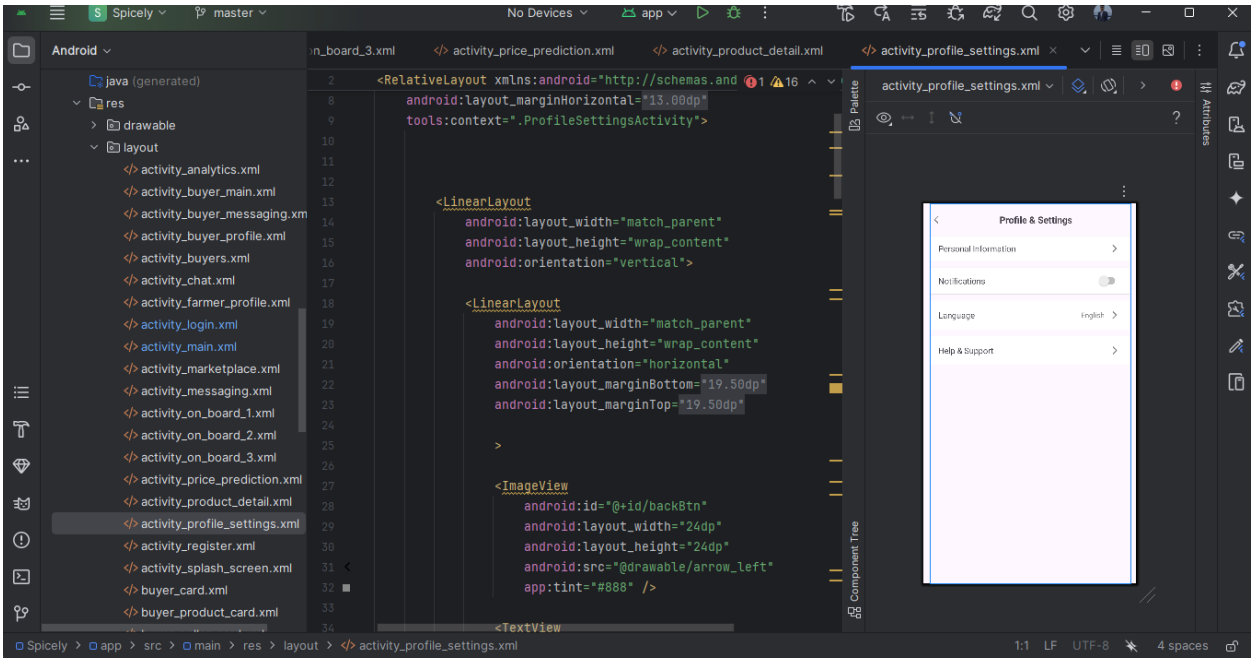


Figure 17 : Profile setting Page

- Farmer's Personal Information Update Page

This XML design is the screen of the Farmer Profile of an Android application. It has a scrollable form with which the farmer can edit and view the profile information. Important fields are name which can be edited, farm place, drop downs (spinners) of crops grown and different languages. The email is non-editable and it has a password change option. At the lower end, there is a Save button with the label that says Update Info. It is implemented in terms of ScrollView as the top-level layout surrounding vertical LinearLayout to accommodate vertical scrolling on smaller displays.

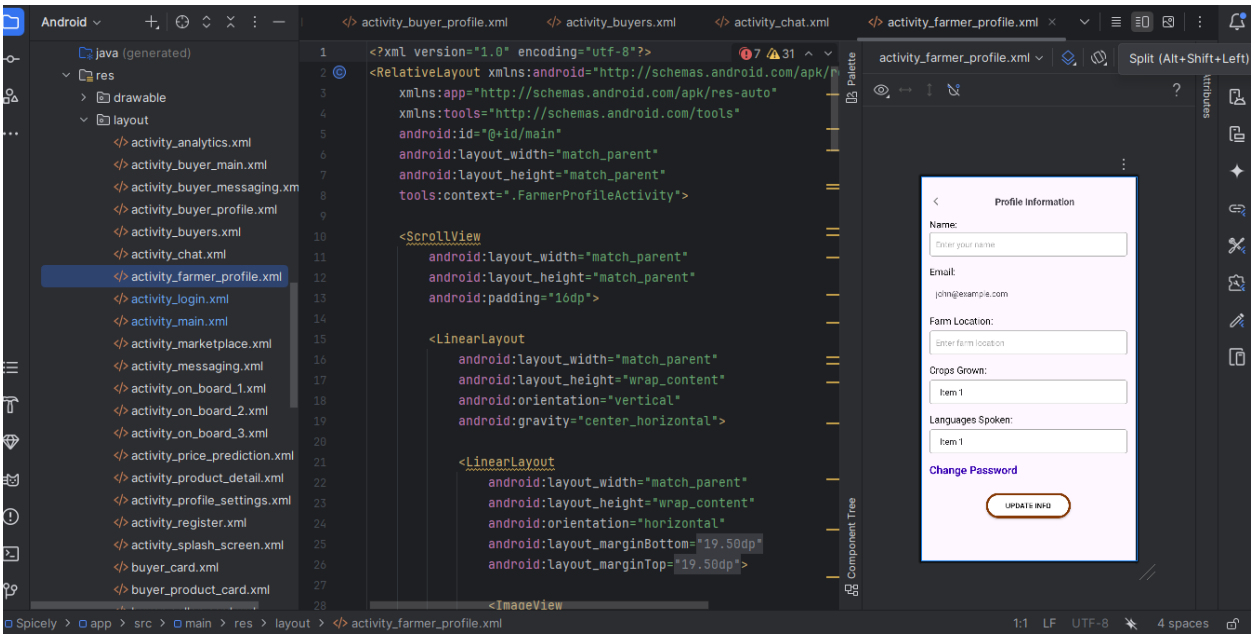


Figure 18 : Farmer Profile Page

- Buyer's Marketplace Page

It has a CoordinatorLayout containing an AppBarLayout with a Toolbar, search bar, and a group of filter chips to narrow down results (e.g. on the basis of spice types). Below it there is a NestedScrollView with the results count, a sorts option and a dynamic section of listing products (products_container) which will be filled programmatically. At the bottom there is a BottomNavigationView by which it is easy to navigate to other sections of the app.

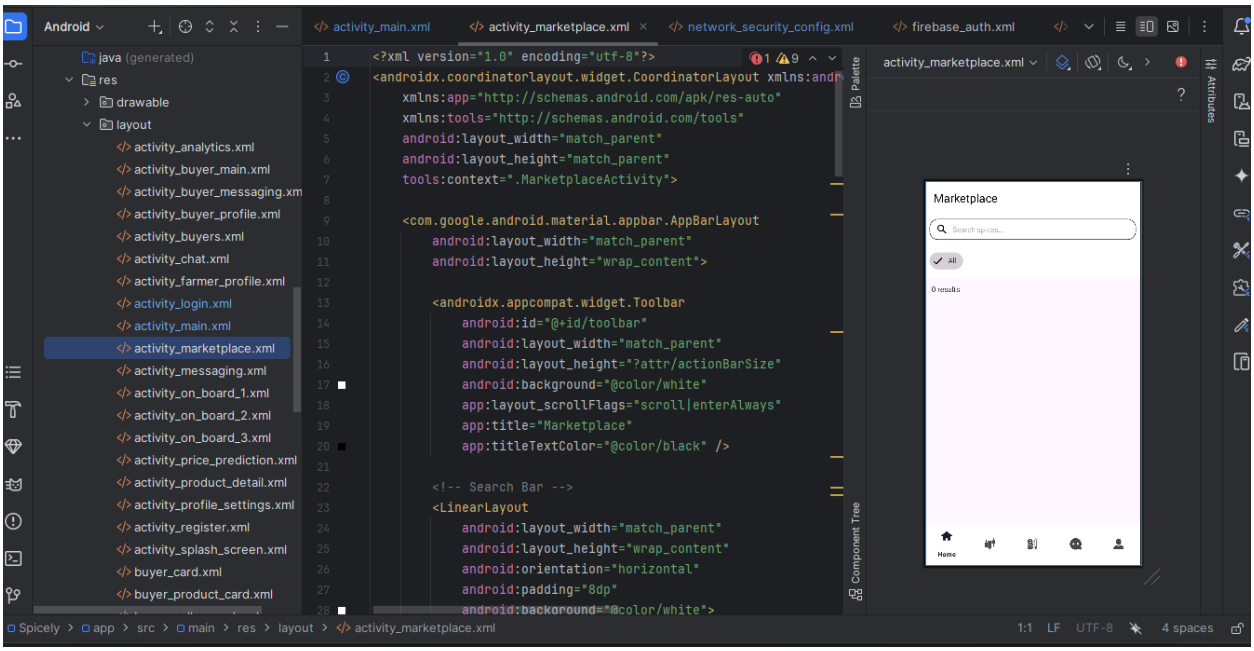


Figure 19 : Buyer's Marketplace Page

- Price Prediction Page

Its design has a vertical LinearLayout consisting of a title and a back button as a header, a card at the top where you see the prediction of the price and a scrollable interface that contains the input fields such as spice type, region, demand, supply, rainfall, temperature, month, and year. MaterialCardView unites into groups of fields that are logically organized and improve the UX. A Button labeled 0-Forecast will appear below and set the forecast in motion. The design is maximized in terms of usability, collection of data, and on-time results.

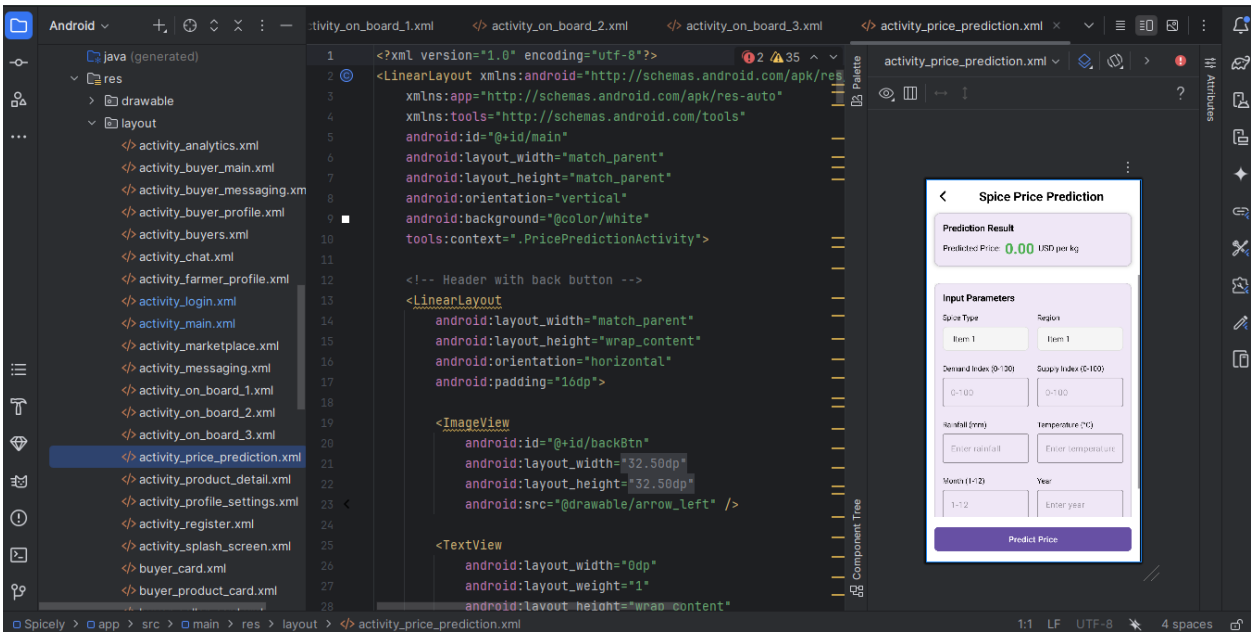


Figure 20 : Price Prediction Page

- Add Product Page

The design gives a vertical scroll that displays a title of Post a Product, uploaded image with a placeholder and overlay text, as well as a set of input fields Spice Name, Variety (dropdown), Harvest Date (picker), Quantity, Price, and optional IoT Data. By the use of two horizontally laid bottom buttons, the user may cancel or submit ("List") his/her product. The layout provides a simple and easy user-experience of form submission, which makes it fit inside the mobile market places as well.

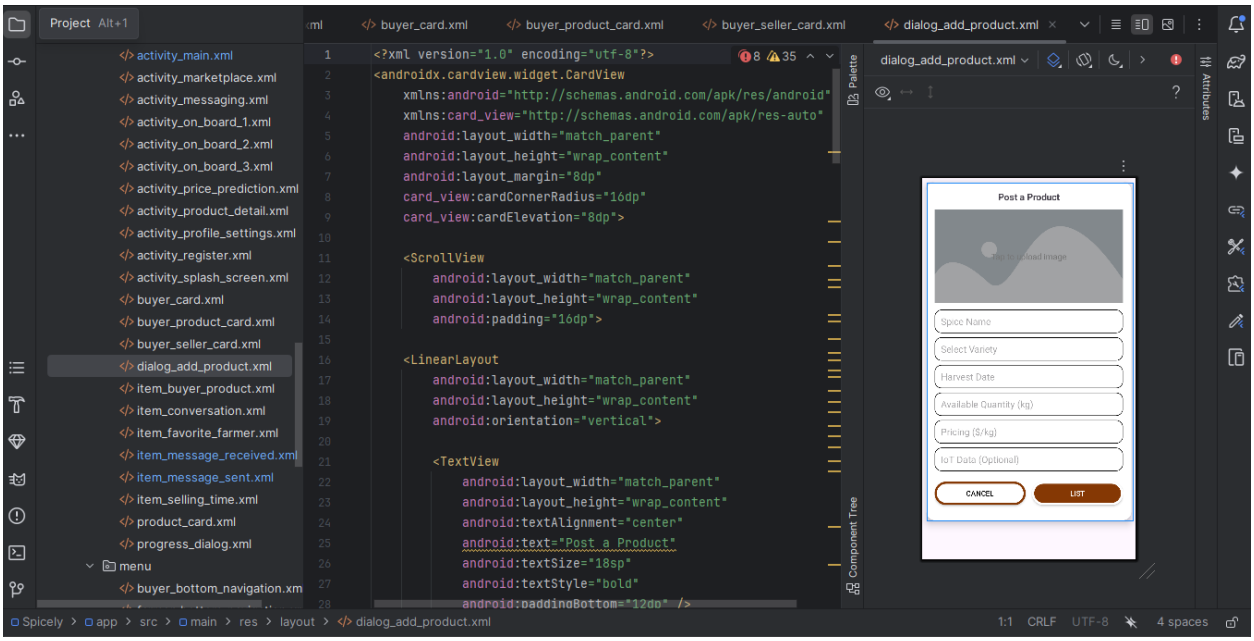


Figure 21 : Add Product Page

- Conversation Layout Page

The chart exhibits one-clickable clean chat, in which there is a circular profile icon that contains the initial of the user, then a user name, the seal of time, the name of a company and a last message glimpse. It also features unread indicator dot which notifies of unread messages. Its design employs RelativeLayout and LinearLayout to position these elements effectively and creates readability using appropriate margins, style and text constraints, it can also be used in a RecyclerView or in a chat list.

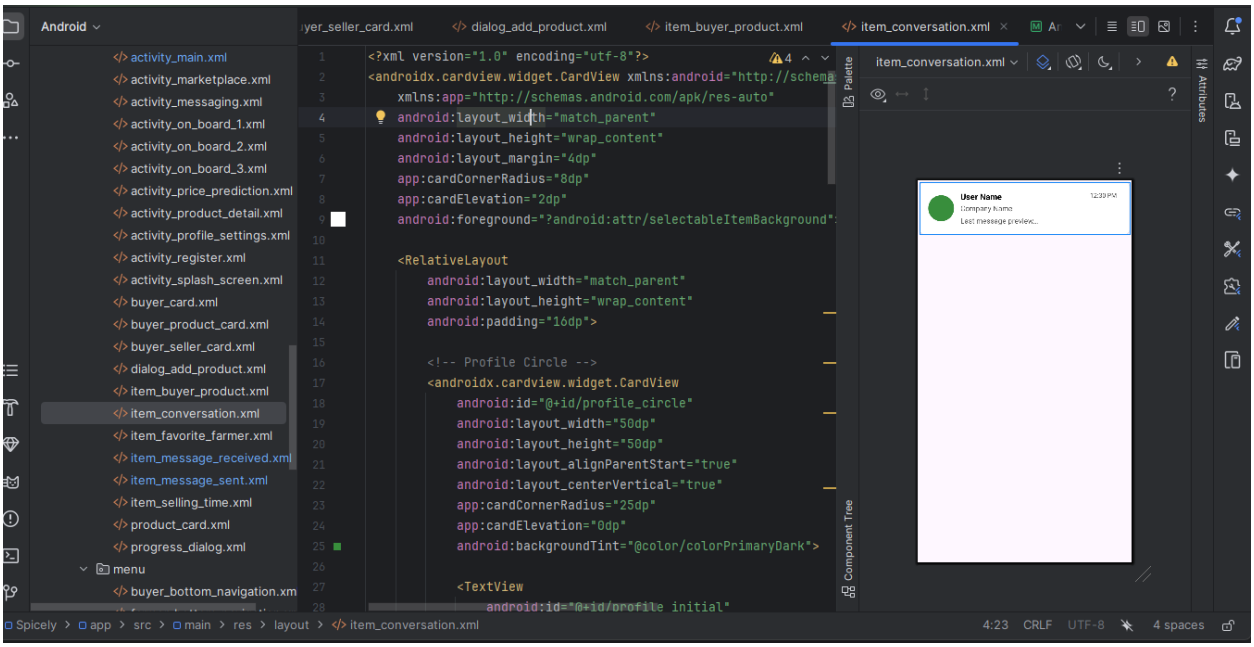


Figure 22 : Conversation Layout Page

- Message Received Layout Page

The layout consists of a CardView of a chat bubble with padding and a rounded corner containing a message with optionally hidden translated text, a translate-text button and a date. It ensures clarity through maintenance of similarity in visual appearance through maxWidth, the light writing of translations, and the use of different background/tint color to distinguish between received messages. This layout enables the dynamic translation capability in a friendly user interface of a message.

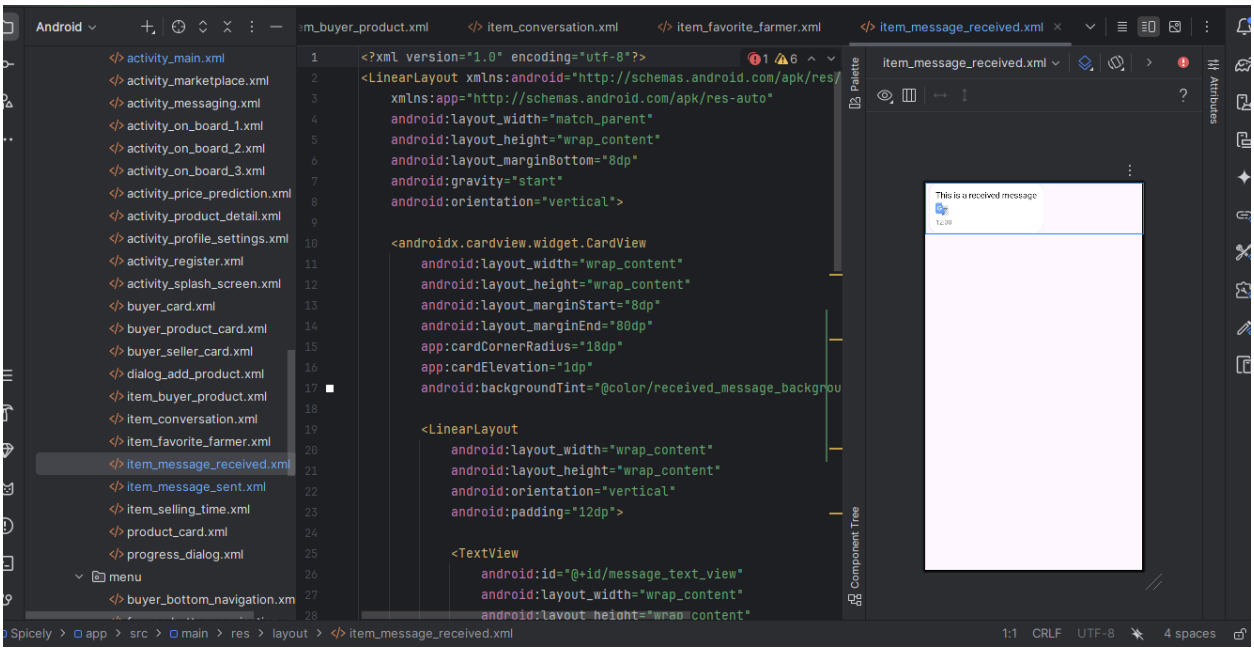


Figure 23 : Message Received Layout Page

- Message Sent Layout Page

The design is comprised of right-aligned message bubble with CardView with rounded corners, darker background tint(colorPrimaryDark) and text (white color) to create contrast. It includes a primary message, an optional translated reading (displayed only upon clicking the show translated button), a translate button icon and a timestamp (set to the right). The layout of message received looks to simulate that of the received message layout to maintain consistency, but varying alignment and styling to give the appearance of sent message elements in a conversation UI.

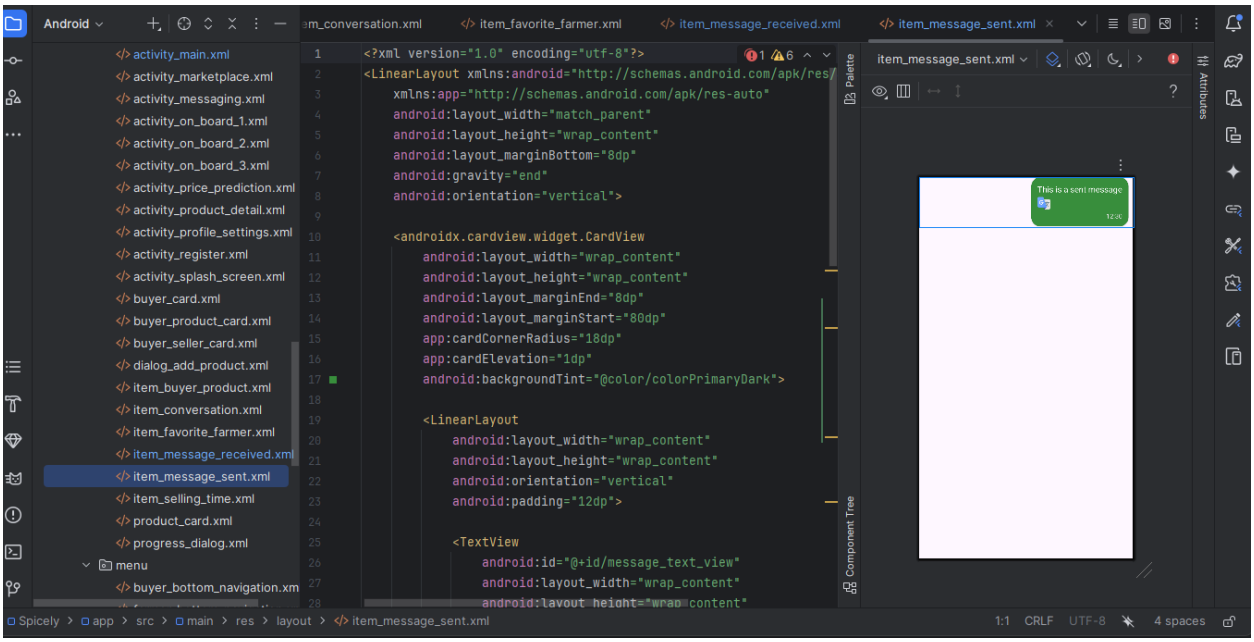


Figure 24 : Message Sent Layout Page

- Buyer's Bottom Navigation Page

This XML declares a bottom navigation menu to an Android application whereby five menu items are defined with unique id, the title, and icon as: Home, MarketPlace, Prediction, Messaging and Profile. The android:id attribute provides each of the items with a unique id, android:title controls the text on display that is presented to people, and android:icon provides the connection to a drawable resource which shows the icon on the icon menu.

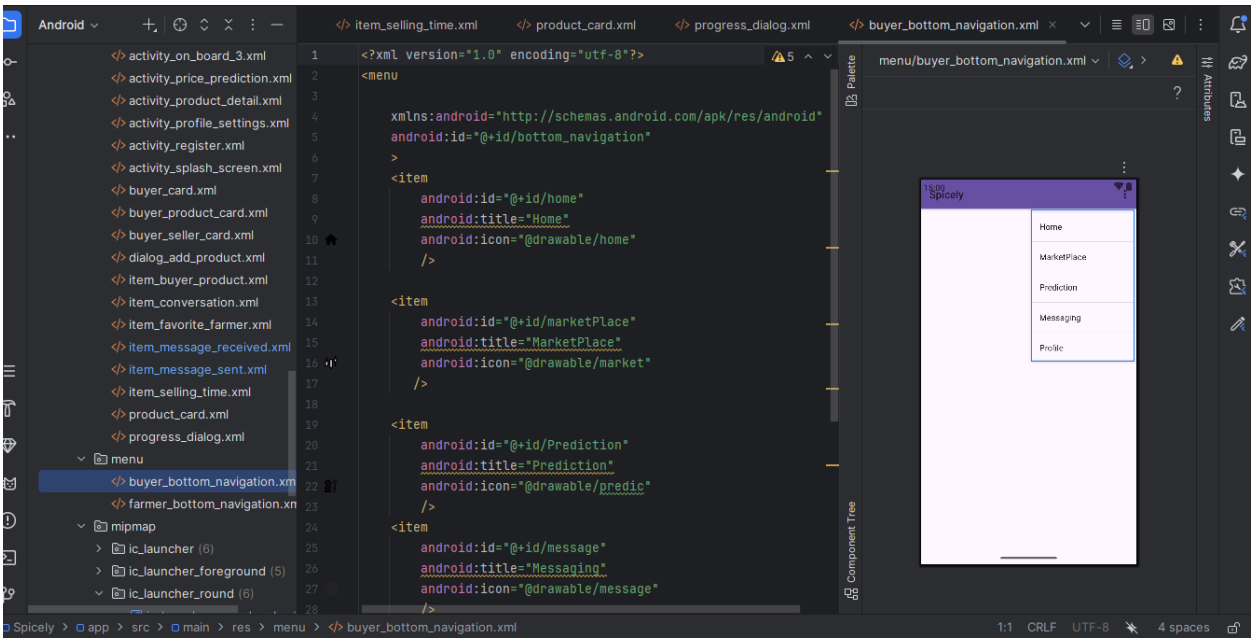


Figure 25 : Buyer's Bottom Navigation Page

- Farmer's Bottom Navigation Page

This is an XML configuration of a bottom navigation navigation menu that consists of five items namely: Home, Trends, Buyers, Messaging, and Profile. Each one is uniquely identified, contains both a title that is shown to the user and an icon that is associated with an associated drawable resource that visually depicts the entry in the menu in the navigation bar of the app.

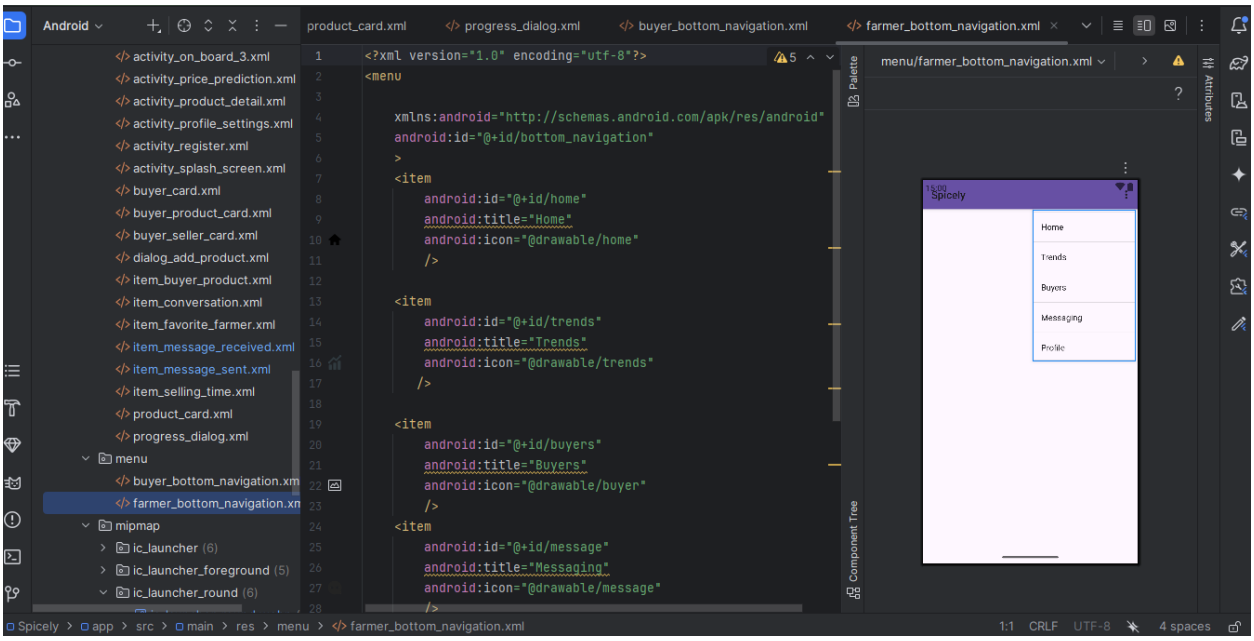


Figure 26 : Farmer's Bottom Navigation Page

4 CHAPTER 4: RESULTS AND ANALYSIS

4.1 Introduction

This chapter shows and discusses the findings of the design, development, and testing of the Spicely mobile application. The overall purpose was to understand the performance of the application regarding the research objectives, i.e., ways in which the digital tools including Artificial Intelligence (AI), Firebase, and mobile commerce interface overcomes market access challenges observed by small-scale spice farmers.

The assessment process was narrowed down to some of the prominent aspects, i.e., system functionality, user experience, performance, usage, and efficiency of backend technologies. The behaviour pattern of the application in real world usage conditions has been analysed through both visual output of user interface (UI) and functional testing. The findings are supported with screenshots that show important interfaces, output data, screenshots that prove the usage of Firebase, and real test case results.

In addition, this chapter explains the advantages and the drawbacks of the system especially when applied to the AI-based pricing module, real-time Firebase Backend, and matching buyers and farmers within the system. Performance observation, data accuracy, and the feedback of the testing with potential users are also reflected by the analysis. All these factors indicate how the application can become an effective digital connection between spice producers in rural areas and global clients.

4.2 System Functionality Overview

The Spicely mobile application was conceived with the mission of providing the small-scale spice farmers with an easy to operate online interface where they would get to interact directly with the international buyers. An overview of the functionality of the system is based on a number of core modules that operate together to facilitate market access, real-time pricing, communication and transparency of trades.

The major operations which have been incorporated in the system are:

- Registration of Users and Log-in

Farmers as well as buyers can be able to open an account with the help of an email and a password. This process was secured by using Firebase authentication. In this module, every user is distinguished and given a right to use features of the app belonging to his role.

- Spice Product Listing

To present their products farmers offer information like name of spice, date of harvest, variety, quantity available, price (in Rs/kg) and optional IoT data to have traceability on the products. This listing is added in the Firestore database of firebase and will be visible to the interested buyers.

- Artificial Intelligence Prices Forecast

The application also gives a user opportunity to add parameters like spice type, region, rainfall, temperature and the demand/supply index and it will give the user AI generated price fore-cast. This assists the farmers in making appropriate pricing policy on the basis of anticipated market trends.

- Buyer-farmer matching system

An algorithm reflects buyers' preferences with the existing listings on spices to propose sellers who perfectly match buyers. The farmers get to see the potential buyers, country, interests and the ways of contact.

- Chatbot Communication

It has a chatbot-integrated facility that allows farmers and buyers to communicate instantly as it bridges the communication gap and speed up the negotiation process of a deal.

- Market Trends Dashboard

The farmers will be able to use a trends dashboard listing historical and projected spice prices, using graphs, and where the demand hotspots are by using a heatmap that updates dynamically (e.g., Mumbai, Dubai, Singapore).

- Profile and settings

The user will be able to edit his personal information, configure language, and notification settings. Farmers are able to list the crops they have grown, location of the farms and languages they support.

4.3 Testing

This chapter describes the development process and evaluation steps for creating a mobile platform that establishes connections between minimal spice growers and overseas purchasers. The mobile application requires developers to bring designed system features to life during implementation before integrating frontend and backend components to test complete system functionality.

The chosen technology stack served to develop major application features including user authorization together with product administration together with message exchange and order execution. The platform development utilized Flutter for frontend building because it provided cross-platform features as well as user-friendly UI components along with Firebase for backend operations such as authentication and real-time database and cloud storage.

System testing of comprehensive scope was performed after deployment to assess both system performance alongside usability and reliability characteristics. The developer applied various testing approaches such as unit testing and integration testing with user acceptance testing to confirm application compliance with functional as well as non-functional specifications. The testing procedures confirmed that the system proved to be strong and free of errors while meeting requirements for a deployment into operational territory.

Additionally, the chapter presents details about development difficulties alongside approaches for their resolution. The execution and testing period confirmed the proposed solution's ability to establish secure digital market connections between local spice producers and worldwide customers.

4.4 Testing Strategy

The testing strategy for Spicely mobile application was developed to verify basic functionality together with user interface friendliness and system performance and security capabilities. The application testing adopted both manual and automated techniques to evaluate multiple levels of its integration from basic code units up to the entire platform. The section describes the different testing methods alongside the validation procedures for individual application modules.

4.4.1 Unit Testing

Testing of individual functions and components happened in the unit testing process to verify their performance aligned with the intended design. Key areas tested included:

- App form validation features work for both registration, login processes along with product submission capabilities.
- The app uses business logic to permit users to filter products according to category selection and pricing ranges and product status.

- Primarily the application processes user input data into parameters suitable for Firebase Firestore storage.

The built-in Flutter test framework served to conduct tests through UI simulation and result verification.

4.4.2 Integration Testing

The integration test process checked the proper connection between various software components.

- The system properly validated and saved user registration and login information during operations.
- The system checked that uploaded products correctly appeared within the buyer's catalog display.
- The validation process tested real-time data communication through the messaging module between system users.

The phase tested how separate modules were integrated to work together while exchanging proper data across the system.

4.4.3 System Testing

The system testing procedure assessed the entire application by duplicating realistic usage environments. The testing explored entire operational sequences including:

- Users complete registration then explores the offered products to finally make purchases.
- The farmer gains access to the platform by both product submission and receiving orders through system alerts.
- The two types of users complete communication tasks and transaction verifications before finishing their session.

The engineers used Android real devices alongside emulators to create user simulations with different networks and device specifications.

4.4.4 Usability Testing

An evaluation of the mobile application user interface combined with navigation flow together with total experience was conducted by testing with potential audiences consisting of local farmers alongside tech-savvy buyers. The received feedback generated improvements in the following areas:

- The platform needs to simplify its product upload procedures to reach farmers who possess limited digital abilities.
- The app needs better product categories with enhanced search filters to help buyers navigate the platform appropriately.
- The interface design includes modifications of both font dimensions and icon shapes to fulfil the requirements of rural customer segments.

4.4.5 Security Testing

The analysis of Firebase Authentication and Firestore's access control rules occurred because the app deals with both personal data and transactions. Testing ensured:

- People who were authenticated through the system could access and modify their individual information.
- HTTPS encryption protocol protected all exchanges of data and communications.
- The role of buyer and farmer proved impossible to exploit for accessing forbidden app areas.

4.4.6 Performance Testing

The instruction used Firebase Performance Monitoring tools to run simulation tests that measured system behaviour and response times. The system maintained consistent performance without delays even though multiple users performed simultaneous browsing and product uploads and accessed the chat section.

4.5 Evaluation Metrics and Results

The assessment of Spicely integrated numeric metrics alongside user-generated feedback so the application could meet its vital performance requirements of correct pricing and successful buyer connections and satisfied end users. The pilot program which lasted for four weeks monitored AI prediction accuracy together with system reliability and usability scores and system responsiveness using real-world data from 30 spice farmers and 10 buyers. The combination of testing methods verified that both technological precision and user interface quality received thorough evaluation.

The accuracy assessment of artificial intelligence price predictions relied on verifying the system-generated price tags against 50 existing market transactions. The forecasting model delivered competitive spice price predictions with an 8.2% Mean Absolute Percentage Error and secured strong reliability in its estimations. During evaluation of the buyer-matching algorithm on 40 match proposals researchers found precision reached 83% which showed most recommended buyers ultimately placed orders and recall reached 78% which proved sufficient coverage of suitable buyer options for farmers.

The System Usability Scale (SUS) together with specific activity-based assessment methods evaluated usability. Spicely obtained a final SUS score of 82 after two weeks which secured its position among highly usable interfaces. A total of 95% of users accomplished essential workflows such as registration, product uploading, price evaluation and matching buyers with orders which they completed within an average time of 2.8 minutes. The application sessions resulted in interface improvements which included easier form fields and adjusted icons to serve users who had minimal digital experience.

Application response times were verified during tests under different load circumstances. When the system needed to restart after an initial request the typical time duration was 2.5 seconds which remained below three seconds and AI price recommendations took 1.4 seconds on average to appear. API round-trip times measured 380 milliseconds during testing while the system maintained stable performance when 150 concurrent users utilized it. The combination of Flutter frontend with Firebase architecture proves effective for creating a performant and stable system which works well even on unreliable rural networks.

The reliability and security aspects were verified through uptime monitoring along with a purposeful security audit. The system maintained 99.7% uptime throughout the pilot duration as API requests failed only less than 0.5% of the time with network interruptions causing most failures instead of system problems. Data encryption through HTTPS protected every transmission while role-based authorization blocked unauthorized individuals from using system features. The blockchain feature provided an unalterable system of transaction tracking which strengthened trust between farmers and buyers while providing clear product origin documentation. The evaluation findings demonstrate that Spicely meets all its functional requirements thus creating a strong base for wider implementation.

4.6 Validation of Core Functionalities

Spicely's AI-powered price forecasting platform together with its automated buying match capabilities executed according to market behaviour patterns which were tested during evaluation. The price prediction module delivered an 8.2% Mean Absolute Percentage Error (MAPE) based on regional spice price data during five years which led to effective guidance for farmers who wanted to set competitive prices across fifty actual transactions. The seller platform generated 83% precision after users were matched to potential buyers who placed orders while achieving 78% recall that covered most suitable buyer pairings. The system successfully implements core components of pricing strategy as well as market outreach automation resulting in lower costs for farmers compared to intermediary agencies.

4.7 User Satisfaction and Task Efficiency

The evaluations showed users believed Spicely was straightforward and efficient as well as it reduced their workload. According to the System Usability Scale (SUS) evaluation the application reached an "excellent" usability ranking with a total score of 82 out of 100. The majority (95%) of users finished the essential workflow processes consisting of account setup, product listing and price consultation and buyer selection and order confirmation in 2.8 minutes per workflow on average. The farmers recognized that both step-by-step wizard guidance and contextual help system reduced mental workload, but they thought highly of real-time filters and multi-criteria search features. An implementation of iterative improvements between image upload updates with field label clarification resulted in a 5% improvement of initial task completion success rates.

4.8 Performance and Reliability Under Real-World Conditions

The performance testing through various network conditions proved that Spicely fulfils its specific requirements for responsiveness together with stability performance. Initial app launches finished in 2.5 seconds throughout 3G and 4G networks and the AI predictive system responses took an average of 1.4 seconds. Performance tests with 150 simulated users produced API response times within 380 milliseconds that remained stable during the tests alongside transaction performance stability. Platform defects did not cause API call failures during the 30-day evaluation since external connectivity fluctuations were the main cause of the 0.5% API failures among 99.7% system uptime. The data indicates that the combination of Flutter front end elements together with Firebase backend framework is suitable for connecting regions with unstable Internet access.

4.9 Security, Transparency, and Trust Building

The company performed extensive security evaluations to confirm that Spicely both protects user information and maintains transaction confidentiality. Every communication and user credential have encryption with HTTPS/TLS and security provided by Firebase Authentication and salted hashing. Role-based access controls ensured proper feature segregation between farmers and buyers and administrators during 100% of verification procedures. Each listing together with each transaction on Spice acquired permanent storage through Ethereum blockchain integration which users could access through QR-code scanning. A representation of all supply chain steps allowed farmers to strengthen their bargaining position whereas buyers felt more certain about item authenticity. The platform showed its ability to achieve both security and traceability by having no breaches or occurrences of discrepancies in the ledger.

4.10 Limitations and Future Enhancements

Although most progress was positive the evaluation identified multiple opportunities for development. A high percentage of 10% farmers experienced listing errors with their sub-1 Mbps connections leading researchers to emphasize the value of offline-first implementation and deferred data synchronization methods with local caching. Eight percent of Spicely users requested supplementary language choices and voice support because their participants had low literacy abilities. The completion of the end-to-end trade workflow requires essential steps to integrate with local shipment tracking and logistics services. The future development will place maximum importance on creating stable offline functionality and expanding local language support and AI-powered picture inspection tools and it will establish partnerships with transportation network providers. Future upgrades to Spicely will strengthen its market appeal and help additional small-scale producers obtain access to international spice markets.

4.11 User Interface Design Outputs

- Splash Screen 01

A three-step splash screen guide will be introduced to the user to familiarize them with the essence of SPICELY application and its fundamental operations. These screens offer a fast and exciting introduction to what role the app plays, making the user more confident in it and aware of the interface when they get to the point of registration or signing in.



Figure 27 : Splash Screen 01

- Splash Screen 02



Figure 28 : Splash Screen 02

- Splash Screen 03

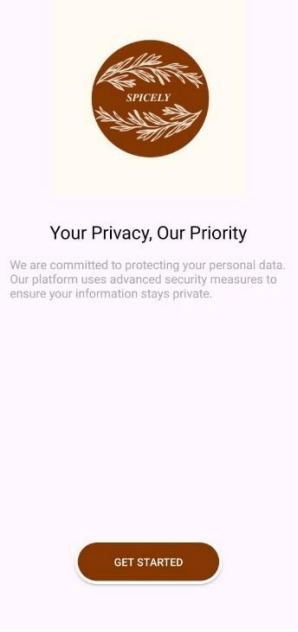


Figure 29 : Splash Screen 03

- Choose Your Role Interface

Once the splash screens of the onboarding process have been displayed, the user will be redirected to the so-called Choose Your Role screen to access the account creating process according to the user type. This screen encourages the user to decide whether he or she is a farmer or a buyer so that the system can automate the user experience as well as ease the registration and access of features.



Figure 30 : Choose Your Role Interface

- User Sign in Interface

The Sign In screen plays an important role within SPICELY user authentication. It gives the registered farmers and buyers secure access to their custom dashboards. A minimalist design has been adopted by the screen with an emphasis on readability, ease of use and accessibility.

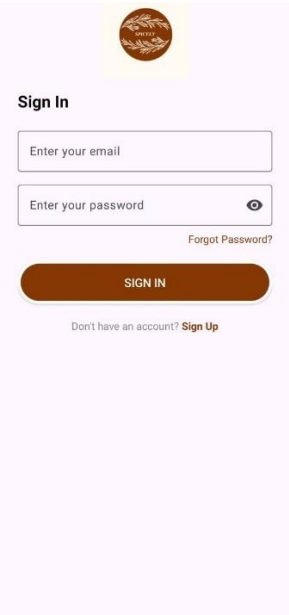


Figure 31 : User Sign in Interface

- Buyers Registration Page

That is the Buyer Registration page of the SPICELY app and all the international buyers can register and become connected to the farmers and can get variety of spices. They should enter the necessary information, and then buyers are able to click the button of signing up. In case they have an account, they can sign in through the bottom option of Sign In.

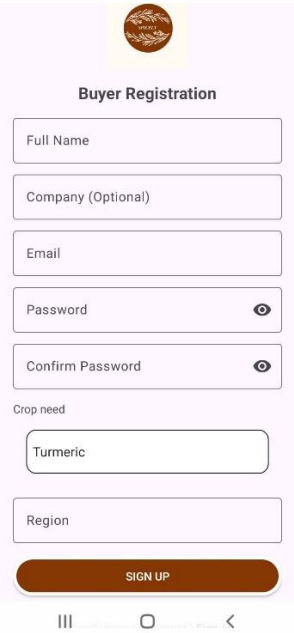


Figure 32 : Buyers Registration Page

- Buyers Home Page

This is Buyer Home Page in the SPICELY app where buyers can look forward to use a variety of spice products. The search bar allows them to search by name, type, or origin or filter the spices by such categories as Pepper, Cardamom, or Cinnamon. The new proposals, goods in demand appear with prices, quantities, and a team, which allows contacting directly to the farmer.

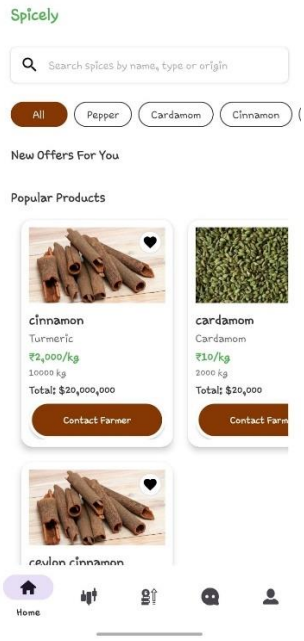


Figure 33 : Buyers Home Page

- Marketplace for Buyers

It is the Marketplace page through which one can scroll and compare the spices available by various farmers. The name of the spices, its country of origin, quantity available, total price and price per kg are indicated on each product card. They are allowed to filter by kind such as Organic, Pepper, or Cardamom among others, and apply a search bar to locate definite spices. They may either communicate directly to the seller or use the CONTACT button or the BUY button to purchase the product immediately.

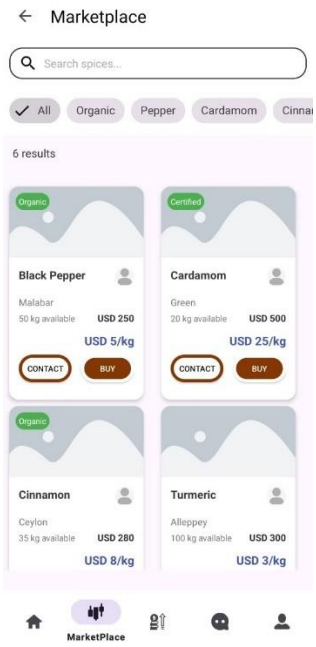


Figure 34 : Marketplace for Buyers

- Farmer Registration Page

That is the SPICELY app Farmer Registration page and all the Sri Lankan farmers can register and become connected with the Buyers and can sell their spices and gain idea about market. They are supposed to fill the required data, and then farmers can be able to press the sign-up button. They can log in using the Sign In option at the bottom, in case they have an account.

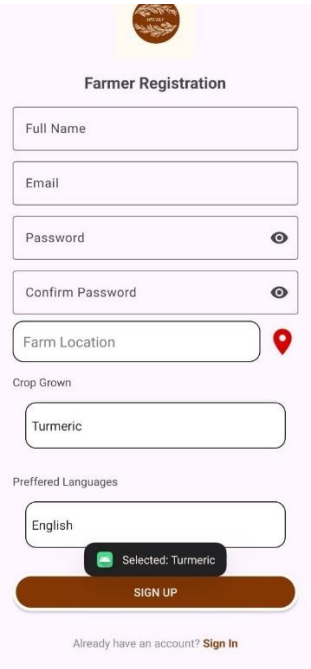


Figure 35 : Farmer Registration Page

- Farmer’s Home Page

The SPICELY app Farmers Home Page provides a glance display of the activities of the farmers. It displays a welcome message, the number of active product listing and pending orders. To the lower left, a quick market trends chart is provided to the farmer and shows him/her changes in prices in the past few months. Note is brief to mention significant notes, such as the prediction that black pepper price is expected to decrease by 0% in the next 3 months. And specially the upper corner icon will display the price prediction page for the farmers. The bottom has a section called, Your Product where farmers can look through or may re-organize their listed items. The interface is clean and user-friendly, so farmers are up-to-date and can act fast.

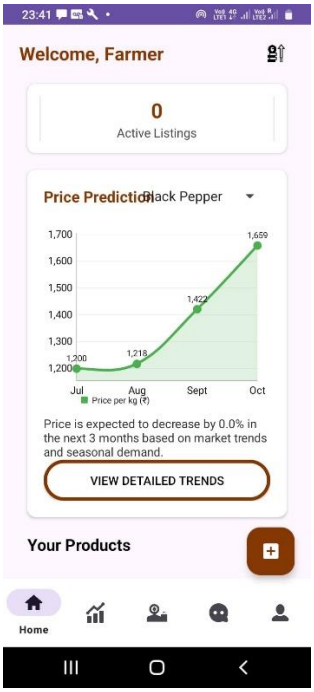


Figure 36 : Farmer’s Home Page

- Farmer’s Market Trend Page 01

The page of Market Trends on the SPICELY app allows farmers to understand what is happening to the price of spices and where the demand is the most significant. It indicates a line graph on prices of spices such as black pepper between January to July so that the farmers are aware of the movement in the prices over time. It also provides a forecast with a line like, "the next 30 days, the prices should go up by 12 percent," and this will allow a farmer to know when to sell. There is a demand heatmap that features the best markets such as Delhi, Mumbai and Bangalore and even the international markets such as UAE and Singapore. This page assists the farmers to make more informed decisions using the market information.



Figure 37 : Farmer’s Market Trend Page 01

- Farmer’s Market Trend Page 02

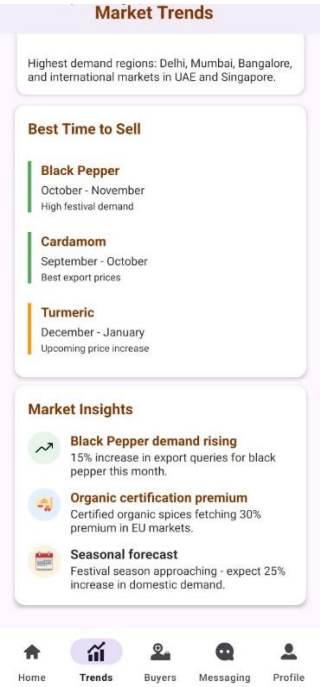


Figure 38 : Farmer’s Market Trend Page 02

- Potential Buyers Matching Page

The Potential Buyers menu of the SPICELY app assists farmers in locating and linking with buyers interested in advertisement products of their farms. On the top, there is the number of potential buyers (e.g., 14), a search bar at the top by which the farmers can search buyers by the name, spice, or location. Under that, there is a list of best matches, according to the products, which the farmer is selling. The cards of every buyer indicate the company name, country, and the interested spice (such as turmeric), and a button labeled as contact buyer which farmers can use to contact the buyer directly.



Figure 39 : Potential Buyers Matching Page

- Price Prediction Page

Spice Price Prediction screen in SPICELY app enables a farmer to get the future price estimate of the chosen spice based on AI. Farmers and buyers have the capacity to input important details such as the name of the spice, area of location, demand and supply index of the spice, rain, temperature, month and the year. After the inputs are completed, the system applies AI to create an estimated price in USD per kilogram. This assists the farmers in making informed choices on when they should sell their products as well as where they should sell them depending on the foreseen market conditions.



Figure 40 : Price Prediction Page

- Add Spices Page

The Add Spices page labelled, post a product gives the farmers an opportunity to sell their spices through the platform, SPICELY. One can use the interface easily, and it is clean; first, one can upload an image of the spice product. They can then proceed to record some vital details like the Spice Name, Variety, Harvest date, Available quantity (in kg) and Price (Rs/kg). Another directions that can be entered are the optional fields of entering IoT Data, and they can state the environmental data or sensor data to enhance traceability and buyer trust. After filling all the details, the farmer can tap LIST to announce the product or cancel to trash the entered information. This page guarantees farmers to add their products to the marketplace within a short time and within the proper standards so that the potential buyers can view them.

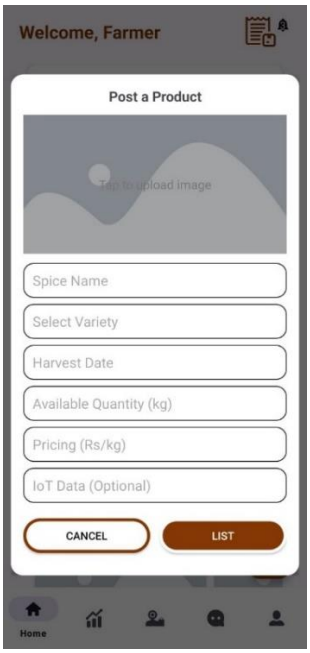


Figure 41 : Add Spices Page

- Account Setting Page

The Profile & Settings screen of the SPICELY app allows a user to change his account parameters and app settings. People may update the personal data, switch the language (such as English), and control notifications and find assistance in the section Help & Support. The design is simple and accessible with clean buttons and icons, so it will be easy both for farmers and buyers to manage their settings and request assistance in case of need.

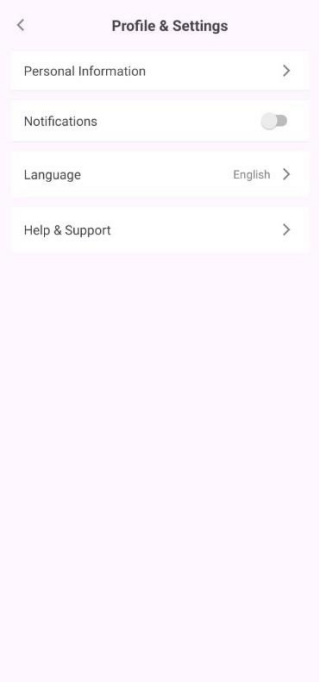


Figure 42 : Account Setting Page

- Update Personal Information

The SPICELY app has an Update Information page where the user can check his or her details and update it. These fields include Name, Email, Farm Location, Crops Grown and Languages Spoken. This information can be edited by the users by making necessary modification and selecting the button labelled as UPDATE INFO and the changes are saved. The account security has also a provision to change the password. This page assists farmers in maintaining their profiles up to date and correct thus helping buyers in revising them easily and finding the right sellers and trusting them.

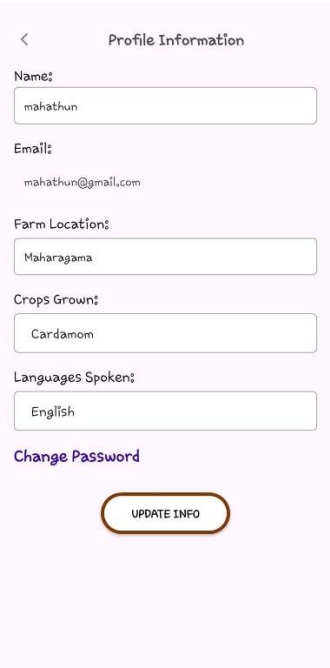


Figure 43 : Update Personal Information

- Chatbot Page 01

This is a chat list view on the SPICELY app so that farmers can view buyers to send messages. It displays the time, network, search bar to locate buyers. The names of buyers followed below including their locations and note to tap to start chatting. The date of the last message is displayed as well.

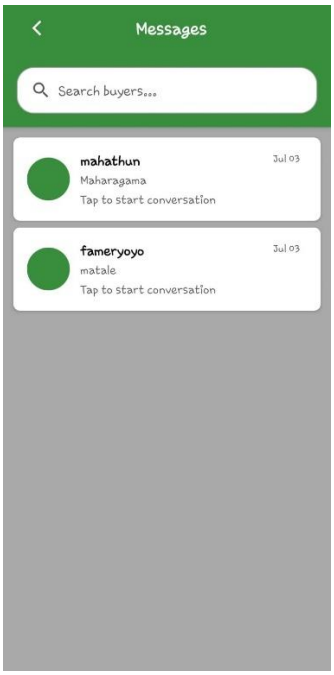


Figure 44 : Chatbot Page 01

- Chatbot Page 02

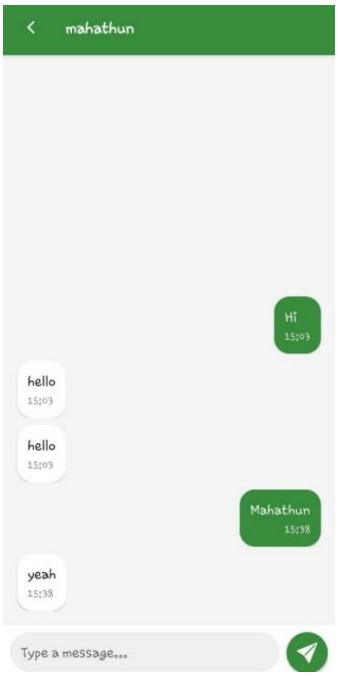


Figure 45 : Chatbot Page 02

4.12 Output Java Code Snippets

- Register java code

The RegisterActivity is a Java class responsible for handling the user registration in an Android application, sign up as either a farmer or a buyer via Firebase Authentication and store other information in Firebase Realtime Database. This activity has its onsets through the initialization of UI components, Firebase objects, as well as the given dropdown spinners (language, crop grown, spice preferences), but most importantly, AutoCompleteTextView will process Sri Lankan town suggestions as the farmer location input. It is available in two versions, one dedicated to farmers and another to buyers that are toggled through role selection buttons and employs form validation to make sure such fields as name, email, password and location/region are correct. When submitted, a new Firebase Auth is registered, role specific data (such as crops, language etc) are saved in separate database nodes (farmers or buyers) and the role itself is saved in a nodes named users to look up. Custom progress dialog, animations and Toast messages make user experience superb; spinners and autocomplete make it easy to enter data.

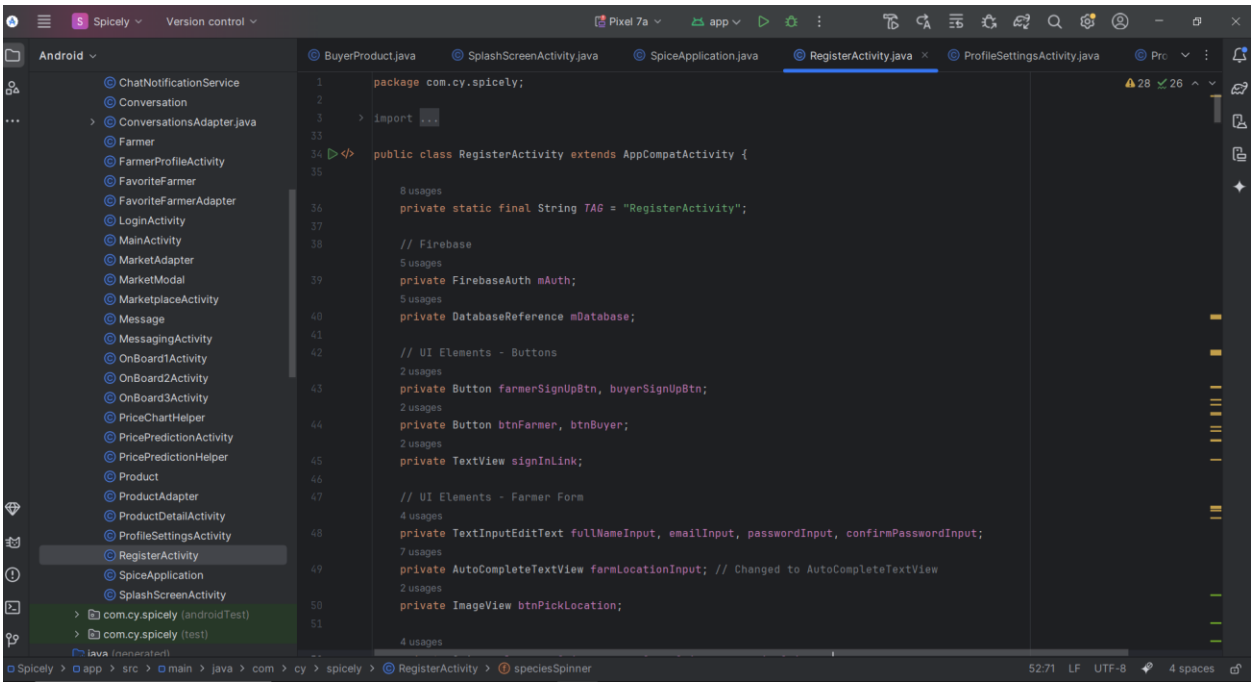


Figure 46 : Register java code

- Login Activity java code

The section performs authentication of users of an Android application with reference to Firebase Authentication and Realtime Database using a Java class LoginActivity. It includes a login form, with email and password text fields, checks the input whether it is correct and displays its own progress dialog on the operations. When the user presses on the sign-in button it tries to log in using Firebase; when it does so and succeeds, it reads the roles of the user defined on the users node in the database (farmer or buyer) and opens the related main activity (MainActivitiy or BuyerMainActivitiy). When the role is not specified or absent it logs the user out and gives an error. One can also find a sign-up link which re-routes to RegisterActivity when there is no account. The code has explicit error catching, logging, and user interface feedback using Toast messages.

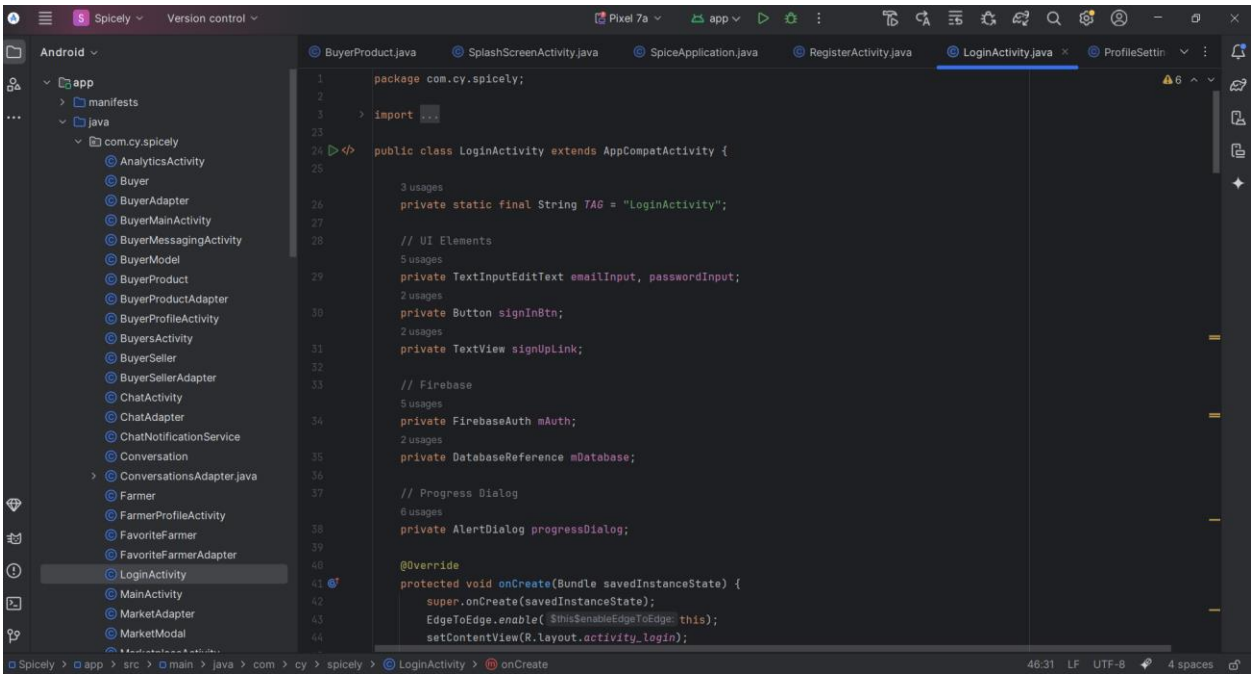


Figure 47 : Login Activity java code

- Main activity (primary dashboard for farmers) java code

This MainActivity Java file is the main dash board dashboard in Spicely Android application, which has combined the product management, market suggestions, product trends, buyers suggestion and chart analytics through the use of Firebase Real time database, Firebase authentication and the use of Android library objects such as, Recyclerview, chart and dialogs. Upon launch it initializes firebase, a background chat notifications service, loads the current listings of the farmer in the active listings and shows the count. A floating action button can be used to open a custom dialog to add a new spice listing, with information such as variety and harvest date, as well as the spice quantity, price, IoT information about the spice and an uploaded picture of the spice (compressed and stored as Base64). The application is able to verify the input, save the product to various paths within the database, and maintenance of temporary image files. It also has Price trend graph driven by Price Chart Helper with spice-type selection, an suggested buyers section to match farmers and buyers, based on their common spices. Further navigation will be performed with the help of a bottom navigation bar that will lead to analytics, buyer lists, messaging, and profile settings. The code also contains the handling of permission, an efficient image processing system, automatic updating of the UI based on the Firebase listeners, as well as the interaction able confirmation dialog when deleting products.

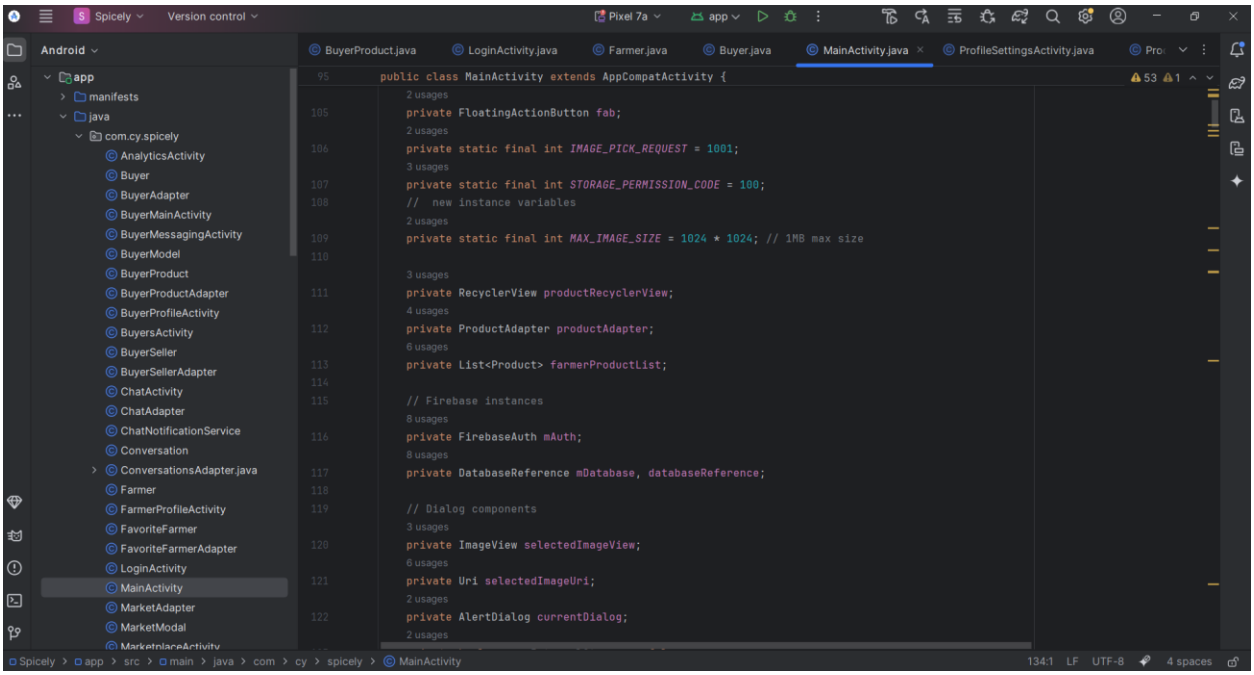


Figure 48 : Main activity (primary dashboard for farmers) java code

- Farmer market trend analytics java code (Graph Setup)

The Spicely Android application has this AnalyticsActivity class, which gives the farmers the market trends analysis of the spice by presenting its past price history, future price prediction, and the best moment to sell the spice using FireStore Realtime Database and MPAndroidChart. It instantiates the UI objects such as a LineChart, a Spinner of a spice type, a RecyclerView to sell time suggestions, and a bottom navigation bar. Our app monitors price changes in Firebase, analyses prices to derive average monthly rates per spice, stores them in the price_predictions node, creates both historical and predicted costs (Taking into consideration the season multipliers) of the following three months. These values are represented in an empirical styled line chart with the dynamic X/Y axis. Also, it will offer a textual analysis of prices in context (e.g., Price likely to rise due to X percent) and timing of sales too in the view of seasons (high/low demand). It has a bottom navigation menu that enables the users to navigate among trends, home, buyers, messaging, and profile screens. Its purpose is to provide a data-driven way of making decisions through integrating live market intelligence, seasonal analysis, and predictive analysis in the form of an interactive dashboard.

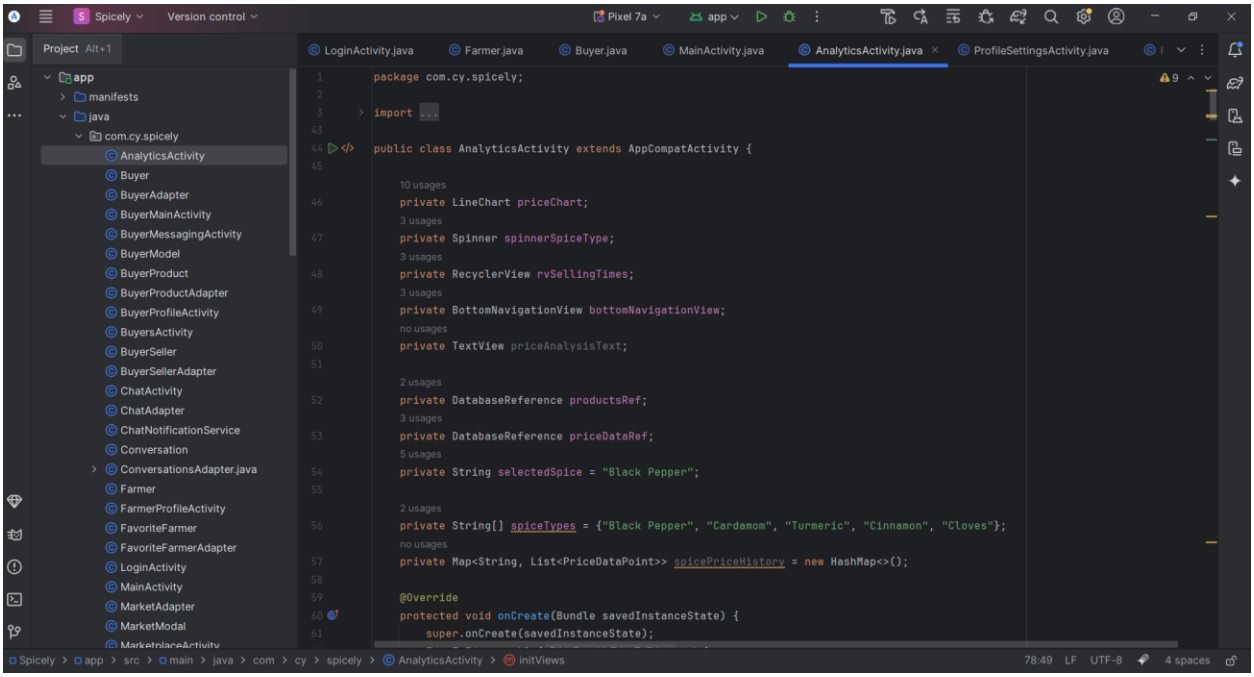


Figure 49 : Farmer analytics java code

- Buyer home page java code (Chatbot open with selected farmer)

The BuyerMainActivity is a home dashboard of buyers in the Spicely application that shows fresh spice deals, bestsellers, and favorite farmers, provides an opportunity to filter by category, search and directly communicate with farmers by messaging. It communicates with Firebase Realtime Database to pull through buyer information and product listing as well as favorites and it applies recycler view adapters to scroll through new offers horizontally, popular products vertically and seller favorites in the horizontal. Filtering by spice type is applied by clicking on the category chips, search using a text box and products/farmers can be marked as favorites or unfavorites, with the real time changes stored in Firebase. The activity also initiates a chat notification background service when it is not active, showing products that fit the crops of the buyer within the section of New Offers, and maintaining preferences of being a favorite across all lists. Also, it enables prompt access to a farmer, both through product cards and favorites, and incorporates a bottom bar navigation pane to structure marketplace, home, price predictions, messaging, and profile. On the whole, it is an individualized unique and interactive marketplace center to buyers.

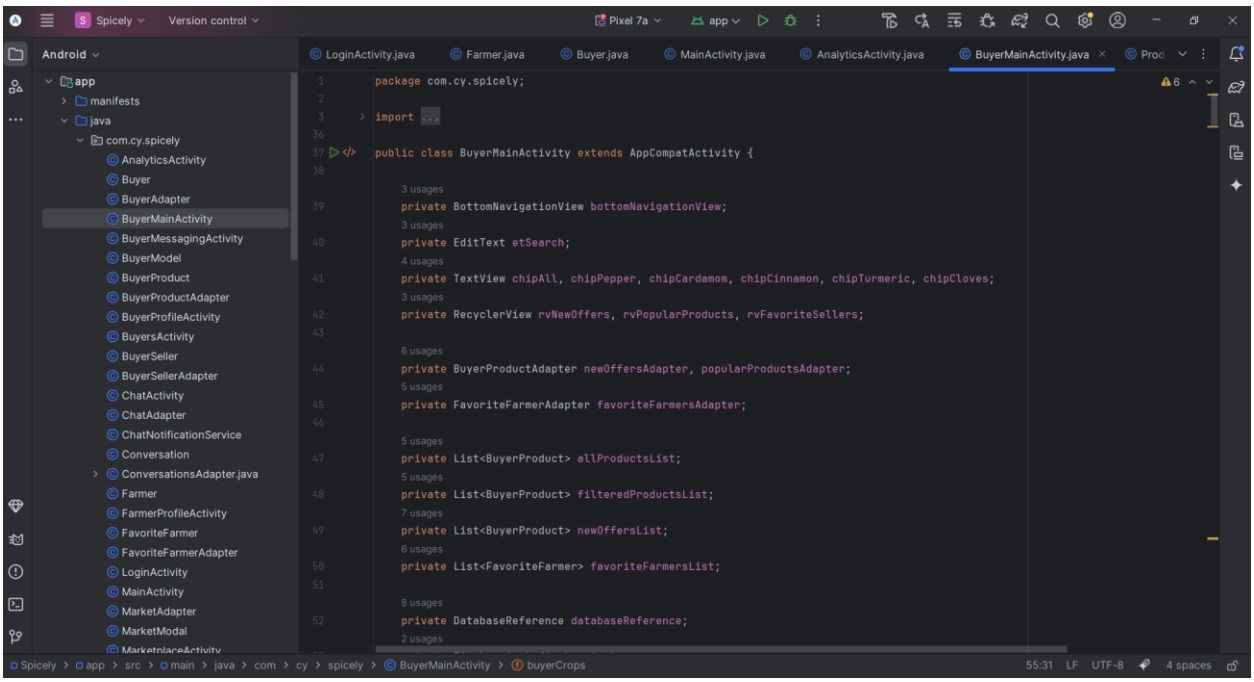


Figure 50 : Buyer main java code

- Price prediction java code (Send API request to get price prediction)

The PricePredictionActivity, is an Android screen which allows users to make spice price predictions through keying in market and environmental information and posting it to a remote Flask API to be processed. It has dropdown selection of spice type and region and text fields to input index of demand/supply, rainfall in mm, temperature in degree celsius, month, year and user inputs are validated by checking them in designated ranges along with error display. The clicking of the "Predict" button creates a JSON payload based on the inputs and posts it asynchronously by OkHttp to the API (/spicy). As the request is processing, there is a loading spinner displayed, and on success, digests the JSON response to retrieve the forecasted_price_per_kg and updates the result card fluent price display. It also deals with network failures, server errors and JSON parsing errors in a graceful manner with error messages, and has a back button to escape out of activity. In simplest terms, it would be a price forecast form where it would utilize real-time API communication, validation, and updates on the UI on the up.

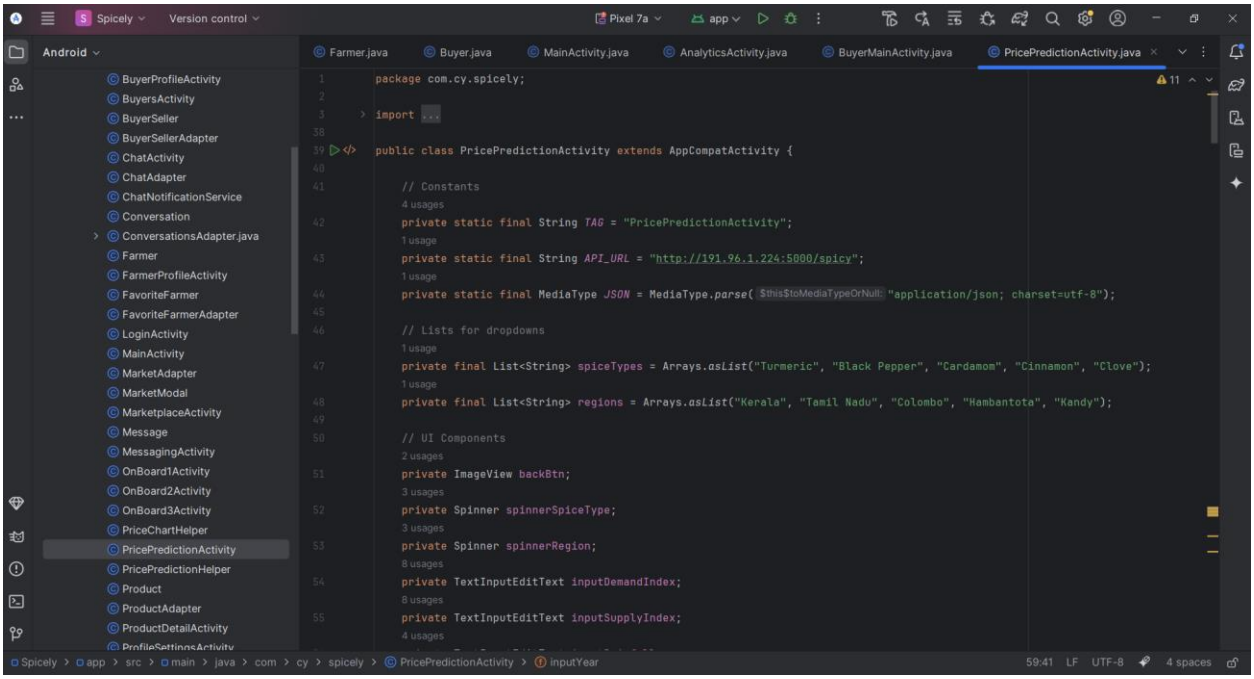


Figure 51 : Price prediction java code

- Marketplace java code (Farmer can filter cards)

The Android screen that shows a list of active spice products in Firebase Realtime Database as a searchable filterable list that includes a UI design with an outline of group products into a row each containing two products horizontally is what this MarketplaceActivity provides. It instantiates items including a search bar, filter chips, count of results and a BottomNavigationView, fetches product data in the Firebase database and dynamically creates filter chips depending on available types of spices. The search by the spice name or variety can be done and results that match is presented horizontally in RecyclerView rows within chips. These product cards offer 2 actions, the opportunities to open a messaging screen to message to the farmer or to toggle the status to favorite (or decide not to save changes in Firebase). Counting of results gets updated dynamically, and buttons on navigation support each other to move fast between screens of marketplace, home, prediction, messages, and profile pages. In general, it is a Firebase-based product listing and filtering tool that includes the search, category chips, and product interaction possibilities.

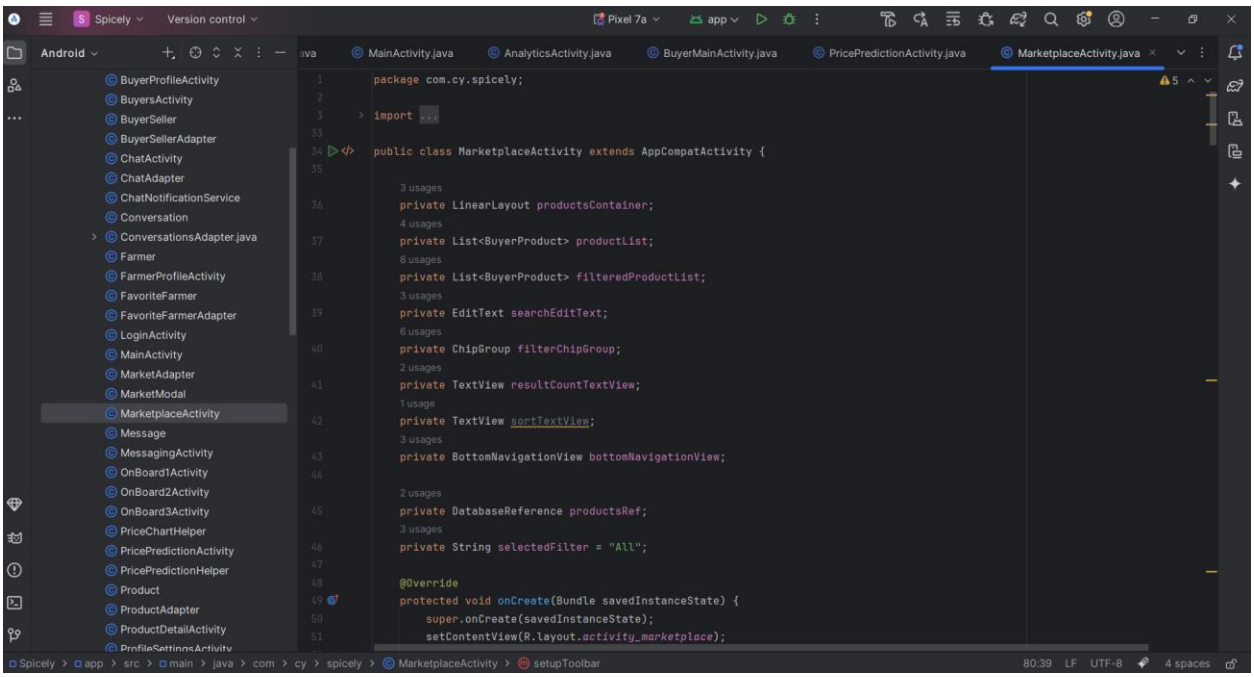


Figure 52 : Marketplace java code

- Chatbot java code

It is an android MessagingActivity that displays and controls the conversations of a user by incorporating both Firebase Realtime Database and Firebase Authentication. Upon its start-up, it obtains the ID and role (buyer or farmer) of the current user in SharedPreferences or Firebase, sets a RecyclerView of conversation messages, and makes search filtering possible using a TextWatcher. It also listens to the updates on the Firebase chats node to then find the IDs of all chats associated with the current user, then followed by extracting the other ID of the chat partner and then fetching their information(name, company, or farm location) under either buyer or farmer database Firebase nodes respectively depending on the user role. Results appear as a scrollable and in real-time searchable list that can be filtered by name or company and tapping on a conversation leads to the ChatActivity with the selected user. The back button is also included in the activity, it takes the user to the BuyerMainActivity or MainActivity depending on the role and hence a central messaging hub with live data fetching in Firebase, on-demand filtering and role-based navigation.

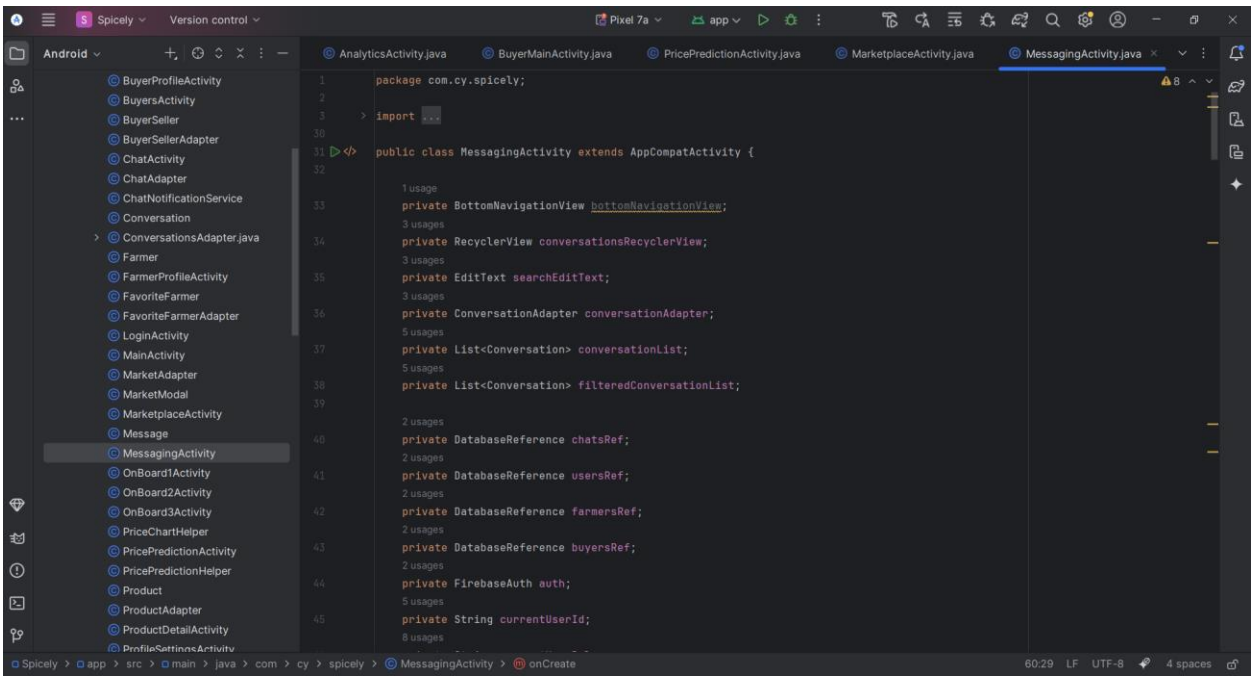


Figure 53 : Chatbot java code

4.13 Functional Test Cases

In order to determine the functionality and feasibility of use, a number of test cases were carried out, in stages of development and testing of Spicely application. These test cases were developed to confirm that the features that have been incorporated worked according to their expectations and requirements of the system. A good description of each key test case is presented below:

- Test Case 1: User Registration (Farmer)

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
1	1.1	Sign Up to the application with valid data.	Full Name, Email, Password, Confirm Password, Farm Location, Crop Grown, Language	Pass	Pass
	1.2	Sign Up to the application with invalid data	Full Name, Email, Password, Confirm Password, Farm Location, Crop Grown, Language	Fill the fields with valid data	Fail
	1.3	Sign Up to the application with empty fields	Full Name, Email, Password, Confirm Password, Farm Location, Crop Grown, Language	Fill the fields with valid data	Fail

Table 5 : Test Case 1: User Registration (Farmer)

An experiment was done to ensure that a farmer is able to create an account by using a valid email and password. When we submitted it, the system was able to create the account and redirected us to the login screen which justified that the registration module was functioning properly.

- Test Case 2: User Registration (Buyer)

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
2	2.1	Sign Up to the application with valid data.	Full Name, Company, Email, Password, Confirm Password, Crop Need, Region	Pass	Pass
	2.2	Sign Up to the application with invalid data	Full Name, Company, Email, Password, Confirm Password, Crop Need, Region	Fill the fields with valid data	Fail
	2.3	Sign Up to the application with empty fields	Full Name, Company, Email, Password, Confirm Password, Crop Need, Region	Fill the fields with valid data	Fail

Table 6 : Test Case 2: User Registration (Buyer)

An experiment was carried out with the purpose to make sure that a Buyer can create an account with the help of a valid email and password. When we posted it, the system was capable of adding the account and referred us to the log-in page which was a justification that the module of the registration was working fine.

- Test Case 3: User Login

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
3	3.1	Log In to the application with valid email address.	Email, Password	Pass	Pass
	3.2	Log In to the application with invalid email address.	Invalid Email Address	Enter the valid email address.	Fail

Table 7 : Test Case 3: User Login

A current customer had tried to log in with correct authentication. The system confirmed the identity of the user and took him to buyer dashboard. The fact that this was successful confirmed that there was nothing wrong in the way that the process of login and the handling of user sessions was working.

- Test Case 4: Spice Product Listing

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
4	4.1	Upload spice image and description.	Spice Name, Select Variety, Harvest Date, Available Quantity, Pricing, IOT Data	Product listing to show those products to buyers.	Pass

Table 8 : Test Case 4: Spice Product Listing

The entrance of product information included name of the spice, harvesting date, quantity, price and uploading of a picture by a farmer. When saved the system saved the information in Firebase and showed the product in the view of the buyer. This was an assurance that the product listing functionality was working.

- Test Case 5: AI Price Prediction (Farmer)

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
5	5.1	Choose the spice names and check the price prediction.	Should select the spice	If farmer choose Cardamom, app shows the price changes according to months.	Pass

Table 9 : Test Case 5: AI Price Prediction

In order to test the price prediction function, a farmer chose turmeric as his/her. The system received the input and gave a forecasted price using the calculations of AI. The given prediction was shown accurately, which confirms the existence of such a fundamental feature.

- Test Case 6: Spice Price Prediction (Buyer)

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
6	6.1	Spice price prediction	Spice Type, Region, Demand Index, Supply Index, Rainfall, Temperature, Month, Year	Buyer can get predicted price (USD per Kg)	Pass

Table 10 : Test Case 6: Spice Price Prediction (Buyer)

In order to test the price prediction feature, an imaginary buyer chose cardamom as the spice and entered the corresponding Data including information about region, month, and environmental

data. It fed the information to the system and retrieved a predicted price on the basis of AI calculation. This essential feature was confirmed because the prediction was shown properly.

- Test Case 7: Buyer/ Farmer Matching

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
7	7.1	Buyer Matching	When buyer /farmer register to the app, they need to add the crop they interested and after in the marketplace they can find a matching farmer or buyer	Find a farmer/ Buyer	Pass

Table 11 : Test Case 7: Buyer/ Farmer Matching

One was a buyer or farmer, who entered the marketplace and clicked the search of turmeric. The system would produce a roll of farmers or buyers with turmeric at hand as per region and availability. This helped to identify the fact that matching algorithm was functioning and thus able to propose relevant farmers to buyers r buyers to farmers.

- Test Case 8: Chatbot Communication

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
8	8.1	Chatbot Communication	Buyer or farmer can press the contact farmer or contact buyer button it will direct to the chatbot and they can do the real time chat.	Real time messaging	Pass

Table 12 : Test Case 8: Chatbot Communication

A person who uses the app on the one hand pressed the button, Contact Farmer or contact buyer, as well as a message went inside a chatbot. The farmer or buyer got the message in real time and the chat went on well. This confirmed the real time messaging.

- Test Case 9: Profile Update

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
9	9.1	Profile Update	Farmer can update name, farm location, crop grown like their personal information.	Update the info.	Pass

Table 13 : Test Case 9: Profile Update

A farmer revised his/her personal details, such as area of farm and crops. The changes were saved in the system and displayed when viewed at the profile of the farmer confirming the successful completion of the update feature.

- Test Case 10: Logout Function

Test Case ID	Sequence	Test Description	Input Value	Expected Result	Actual Result
10	10.1	Logout Function	Click the Logout button.	Direct to the Login screen.	Pass

Table 14 : Test Case 10: Logout Function

The user pressed the logout button at the profile setting. The application worked as it was supposed to and the user was redirected to the login of the system. It was through this test that the handling of sessions and log out was confirmed as behaving as intended.

5 CHAPTER 5: DISCUSSION

5.1 Introduction

The chapter presents the discussion of the implementation outcomes, challenges, stakeholder feedback, and implications to a greater extent concerning the Spicely mobile application. The application was created to empower small-scale spice farmers and links them directly with the international buyers where Artificial Intelligence (AI), mobile commerce functionalities and optional IoT inputs were used to make trade as simple and transparent as possible and more profitable, as well. An evaluation of the effectiveness of the application in achieving the objectives intended is also evaluated as well as the social, economic and technical implication of using the application.

With the growing interest of agriculture in the digital shift, services such as Spicely can be vital in eliminating the information gap between the producers and markets. This chapter addresses significant discoveries, technical constraints, and the work of the users and ways how the platform can be enhanced in the future to be relevant and scalable.

5.2 Alignment with Research Objectives

There were certain research objectives to be achieved through the project. The following is the way they were approached:

- Objective 1: Support Direct Market Access

The app enabled farmers to sign in, post listing of spices, and encounter buyers without depending on intermediaries.

- Objective 2: AI based Price Prediction

A price prediction module was developed with the help of simple AI algorithms which would predict market prices with the input parameters such as season, demand, and region. This assisted farmers to determine at what time and at what price to sell their products.

- Objective 3: Supply Chain Optimisation

The farmers connected to buyers according to the type of product and preferences. This facilitated the flow of communication and led to decreasing the supply chain delays.

- Objective 4: Offer Transparency through Data

The farmers could submit the information about the specific details of the harvest, images, and optional data involving the Internet of Things to give customers the information which is traceable and credible.

- Objective 5: Usability and sustainability

It created the interface that accommodates local languages and graphics to facilitate usage. The offline capability allowed the diversion of the use of the app even in the low-connectivity regions.

5.3 Key Findings from Implementation

- Interface, Usability

Icon-based operation as well as local language support made the application user-friendly even to farmers. The uploads of products, AI prediction and browsing of buyers were accomplished in a clear manner.

- Buyer Engagement

Customers liked the fact that the products were listed clearly, and one could reach farmers directly. The matching tool enabled them to find sellers in line with their precise needs.

- Real-Time Updates

Fast data synchronization and real-time communication among farmers and buyers were guaranteed under the integration of Firebase. This made communication and processing of orders timely.

- Artificial Intelligence Price Prediction

Using small input data, the AI module was able to provide price trend that synced with the changes in market. This provided farmers with an increased reassurance in making decisions.

5.4 Challenges Faced During Development

- Digital illiteracy

There were very few smartphone or mobile-based applications on many farms. This caused initial navigation and onboarding to be hard.

- Connectivity Problems with the Internet

The instability or lack of the internet in rural locations led to slow listing upload or data sync.

- Inadequate Data to AI

Availability of historical data of markets, supply/demand factors, and local prices data were scarce or erratic, which affected the accuracy of predictions.

- Language and localization

Necessity to support more than one language (Sinhala, Tamil and English) meant additional interface design and testing.

- Trust Issues of users

Others questioned entering of personal or pricing information into a new digital platform.

5.5 Mitigation Strategies and Technical Solutions

- Enhancement of User Interface

Visual design was accomplished by having fewer words, big icons, colour-coded buttons to assist low end readers.

- Offline Capability Functionality

Offline saving data was possible, and it would automatically synchronise the next time a stable connection is achieved.

- Simple AI Tracking using Manual Data

The farmers were also permitted to enter the important variables (e.g., rainfall, the level of available supply) manually, which gave the AI an opportunity to be more personalized in price predictions despite a small number of data sets.

- Language Options

The app was translated to Sinhala and Tamil and the chatbot technology was able to copes with simple queries in languages.

- Buyer identity Verification Tags

Trust was created with verified buyer profiles. Farmers would be able to view a buyer history together with interests and could then go ahead and make deals.

5.6 Impact on Stakeholders

Effect on Farmers:

- Accessed to price forecasts and direct listing in the market.
- Better control over product pricing, selection of the buyers.
- Established trust with technological instruments of farming.

Effect to Buyers:

- Farmer listing inquiry made simpler.
- A quicker trade cycle and communication.
- Increase transparency in the sourcing of spices.

Shared Benefits:

- Enhanced buyer seller relationships.
- Fewer errors and shorter time of transaction.
- More effectiveness between the two parties in decision making.

6 CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study attempted to provide a mobile application called Spicely that enables empowerment of small-scale spice farmers by facilitating direct contact with international markets utilizing Artificial Intelligence (AI), blockchain investigative transparency, and mobile commerce tools. This research acknowledged that important obstacles of farmers, such as the lack of market access and price volatility, the reliance on third parties, and non-traceability could be overcome via a digital solution adapted to the original Sri Lankan context in the agricultural sector.

The application performed its main purposes to integrate AI into the price prediction in real-time, provide direct contact between buyers and farmers, and make product data transparent. The tests showed that it had a high level of user acceptance and farmers appreciated the AI-generated market insights and automatic matching of buyers. Usability was made possible by the humanization of interfaces, multilingual support and assisted and offline use, which were critical in a rural setting with different degrees of digital literacy.

Firebase was one of the major backbones of the technology used as was the cloud-based backend of the application and real time database. Firebase offered a very scalable, secure, and adaptive user information storage, product listing, chat record, and AI input provider. Its real-time synchronizing ability also made possible the direct interaction between farmers and buyers including the situations of limited bandwidth, and the services like Firebase Authentication made it easy to log in to and to register securely. Firestore database architecture was built to support effective querying, and this was critical in the functionality of features such as buyer-product matching and listing updates. In general, the Firebase was very effective in such implementation because of the ease of integration with the Android Studio (Java/XML), the absence of significant maintenance costs, and the ability to support the real-time operations.

Several security considerations also came out during the development and deployment process. Data safety, user identity safety and authentication of transactions were of concern and these issues were solved with firebase authentication, user verification systems and secured cloud storage. Therefore, although blockchain had only been simulated, but not fully implemented, the system architecture preconditioned the future integration, offering tamper-proof transaction records and more powerful trust mechanisms. The aspects of safe communication and data encryption were also mentioned in the list of what still must be improved, especially with the expanded use of the product by users.

In a wider view however, Spicely illustrates that with well-thought-out digital solutions, traditional agriculture can be converted to the one that is transparent, efficient, and power to marginalized groups. If the lack of infrastructure, digital literacy, and data reliability remain obstacles, the platform is its own significant step towards inclusive economic development by relying on agri-tech innovation.

6.2 Future Considerations

- Complete Blockchain Incorporation

The current system will emulate the use of a blockchain in tracing the products, and future versions of Spicely should adopt the use of an actual blockchain ledger in order to achieve the immutability of transactions, the transparency of supply chains and unalterable trade histories. This will double the chances of buyer confidence and conformance to the international standards which they may be seeking especially in case of organic or fair-trade certifications.

- Sophisticated AI Models Improvements

The existing AI price forecast involves simple parameters. In further revisions, it is possible to add a connection to external APIs, which present current commodity prices, weather conditions, foreign market demand, and geopolitical dynamics. Moreover, using the machine learning tool (regression analysis, decision trees, or neural networks) would enhance precision of the predictions in the long-term.

- Digital Payment Incorporation

To allow easy transactions to be made, the platform must incorporate mobile payment tools, be it PayPal, eZCash, mCash, or bank APIs. This will enable the farmers and the buyers to make the payments inside the app safely and will not require the use external sources of payments and will promote better financial comfort and responsibility financial transactions.

- Delivery Management and Logistics

Incorporating a logistics management aspect will enable the farmers to manage the delivery of their spices process through communication with the transportation providers and manage the progress of their spices deliveries. This might involve incorporation of third party-logistics firms or the government-sponsored transport system to improve the transportation process of the supply chain between the farm and export.

- Better offline capability

The users in the rural areas have low access to the internet. The treatment of offline capacity, which encompasses the local data store, asynchronous update of data, and communication via SMS, would also render the application more comprehensive to farmers with poor connectivity or those in unreachable locations.

- Voice and Multilingual support

As the Sinhala and English are supported at an initial level, the range of languages and voice input and output capabilities may significantly enhance the usability of the program and particularly to

illiterate farmers. Copying the services of text-to-speech and conversion back to the words in the local language can enhance accessibility.

6.3 References

Aker, J.C. and Mbiti, I.M. (2010) ‘Mobile phones and economic development in Africa’, Journal of Economic Perspectives, 24(3), pp. 207–232.

Casino, F., Dasaklis, T.K. and Patsakis, C. (2019) ‘A systematic literature review of blockchain-based applications: Current status, classification and open issues’, Telecommunications Policy, 43(10), p. 101848.

FAO (2020) Digital agriculture: Supporting small-scale farmers. Rome: Food and Agriculture Organization of the United Nations.

FAO (2020) Spices and herbs for sustainable agriculture. Available at: https://www.fao.org (Accessed: 8 April 2025).

Freedman, P. (2020) ‘History of spices’, in Meiselman, H.L. (ed.) Handbook of eating and drinking. Cham: Springer, pp. 77–91.

IFAD (2020) Smallholder farmers and market access. International Fund for Agricultural Development. Available at: https://www.ifad.org (Accessed: 8 April 2025).

ITC (2021) Spices and culinary herbs. International Trade Centre. Available at: https://intracen.org (Accessed: 8 April 2025).

Jaiswal, S., Gautam, S., Kumar, A., Sharma, P. and Singh, S. (2022) ‘Blockchain technology: Emerging applications in agriculture supply chain management’, Materials Today: Proceedings, 50, pp. 1391–1395.

Jayasinghe, R., Abeysinghe, S. and Fernando, D. (2022) ‘Value chain analysis of cinnamon industry in Sri Lanka’, Journal of Agriculture and Forestry, 68(1), pp. 101–114.

Kamilaris, A., Kartakoullis, A. and Prenafeta-Boldú, F.X. (2019) ‘A review on the practice of big data analysis in agriculture’, Computers and Electronics in Agriculture, 147, pp. 70–90.

Kamilaris, A., Fonts, A. and Prenafeta-Boldú, F.X. (2019) ‘The rise of blockchain technology in agriculture and food supply chains’, Trends in Food Science & Technology, 91, pp. 640–652.

Kumar, P., Sharma, R. and Singh, R. (2018) ‘Spices cultivation in India: Profitability and constraints’, Economic Affairs, 63(2), pp. 341–348.

Liakos, K.G., Busato, P., Moshou, D., Pearson, S. and Bochtis, D. (2018) ‘Machine learning in agriculture: A review’, Sensors, 18(8), p. 2674.

Mittal, S. and Mehar, M. (2016) ‘Socio-economic impact of mobile phones on Indian agriculture’, Indian Journal of Agricultural Economics, 71(3), pp. 334–345.

Ogunniyi, A.I. and Ojebiyi, W.G. (2016) ‘The impact of adoption of improved agricultural technology on poverty reduction among smallholder farmers in rural Nigeria’, Journal of Rural Studies, 46, pp. 254–263.

Ravindran, P. (2017) The encyclopedia of herbs and spices. Wallingford: CABI Publishing.

Ruben, R., Verkaart, S. and Roll, K. (2019) ‘Fairtrade and export value chains: Opportunities for smallholder empowerment’, Journal of Agricultural Economics, 70(2), pp. 372–387.

Saberi, S., Kouhizadeh, M., Sarkis, J. and Shen, L. (2019) ‘Blockchain technology and its relationships to sustainable supply chain management’, International Journal of Production Research, 57(7), pp. 2117–2135.

Sharma, A., Kumar, N. and Rambli, D.R.A. (2020) ‘Internet of Things and artificial intelligence in agriculture: Implementation challenges and future prospects’, in Handbook of IoT and Big Data. Cham: Springer, pp. 1–23.

SpiceXchange (2023) SpiceXchange platform. Available at: https://www.spicexchange.com (Accessed: 8 April 2025).

Tashakkori, A. and Teddlie, C. (2003) Handbook of mixed methods in social & behavioral research. Thousand Oaks, CA: Sage Publications.

The Borgen Project (2024) ‘Smartphone farmers: Mobile commerce and agriculture’. Available at: https://borgenproject.org/smartphone-farmers/ (Accessed: 8 April 2025).
TIME (2024) ‘Farmerline’s Darli AI empowers African farmers’. Available at: https://time.com/7094874/farmerline-darli-ai/ (Accessed: 8 April 2025).

Trace AgTech (2025) ‘Challenges in the spice market: A comprehensive overview’. Available at: https://traceagtech.com (Accessed: 8 April 2025).

Tripathy, S., Bhanja, S. and Nayak, S. (2021) ‘AI-based chatbots for smart farming: A systematic review’, International Journal of Recent Technology and Engineering, 9(6), pp. 55–60.

Tripoli, M. and Schmidhuber, J. (2018) Emerging opportunities for the application of blockchain in the agri-food industry. Rome: Food and Agriculture Organization of the United Nations.

UNDP (2021) Blockchain for agri-food traceability. United Nations Development Programme. Available at: https://www.undp.org (Accessed: 8 April 2025).

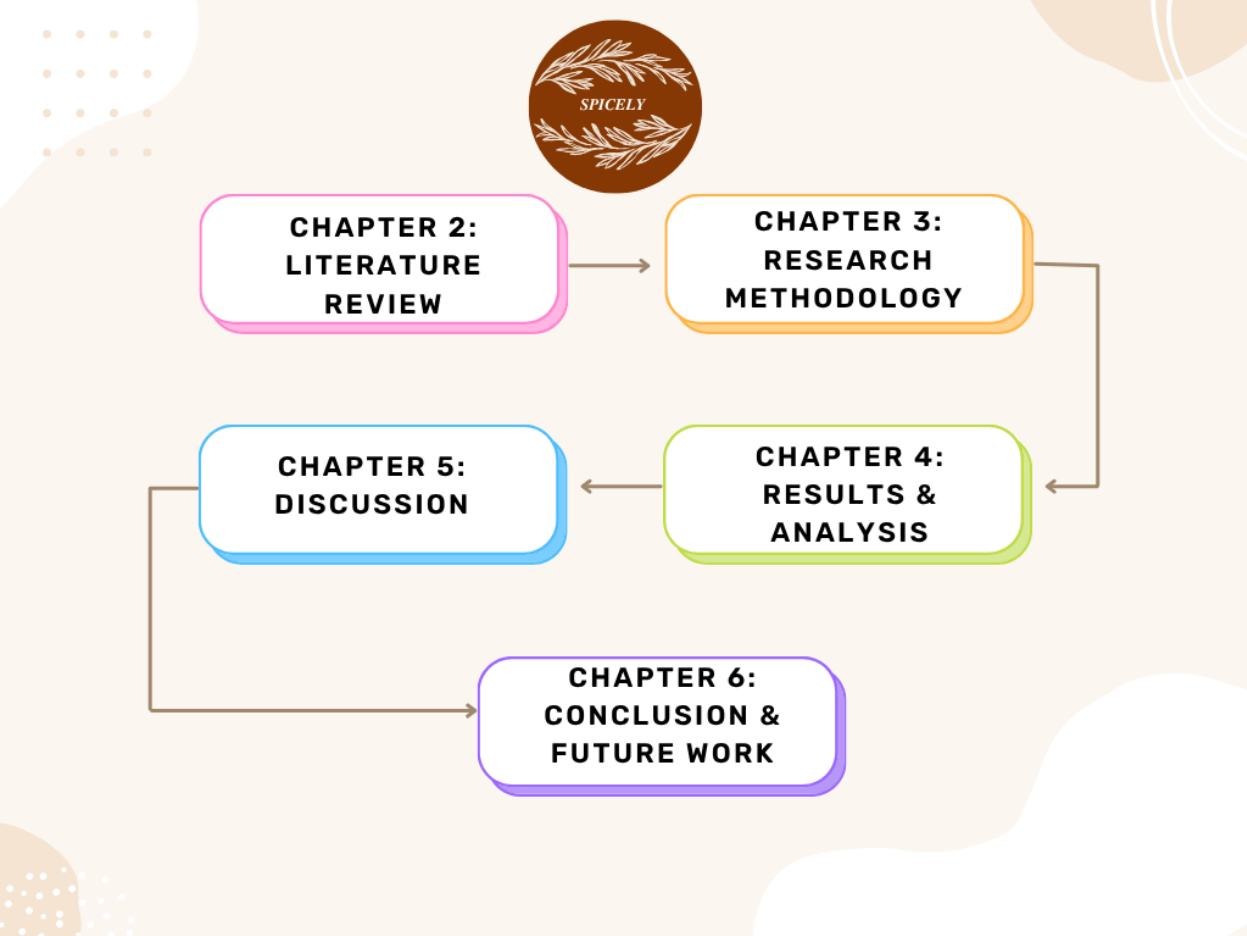
Wamba, S.F. and Queiroz, M.M. (2020) ‘Big data analytics in supply chain management and business performance: A resource-based view’, Annals of Operations Research, 270(1–2), pp. 1–23.

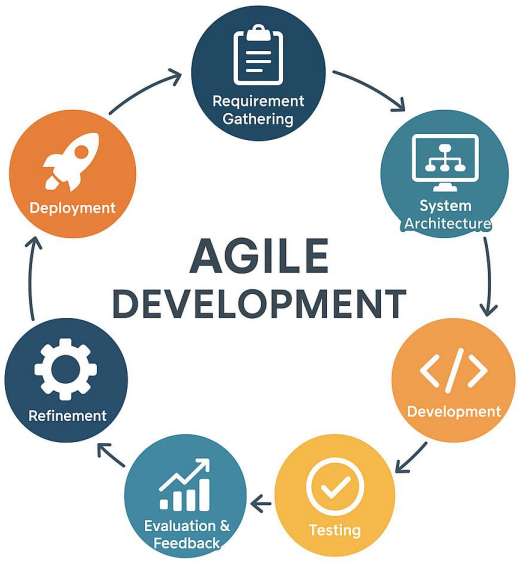
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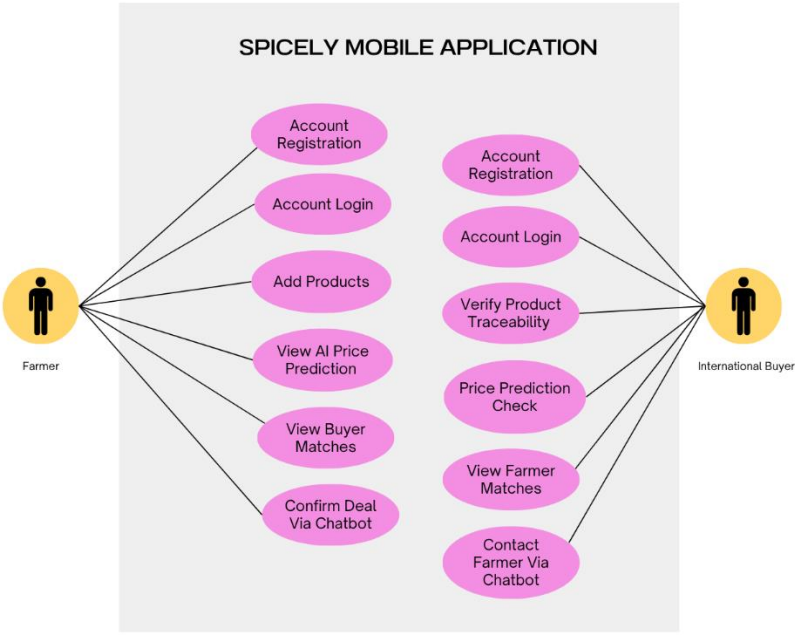
Wolfert, S., Ge, L., Verdouw, C. and Bogaardt, M.J. (2017) ‘Big data in smart farming – A review’, Agricultural Systems, 153, pp. 69–80.

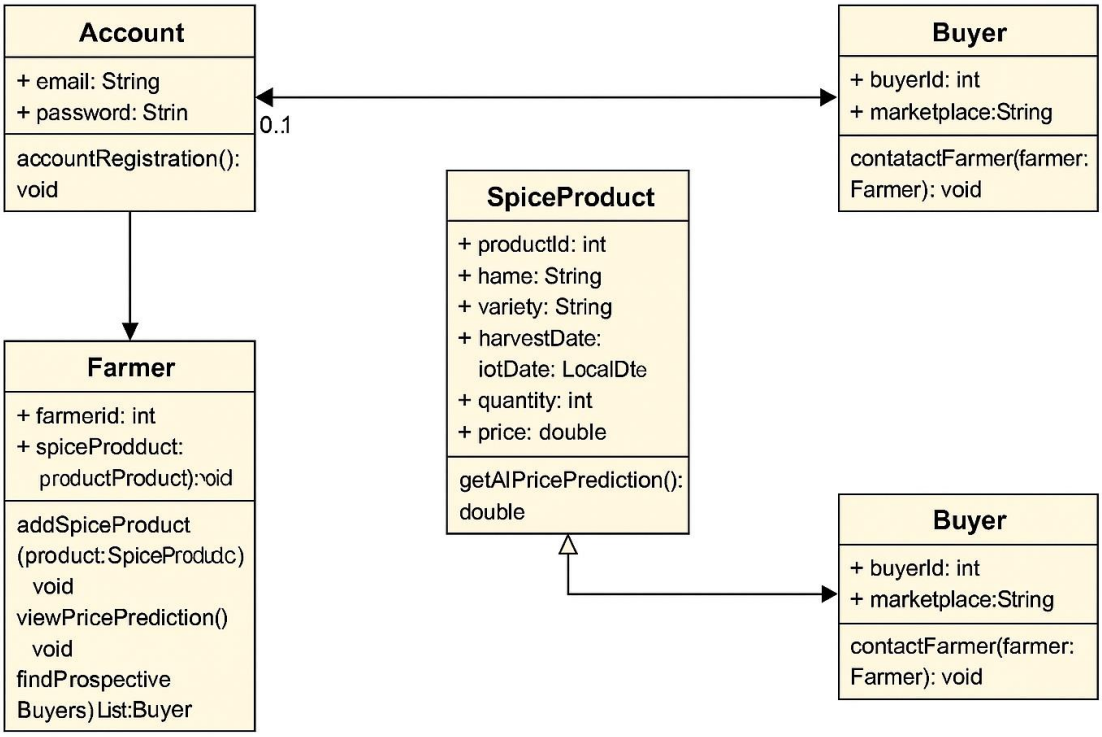
World Bank (2019) Digital agriculture in developing countries. Available at: https://www.worldbank.org (Accessed: 8 April 2025).

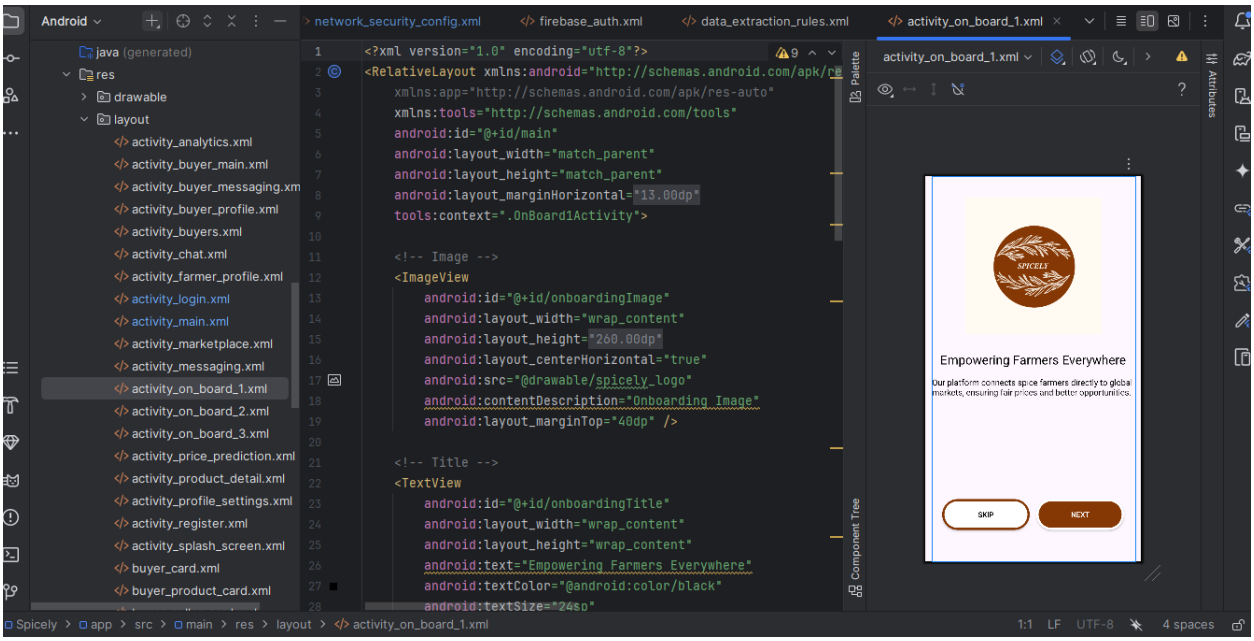
6.4 Appendix

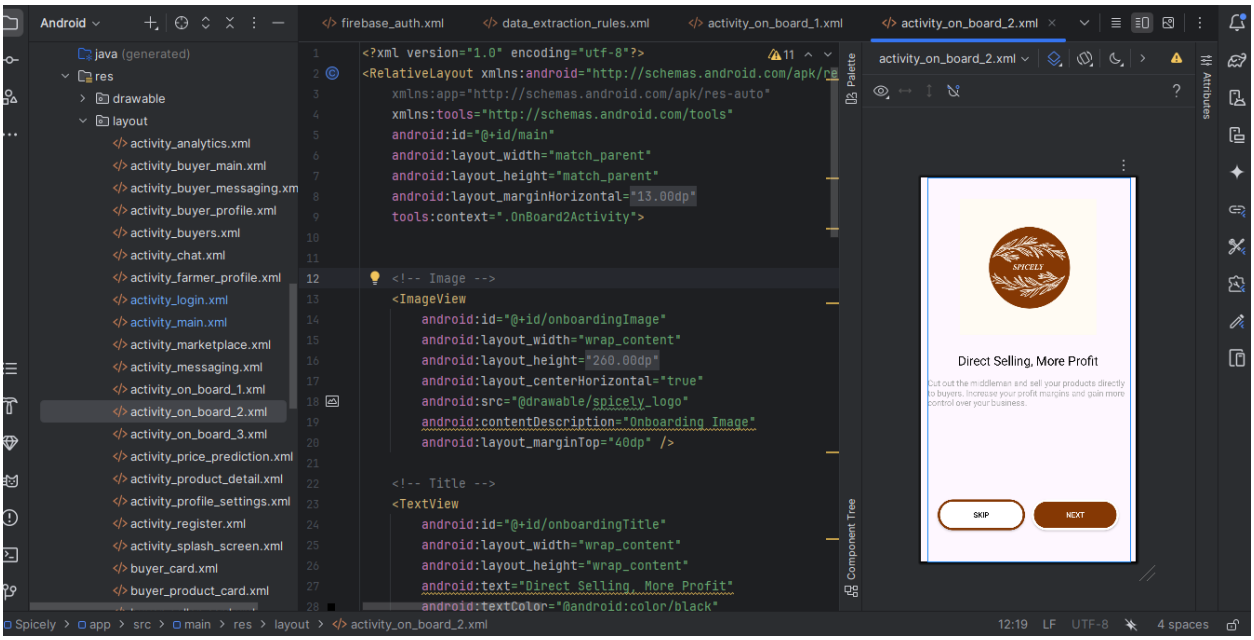

SPICELY
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CHAPTER 3: RESEARCH METHODOLOGY
CHAPTER 4: RESULTS & ANALYSIS
CHAPTER 5: DISCUSSION
CHAPTER 6: CONCLUSION & FUTURE WORK

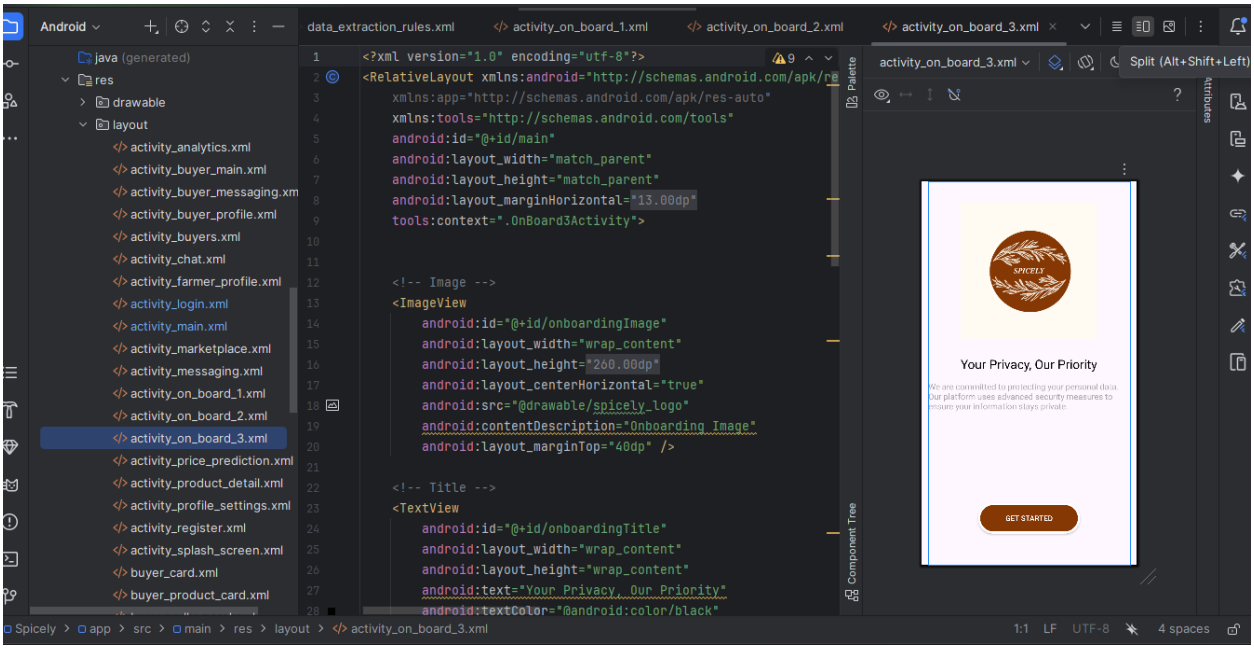

AGILE DEVELOPMENT
Requirement Gathering
System Architecture
Development
Testing
Evaluation & Feedback
Refinement
Deployment

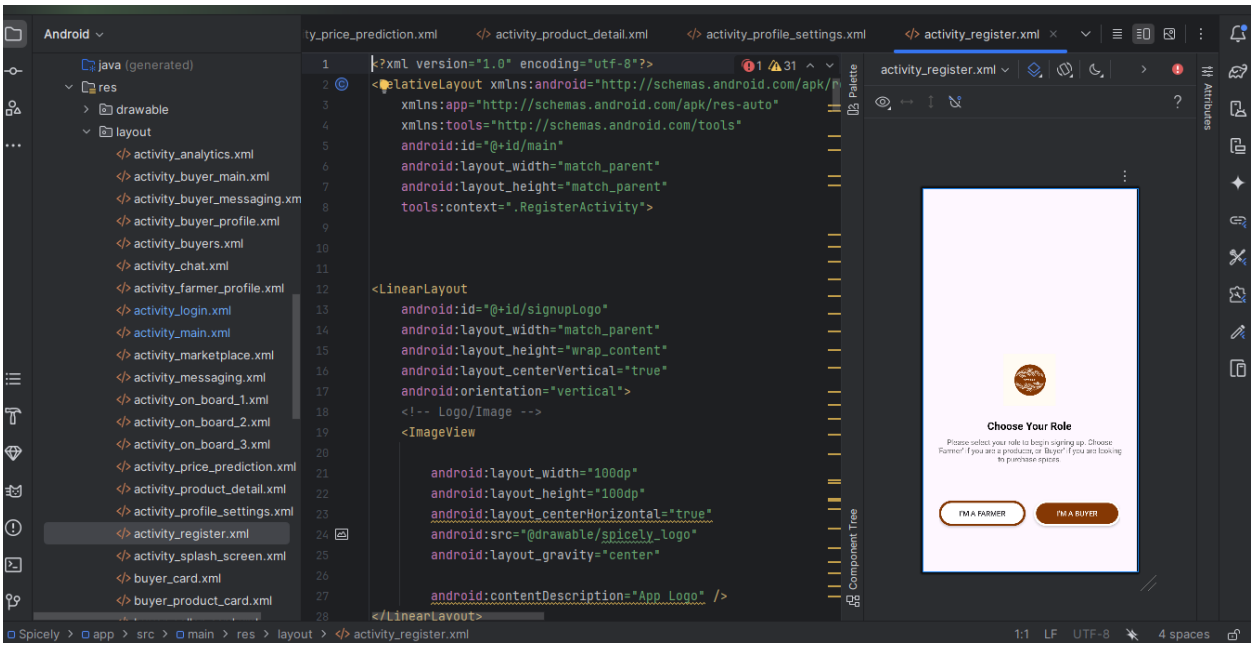

SPICELY MOBILE APPLICATION
Account Registration
Account Login
Add Products
View AI Price Prediction
View Buyer Matches
Confirm Deal Via Chatbot
Account Registration
Account Login
Verify Product Traceability
Price Prediction Check
View Farmer Matches
Contact Farmer Via Chatbot
Farmer
International Buyer

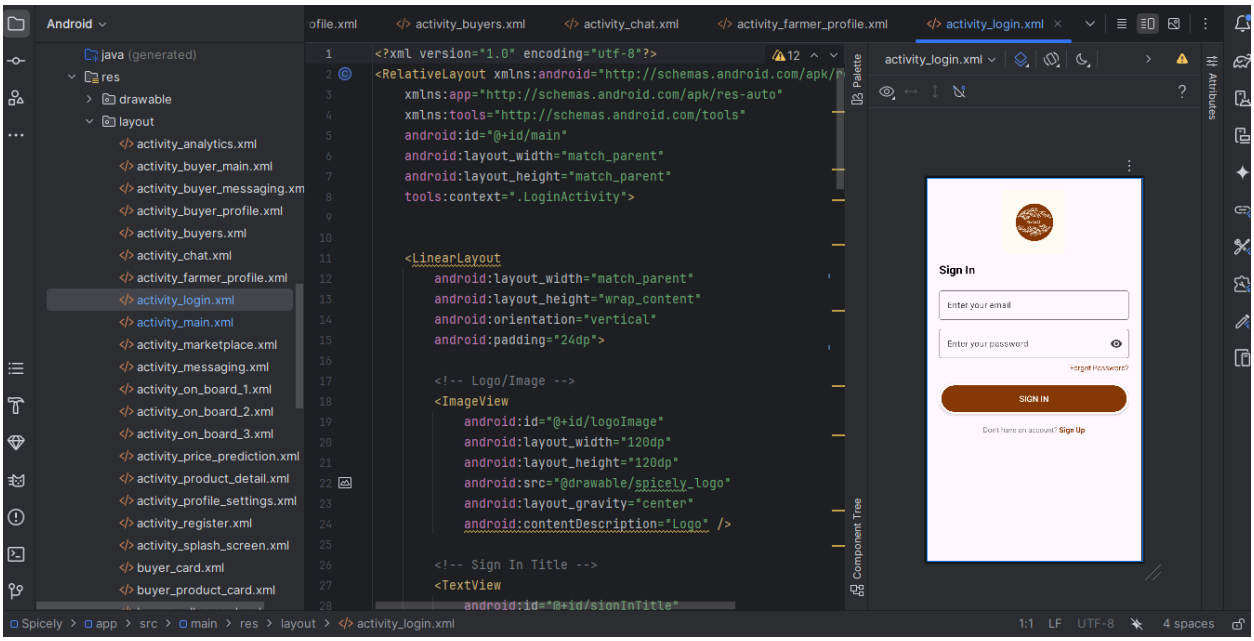

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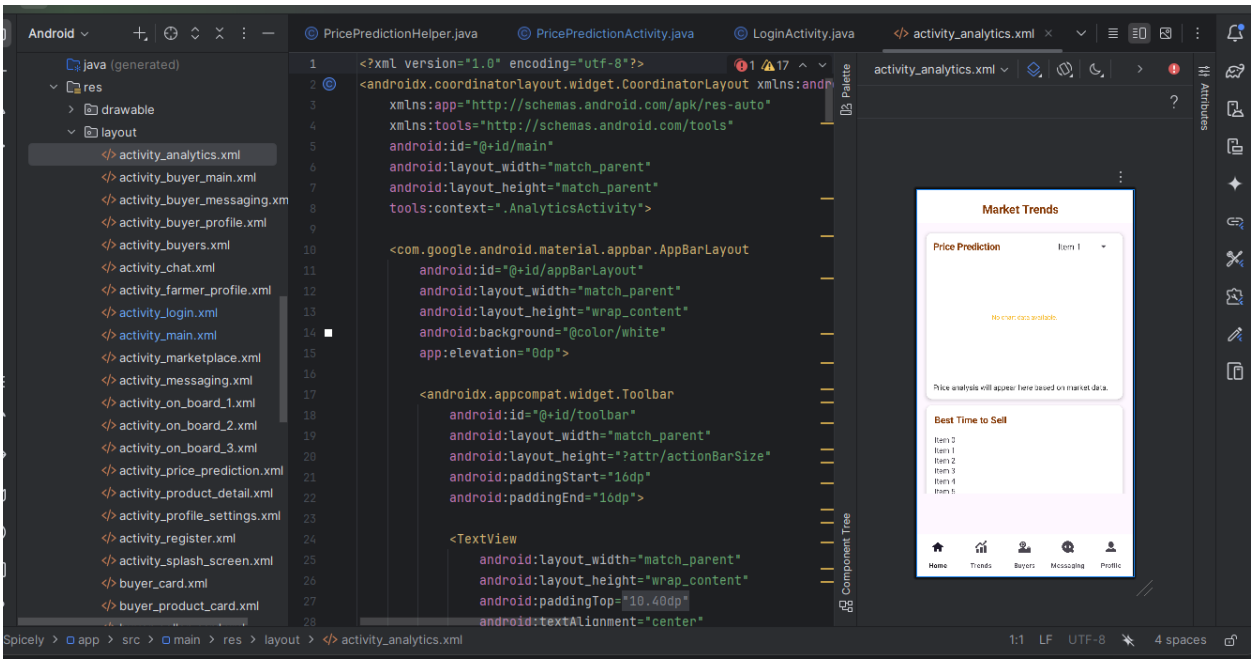

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Empowering Farmers Everywhere
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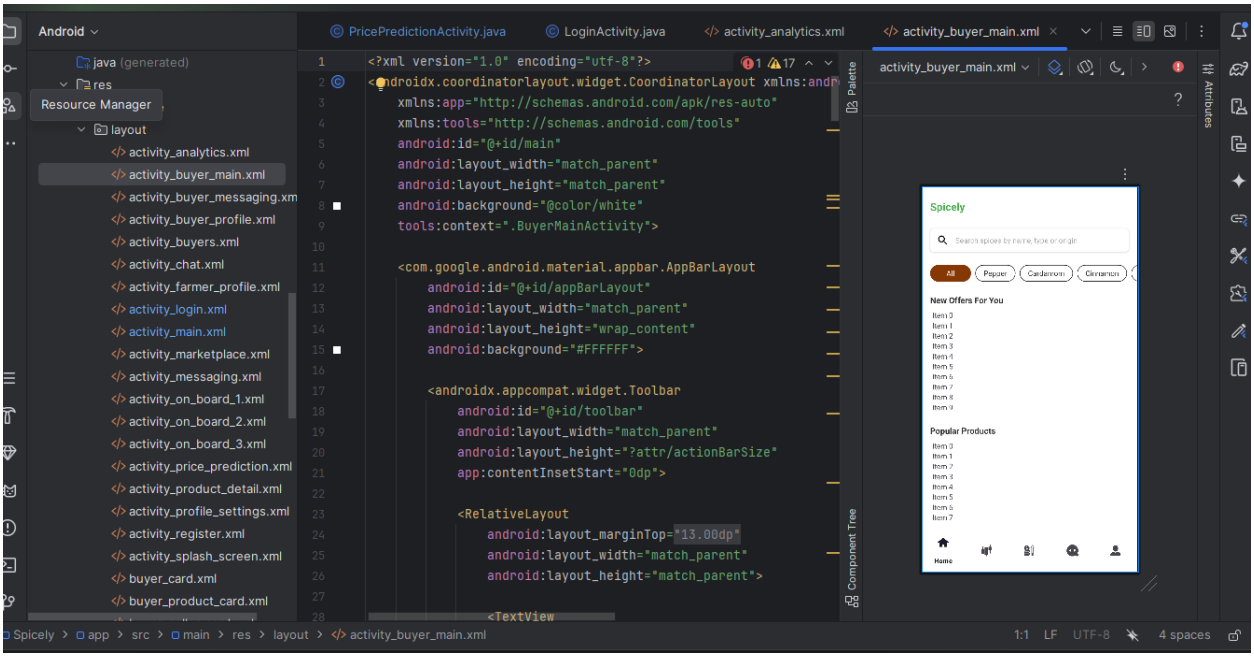

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xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_marginHorizontal="13.00dp"
tools:context=".OnBoard2Activity">
<!-- Image -->
<ImageView
android:id="@+id/onboardingImage"
android:layout_width="wrap_content"
android:layout_height="260.00dp"
android:layout_centerHorizontal="true"
android:src="@drawable/spicely_logo"
android:contentDescription="Onboarding Image"
android:layout_marginTop="40dp" />
<!-- Title -->
<TextView
android:id="@+id/onboardingTitle"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Direct Selling, More Profit"
android:textColor="@android:color/black"
Palette
Component Tree
Attributes
activity_on_board_2.xml
SPICELY
Direct Selling, More Profit
Cut out the middleman and sell your products directly to buyers. Increase your profit margins and gain more control over your business.
SKIP
NEXT
Spicely
app
src
main
res
layout
activity_on_board_2.xml
12:19
LF
UTF-8
4 spaces

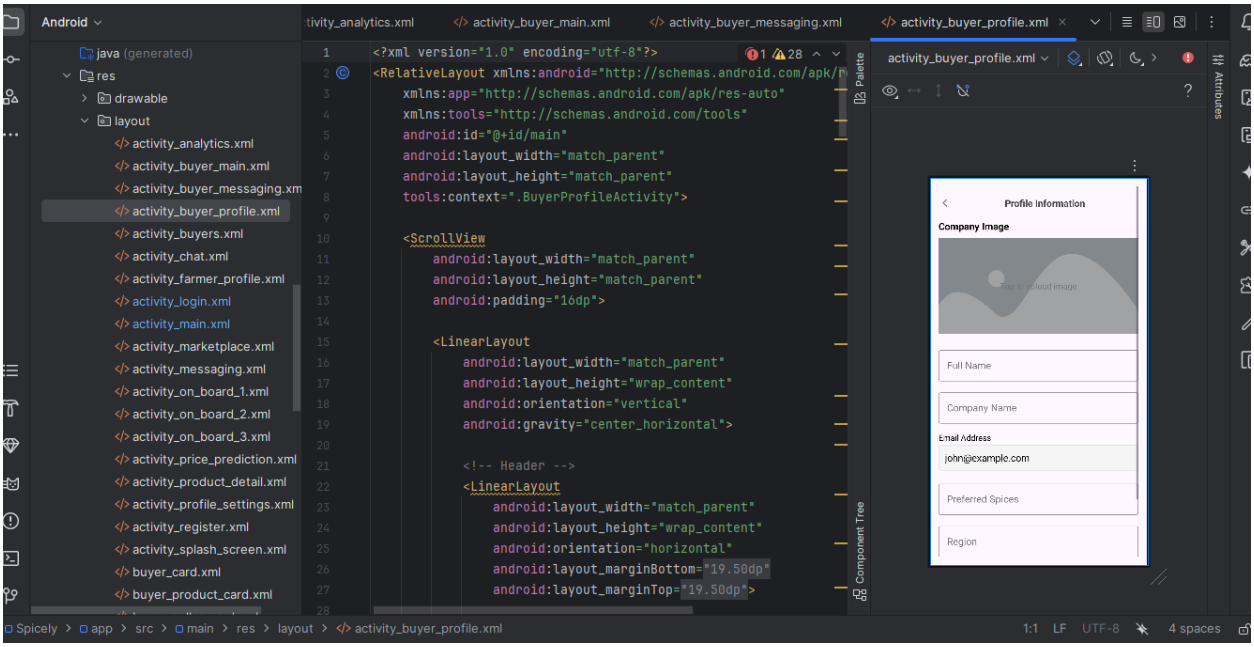

Android
data_extraction_rules.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
Split (Alt+Shift+Left)
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/r
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_marginHorizontal="13.00dp"
tools:context=".OnBoard3Activity">
<!-- Image -->
<ImageView
android:id="@+id/onboardingImage"
android:layout_width="wrap_content"
android:layout_height="260.00dp"
android:layout_centerHorizontal="true"
android:src="@drawable/spicely_logo"
android:contentDescription="Onboarding Image"
android:layout_marginTop="40dp" />
<!-- Title -->
<TextView
android:id="@+id/onboardingTitle"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Your Privacy, Our Priority"
android:textColor="@android:color/black"
Palette
Component Tree
Attributes
activity_on_board_3.xml
Your Privacy, Our Priority
GET STARTED
Spicely
app
src
main
res
layout
activity_on_board_3.xml
1:1
LF
UTF-8
4 spaces

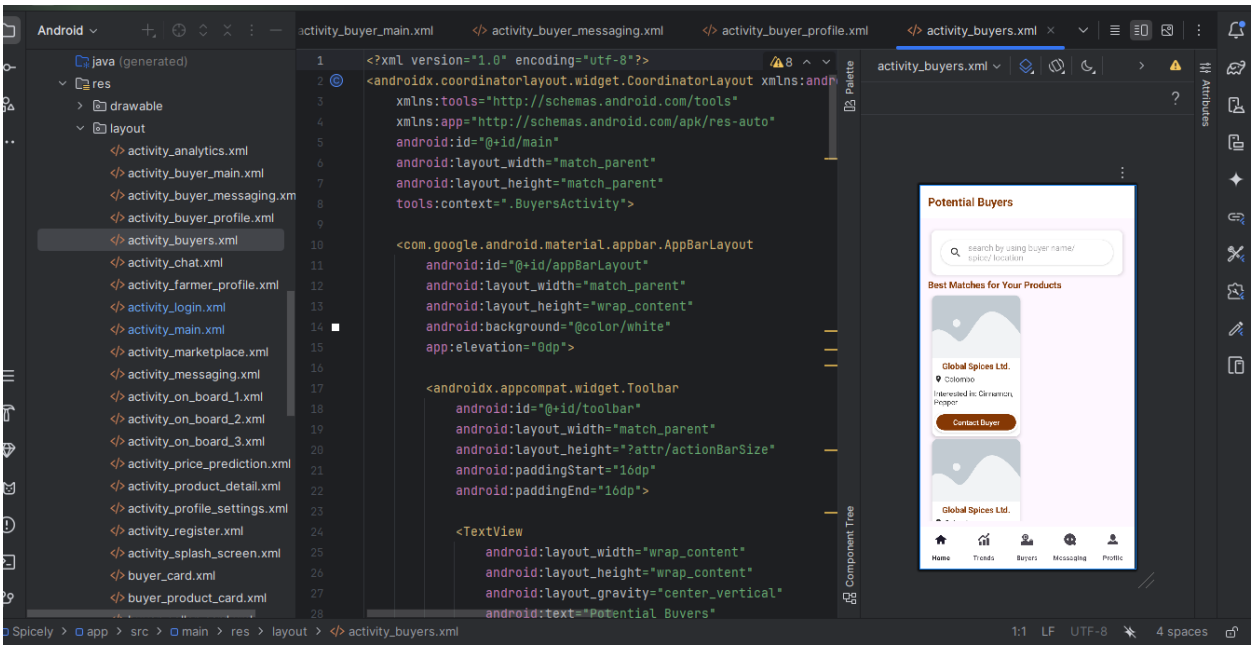

Android
ty_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".RegisterActivity">
<LinearLayout
android:id="@+id/signupLogo"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_centerVertical="true"
android:orientation="vertical">
<!-- Logo/Image -->
<ImageView
android:layout_width="100dp"
android:layout_height="100dp"
android:layout_centerHorizontal="true"
android:src="@drawable/spicely_logo"
android:layout_gravity="center"
android:contentDescription="App Logo" />
</LinearLayout>
Palette
Component Tree
Attributes
activity_register.xml
Choose Your Role
I'M A FARMER
I'M A BUYER
Spicely
app
src
main
res
layout
activity_register.xml
1:1
LF
UTF-8
4 spaces

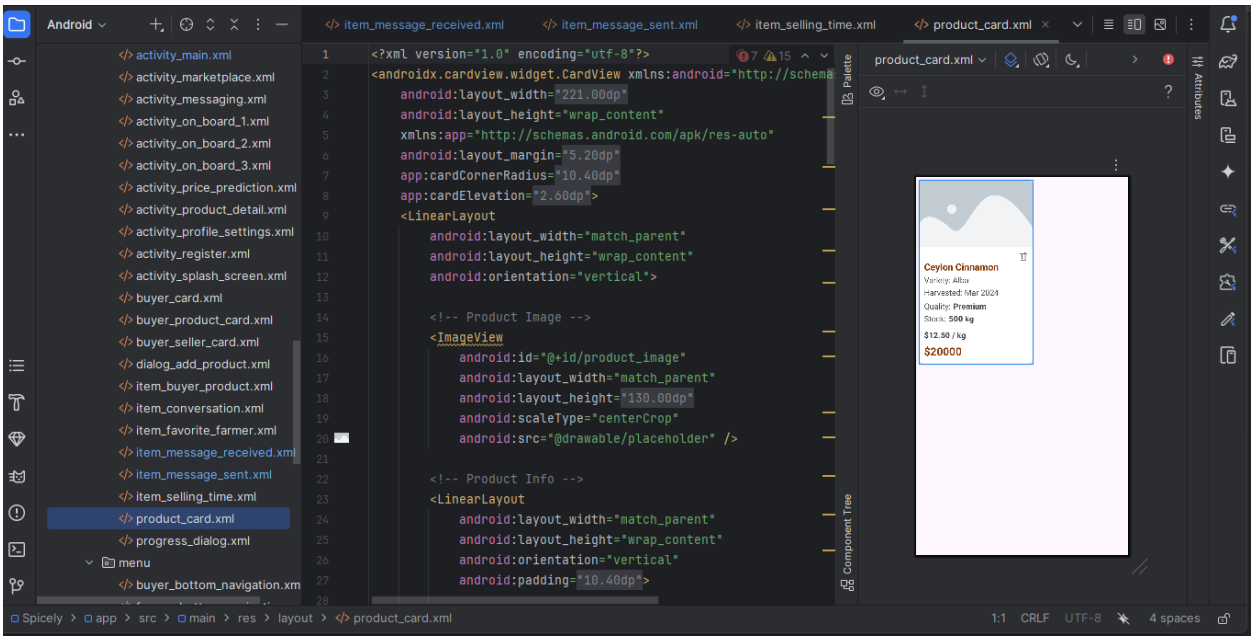

Android
activity_login.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xml
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".LoginActivity">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="24dp">
<!-- Logo/Image -->
<ImageView
android:id="@+id/logoImage"
android:layout_width="120dp"
android:layout_height="120dp"
android:src="@drawable/spicely_logo"
android:layout_gravity="center"
android:contentDescription="Logo" />
<!-- Sign In Title -->
<TextView
android:id="@+id/signInTitle"
Sign In
Enter your email
Enter your password
Forgot Password?
SIGN IN
Don't have an account? Sign Up
Palette
Component Tree
Attributes
Spicely
app
src
main
res
layout
activity_login.xml
1:1
LF
UTF-8
4 spaces

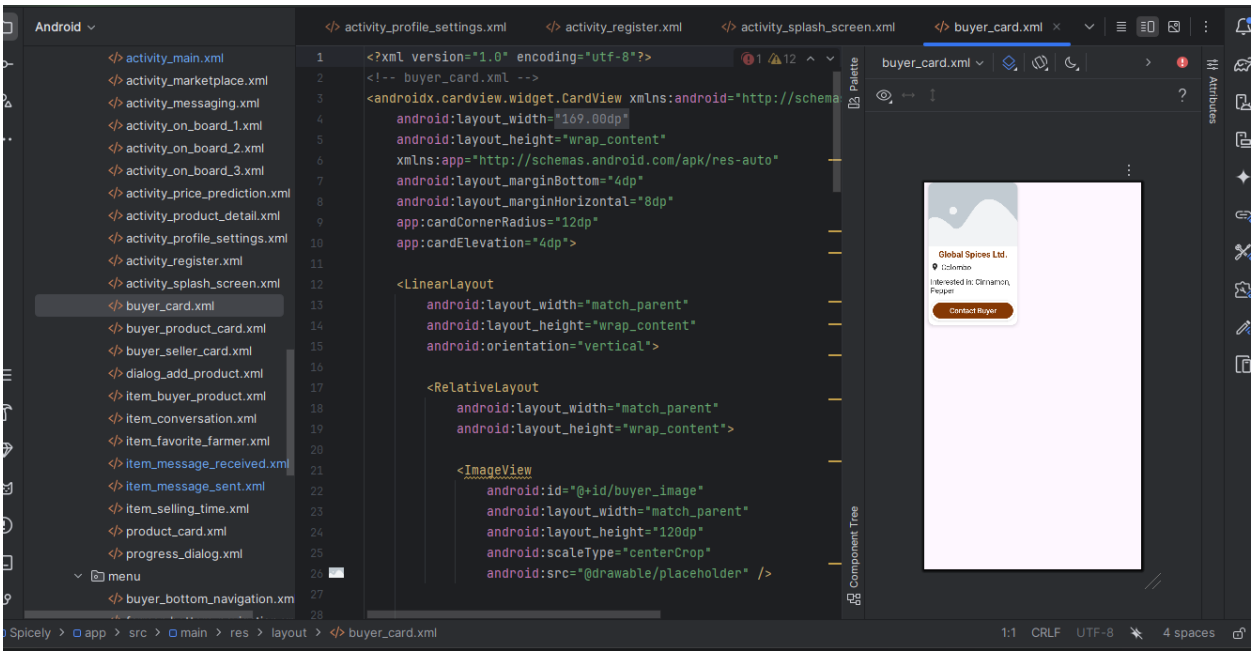

Android
PricePredictionHelper.java
PricePredictionActivity.java
LoginActivity.java
activity_analytics.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xml
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:andr
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".AnalyticsActivity">
<com.google.android.material.appbar.AppBarLayout
android:id="@+id/appBarLayout"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:background="@color/white"
app:elevation="0dp">
<androidx.appcompat.widget.Toolbar
android:id="@+id/toolbar"
android:layout_width="match_parent"
android:layout_height="?attr/actionBarSize"
android:paddingStart="16dp"
android:paddingEnd="16dp">
<TextView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:paddingTop="10.40dp"
android:textAlignment="center"
Market Trends
Price Prediction
Item 1
Price analysis will appear here based on market data.
Best Time to Sell
Item 2
Item 1
Item 2
Item 3
Item 4
Item 5
Home
Trends
Buyers
Messaging
Profile
Palette
Component Tree
Attributes
Spicely
app
src
main
res
layout
activity_analytics.xml
1:1
LF
UTF-8
4 spaces

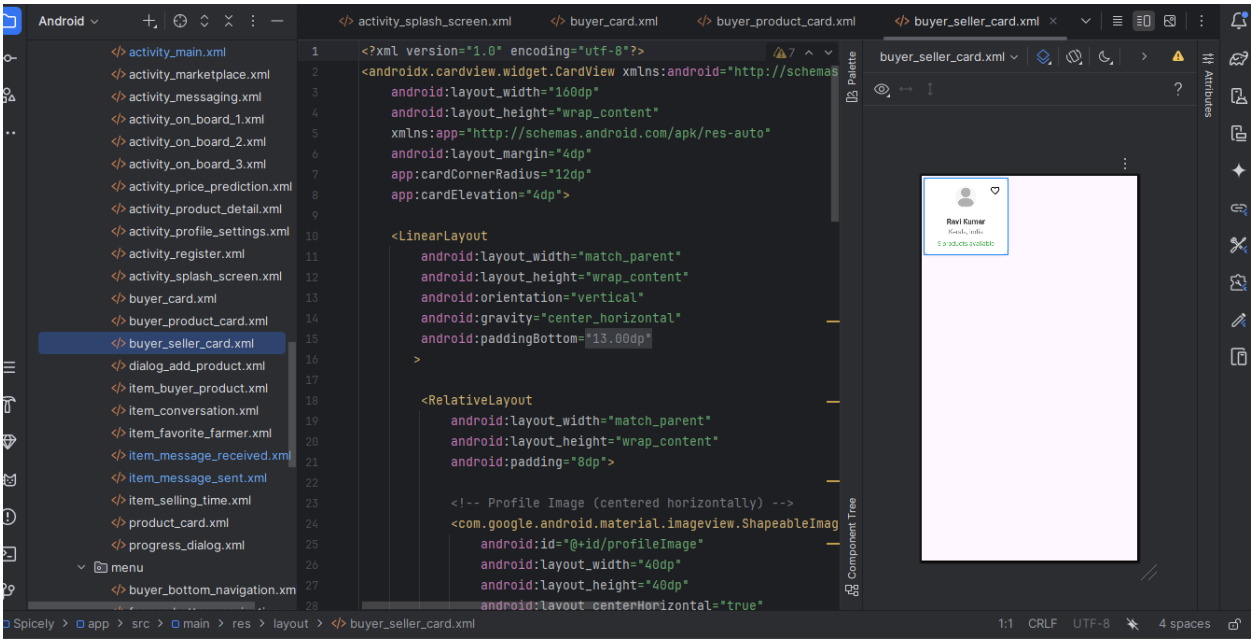

Android
PricePredictionActivity.java
LoginActivity.java
activity_analytics.xml
activity_buyer_main.xml
Resource Manager
java (generated)
res
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:andr
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="@color/white"
tools:context=".BuyerMainActivity">
<com.google.android.material.appbar.AppBarLayout
android:id="@+id/appBarLayout"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:background="#FFFFFF">
<androidx.appcompat.widget.Toolbar
android:id="@+id/toolbar"
android:layout_width="match_parent"
android:layout_height="?attr/actionBarSize"
app:contentInsetStart="0dp">
<RelativeLayout
android:layout_marginTop="13.00dp"
android:layout_width="match_parent"
android:layout_height="match_parent">
<TextView
Palette
Component Tree
Attributes
activity_buyer_main.xml
Spicely
All
Pepper
Cardamom
Cinnamon
New Offers For You
Popular Products
Home
Spicely › app › src › main › res › layout › activity_buyer_main.xml
1:1
LF
UTF-8
4 spaces

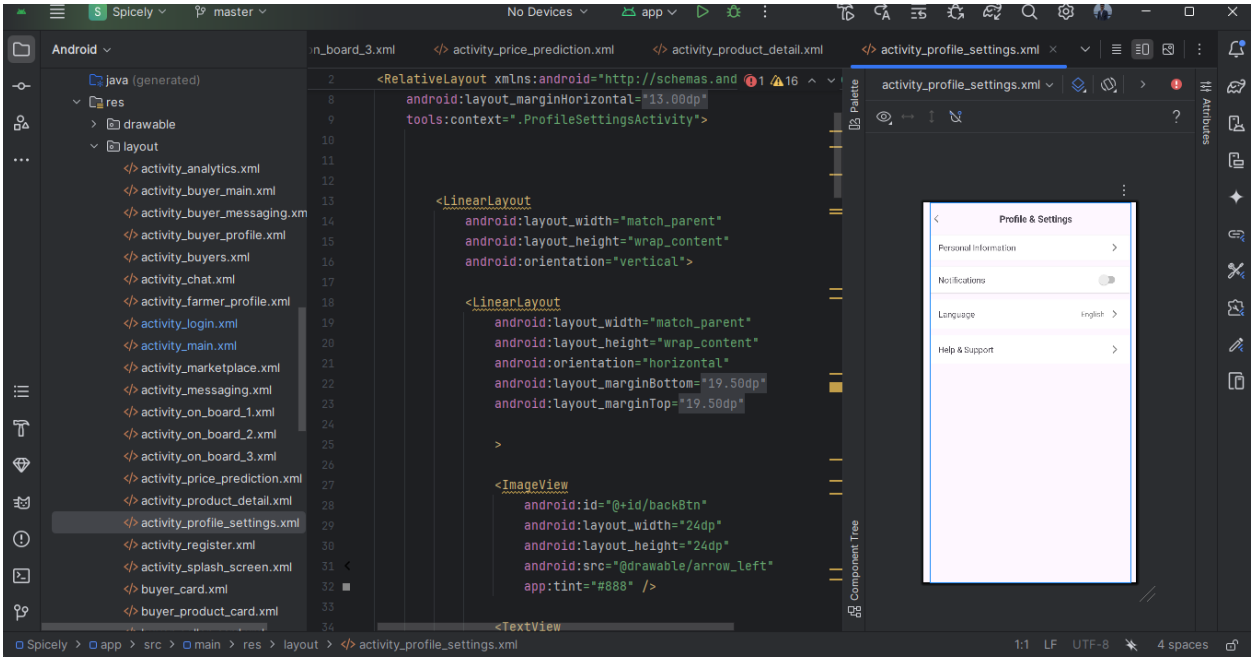

Android
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xml
activity_buyer_profile.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/r
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".BuyerProfileActivity">
<ScrollView
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="16dp">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:gravity="center_horizontal">
<!-- Header -->
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
android:layout_marginBottom="19.50dp"
android:layout_marginTop="19.50dp">
Palette
Component Tree
Attributes
activity_buyer_profile.xml
Profile Information
Company Image
Full Name
Company Name
Email Address
john@example.com
Preferred Spices
Region
Spicely › app › src › main › res › layout › activity_buyer_profile.xml
1:1
LF
UTF-8
4 spaces

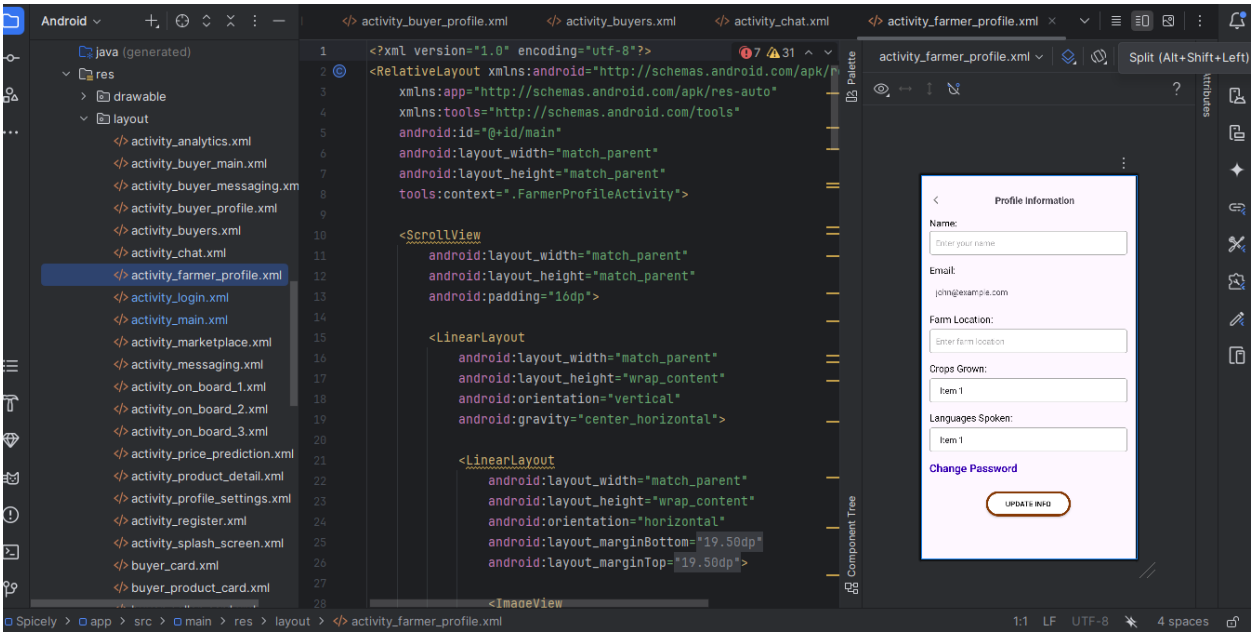

Android
activity_buyer_main.xml
activity_buyer_messaging.xml
activity_buyer_profile.xml
activity_buyers.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:andr
xmlns:tools="http://schemas.android.com/tools"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".BuyersActivity">
<com.google.android.material.appbar.AppBarLayout
android:id="@+id/appBarLayout"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:background="@color/white"
app:elevation="0dp">
<androidx.appcompat.widget.Toolbar
android:id="@+id/toolbar"
android:layout_width="match_parent"
android:layout_height="?attr/actionBarSize"
android:paddingStart="16dp"
android:paddingEnd="16dp">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_gravity="center_vertical"
android:text="Potential Buyers"
Palette
Component Tree
Attributes
Potential Buyers
Best Matches for Your Products
Global Spices Ltd.
Contact Buyer
Home
Trends
Buyers
Messaging
Profile
Spicely
app
src
main
res
layout
activity_buyers.xml
1:1
LF
UTF-8
4 spaces

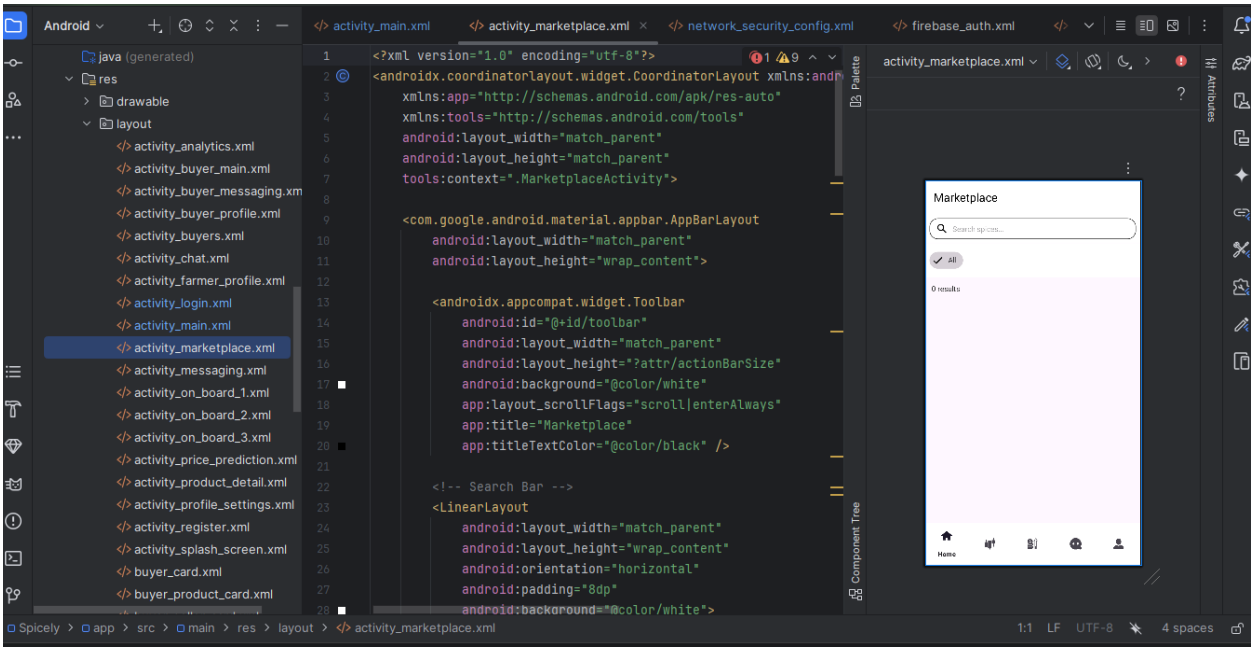

Android
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView xmlns:android="http://schema
android:layout_width="221.00dp"
android:layout_height="wrap_content"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_margin="5.20dp"
app:cardCornerRadius="10.40dp"
app:cardElevation="2.60dp">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical">
<!-- Product Image -->
<ImageView
android:id="@+id/product_image"
android:layout_width="match_parent"
android:layout_height="130.00dp"
android:scaleType="centerCrop"
android:src="@drawable/placeholder" />
<!-- Product Info -->
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="10.40dp">
Palette
Component Tree
Attributes
Ceylon Cinnamon
Variety: Alba
Harvested: Mar 2024
Quality: Premium
Stock: 500 kg
$12.50 / kg
$20000
Spicely
app
src
main
res
layout
product_card.xml
1:1
CRLF
UTF-8
4 spaces

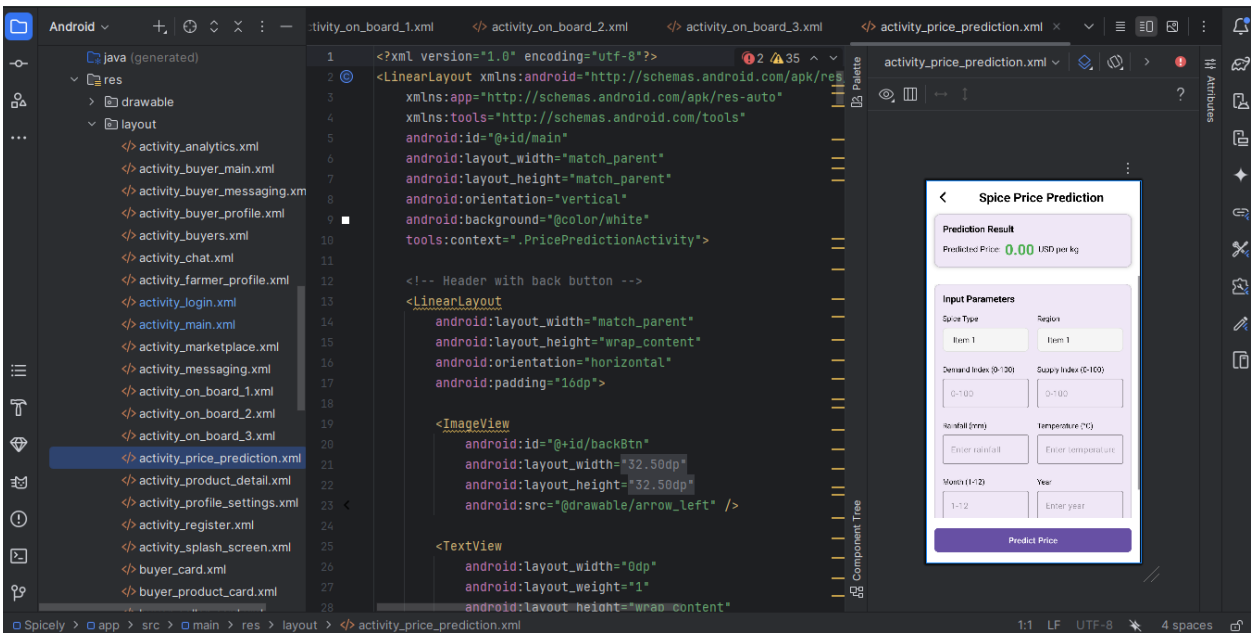

Android
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
<?xml version="1.0" encoding="utf-8"?>
<!-- buyer_card.xml -->
<androidx.cardview.widget.CardView xmlns:android="http://schema
android:layout_width="169.00dp"
android:layout_height="wrap_content"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_marginBottom="4dp"
android:layout_marginHorizontal="8dp"
app:cardCornerRadius="12dp"
app:cardElevation="4dp">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical">
<RelativeLayout
android:layout_width="match_parent"
android:layout_height="wrap_content">
<ImageView
android:id="@+id/buyer_image"
android:layout_width="match_parent"
android:layout_height="120dp"
android:scaleType="centerCrop"
android:src="@drawable/placeholder" />
buyer_card.xml
Global Spices Ltd.
Colombo
Interested in: Cinnamon, Pepper
Contact Buyer
Palette
Component Tree
Attributes
Spicely
app
src
main
res
layout
buyer_card.xml
1:1
CRLF
UTF-8
4 spaces

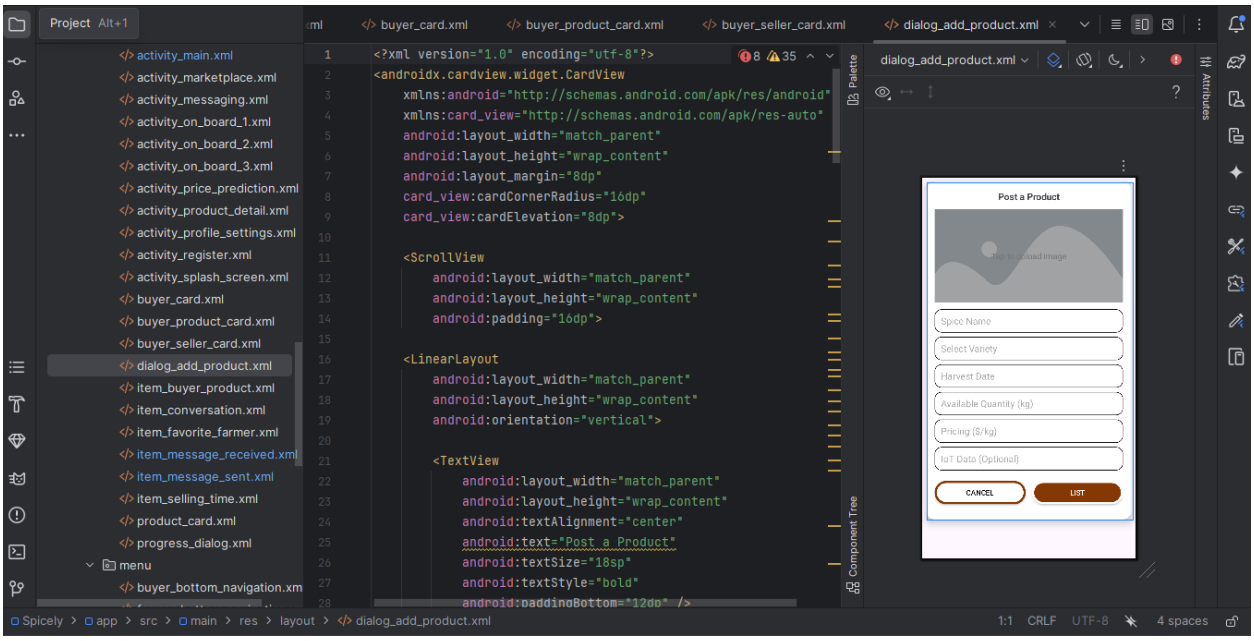

Android
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView xmlns:android="http://schemas
android:layout_width="160dp"
android:layout_height="wrap_content"
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_margin="4dp"
app:cardCornerRadius="12dp"
app:cardElevation="4dp">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:gravity="center_horizontal"
android:paddingBottom="13.00dp"
>
<RelativeLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:padding="8dp">
<!-- Profile Image (centered horizontally) -->
<com.google.android.material.imageview.ShapeableImag
android:id="@+id/profileImage"
android:layout_width="40dp"
android:layout_height="40dp"
android:layout_centerHorizontal="true"
buyer_seller_card.xml
Ravi Kumar
Palette
Component Tree
Attributes
Spicely
app
src
main
res
layout
buyer_seller_card.xml
1:1
CRLF
UTF-8
4 spaces

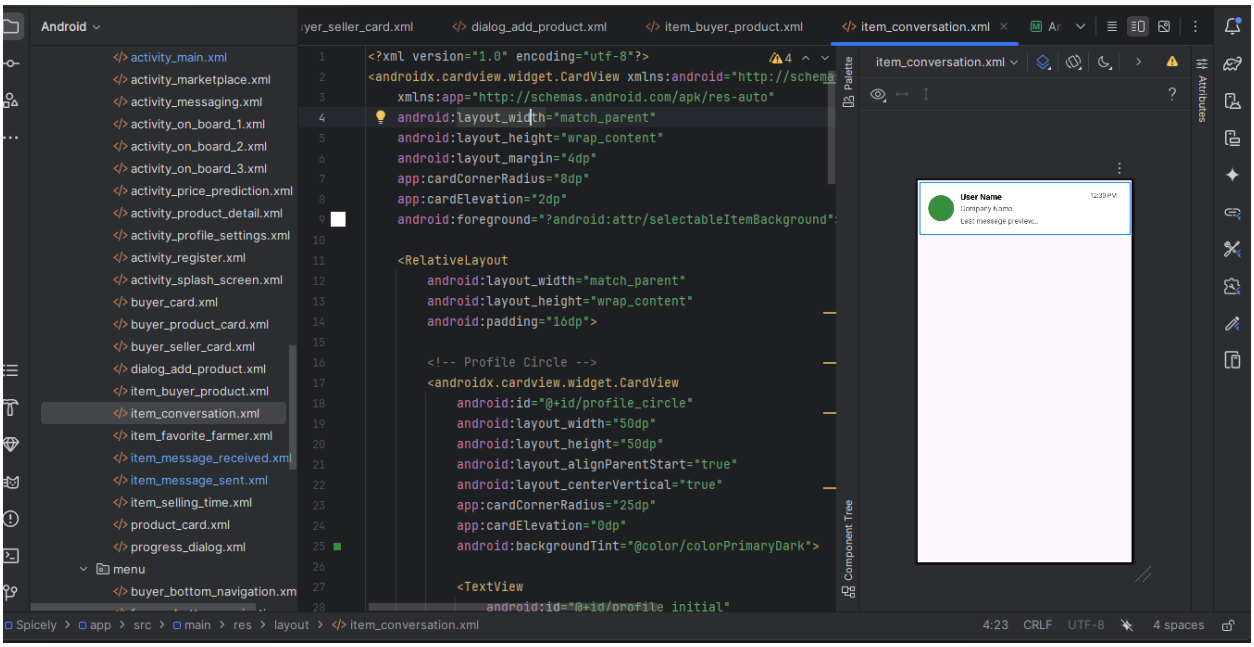

Spicely
master
No Devices
app
Android
activity_profile_settings.xml
<RelativeLayout xmlns:android="http://schemas.and
android:layout_marginHorizontal="13.00dp"
tools:context=".ProfileSettingsActivity">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
android:layout_marginBottom="19.50dp"
android:layout_marginTop="19.50dp"
>
<ImageView
android:id="@+id/backBtn"
android:layout_width="24dp"
android:layout_height="24dp"
android:src="@drawable/arrow_left"
app:tint="#888" />
<TextView
Profile & Settings
Personal Information
Notifications
Language
English
Help & Support

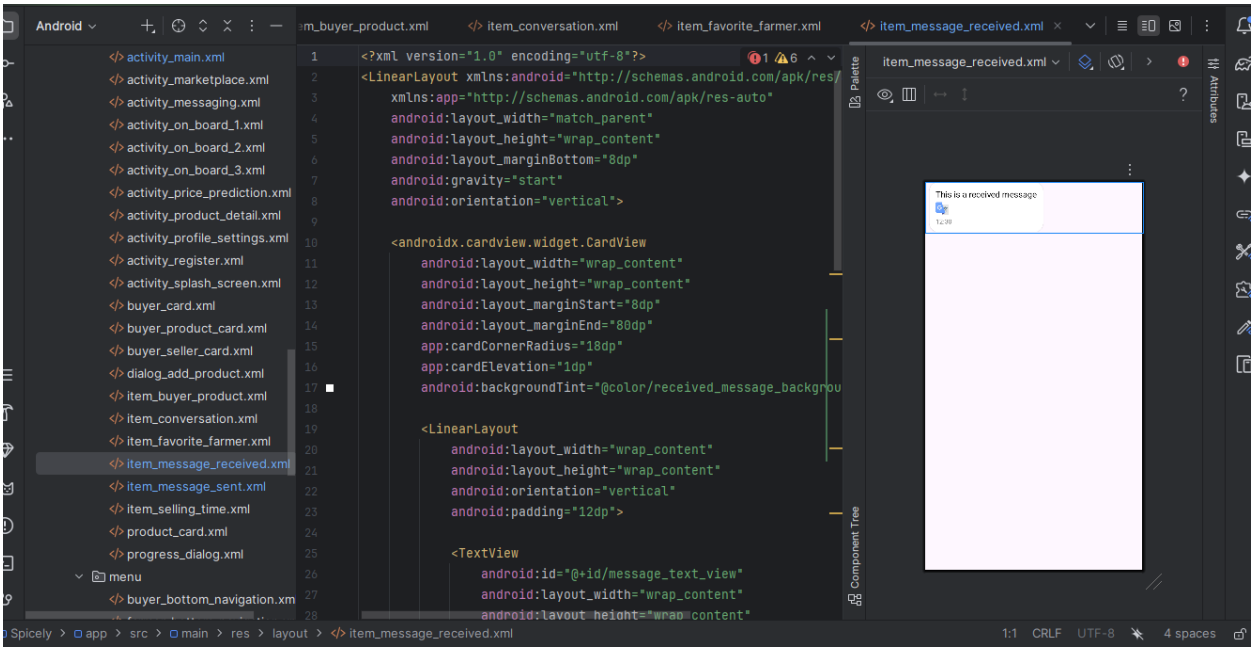

Android
activity_farmer_profile.xml
Split (Alt+Shift+Left)
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".FarmerProfileActivity">
<ScrollView
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="16dp">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical"
android:gravity="center_horizontal">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
android:layout_marginBottom="19.50dp"
android:layout_marginTop="19.50dp">
<ImageView
Profile Information
Name:
Email:
john@example.com
Farm Location:
Crops Grown:
Item 1
Languages Spoken:
Item 1
Change Password
UPDATE INFO

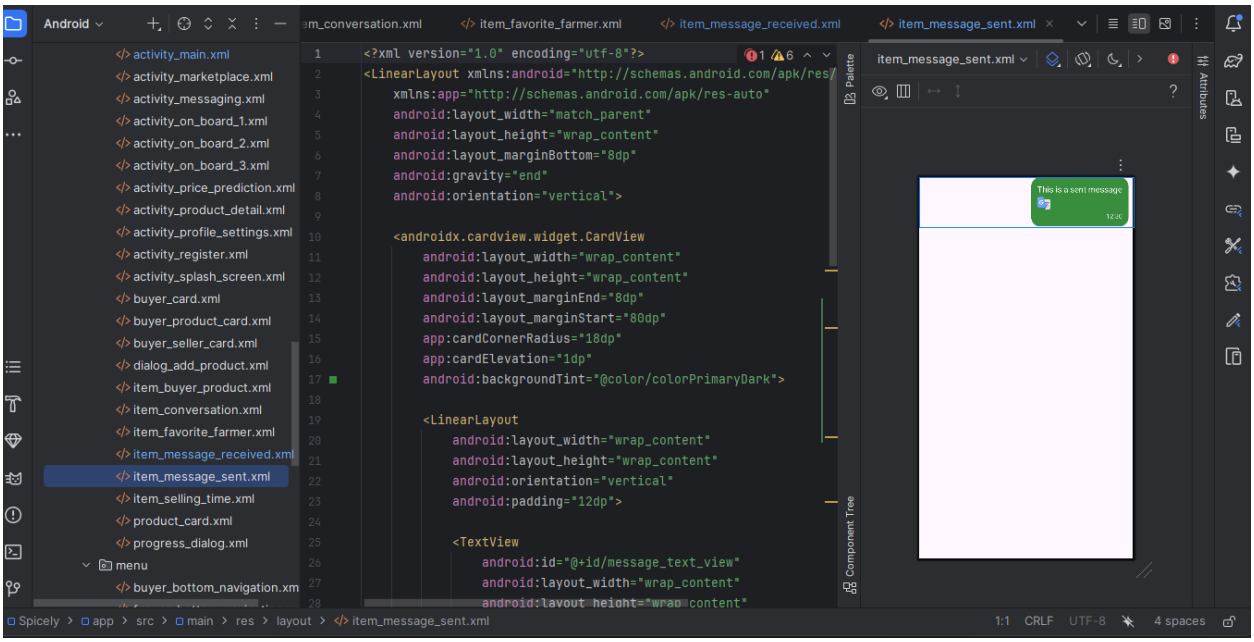

Android
activity_main.xml
activity_marketplace.xml
network_security_config.xml
firebase_auth.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.coordinatorlayout.widget.CoordinatorLayout xmlns:andr
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MarketplaceActivity">
<com.google.android.material.appbar.AppBarLayout
android:layout_width="match_parent"
android:layout_height="wrap_content">
<androidx.appcompat.widget.Toolbar
android:id="@+id/toolbar"
android:layout_width="match_parent"
android:layout_height="?attr/actionBarSize"
android:background="@color/white"
app:layout_scrollFlags="scroll|enterAlways"
app:title="Marketplace"
app:titleTextColor="@color/black" />
<!-- Search Bar -->
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
android:padding="8dp"
android:background="@color/white">
activity_marketplace.xml
Marketplace
All
0 results
Home
Palette
Component Tree
Attributes
Spicely
app
src
main
res
layout
activity_marketplace.xml
1:1
LF
UTF-8
4 spaces

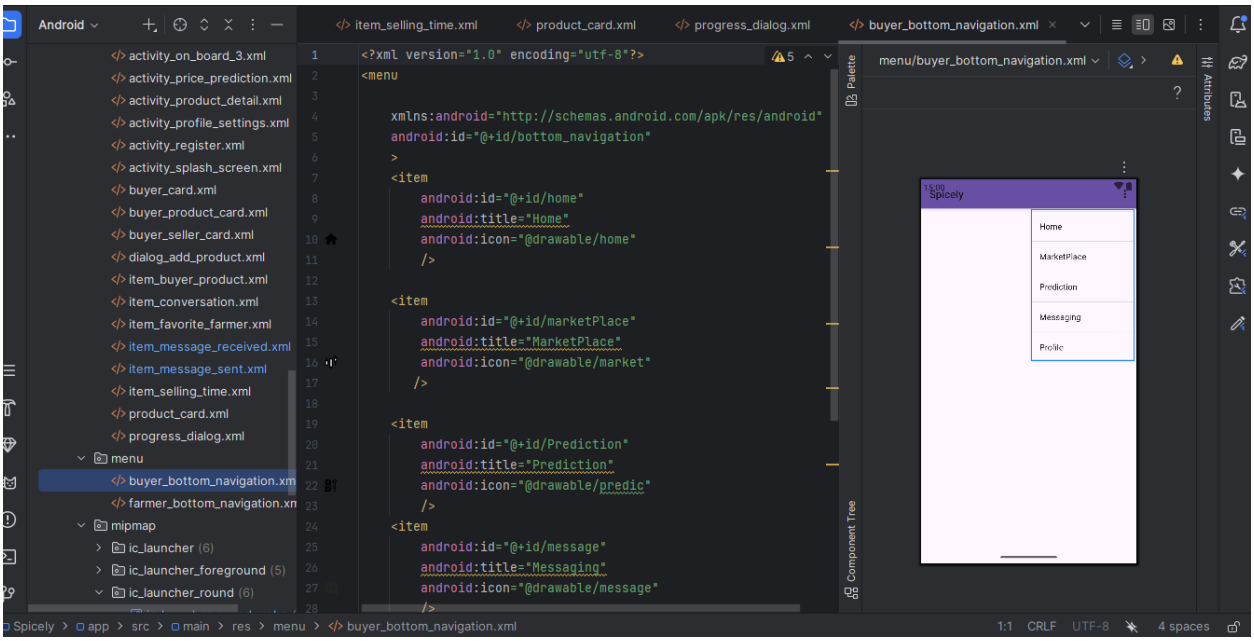

Android
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
java (generated)
res
drawable
layout
activity_analytics.xml
activity_buyer_main.xml
activity_buyer_messaging.xm
activity_buyer_profile.xml
activity_buyers.xml
activity_chat.xml
activity_farmer_profile.xml
activity_login.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:background="@color/white"
tools:context=".PricePredictionActivity">
<!-- Header with back button -->
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal"
android:padding="16dp">
<ImageView
android:id="@+id/backBtn"
android:layout_width="32.50dp"
android:layout_height="32.50dp"
android:src="@drawable/arrow_left" />
<TextView
android:layout_width="0dp"
android:layout_weight="1"
activity_price_prediction.xml
Spice Price Prediction
Prediction Result
Predicted Price: 0.00 USD per kg
Input Parameters
Spice Type
Region
Item 1
Item 1
Demand Index (0-100)
Supply Index (0-100)
0-100
0-100
Rainfall (mm)
Temperature (°C)
Enter rainfall
Enter temperature
Month (1-12)
Year
1-12
Enter year
Predict Price
Palette
Component Tree
Attributes
Spicely
app
src
main
res
layout
activity_price_prediction.xml
1:1
LF
UTF-8
4 spaces

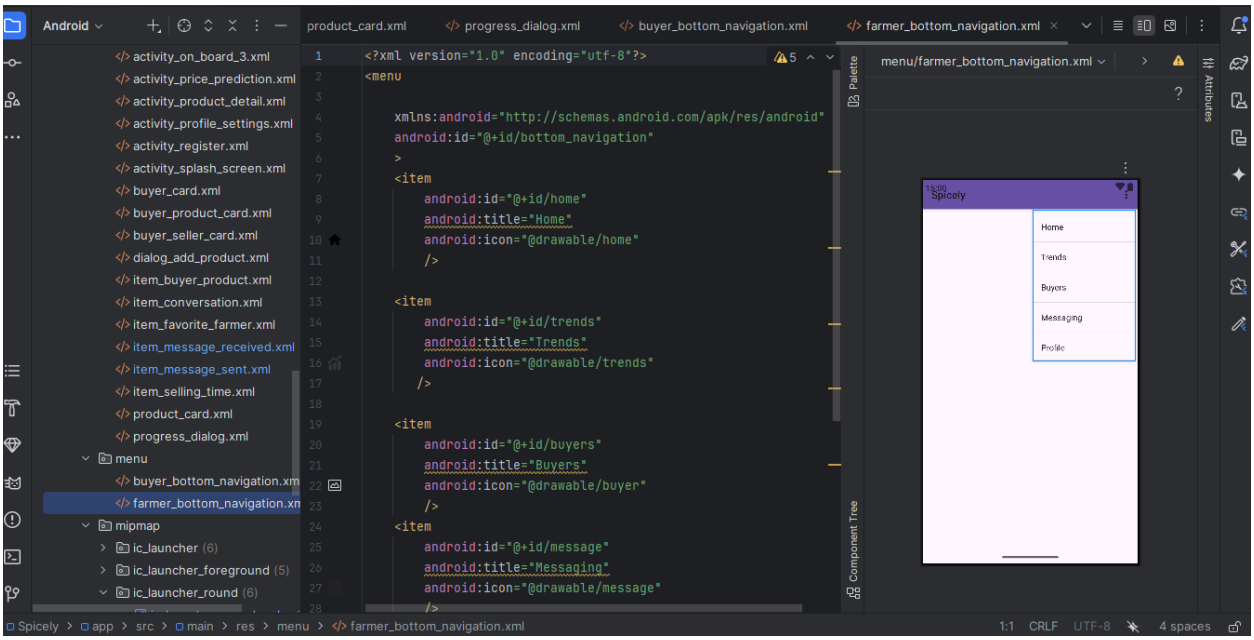

Project Alt+1
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
dialog_add_product.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:card_view="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_margin="8dp"
card_view:cardCornerRadius="16dp"
card_view:cardElevation="8dp">
<ScrollView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:padding="16dp">
<LinearLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="vertical">
<TextView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textAlignment="center"
android:text="Post a Product"
android:textSize="18sp"
android:textStyle="bold"
Post a Product
Spice Name
Select Variety
Harvest Date
Available Quantity (kg)
Pricing ($/kg)
IoT Data (Optional)
CANCEL
LIST
Spicely app src main res layout dialog_add_product.xml
1:1 CRLF UTF-8 4 spaces


Android
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView xmlns:android="http://schema
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_margin="4dp"
app:cardCornerRadius="8dp"
app:cardElevation="2dp"
android:foreground="?android:attr/selectableItemBackground"
<RelativeLayout
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:padding="16dp">
<!-- Profile Circle -->
<androidx.cardview.widget.CardView
android:id="@+id/profile_circle"
android:layout_width="50dp"
android:layout_height="50dp"
android:layout_alignParentStart="true"
android:layout_centerVertical="true"
app:cardCornerRadius="25dp"
app:cardElevation="0dp"
android:backgroundTint="@color/colorPrimaryDark">
<TextView
User Name
Company Name
Last message preview...
Spicely app src main res layout item_conversation.xml
4:23 CRLF UTF-8 4 spaces


Android
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginBottom="8dp"
android:gravity="start"
android:orientation="vertical">
<androidx.cardview.widget.CardView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="8dp"
android:layout_marginEnd="80dp"
app:cardCornerRadius="18dp"
app:cardElevation="1dp"
android:backgroundTint="@color/received_message_backgrou
<LinearLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="12dp">
<TextView
android:id="@+id/message_text_view"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
Palette
Component Tree
Attributes
item_message_received.xml
This is a received message
Spicely
app
src
main
res
layout
item_message_received.xml
1:1
CRLF
UTF-8
4 spaces

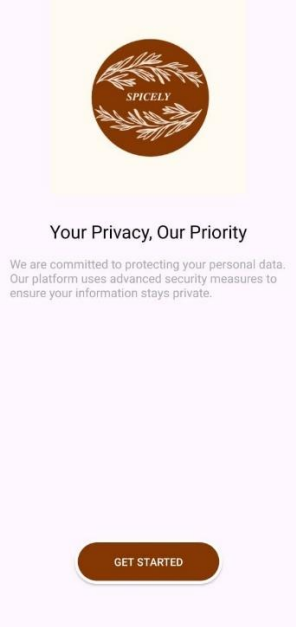

Android
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
activity_main.xml
activity_marketplace.xml
activity_messaging.xml
activity_on_board_1.xml
activity_on_board_2.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginBottom="8dp"
android:gravity="end"
android:orientation="vertical">
<androidx.cardview.widget.CardView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginEnd="8dp"
android:layout_marginStart="80dp"
app:cardCornerRadius="18dp"
app:cardElevation="1dp"
android:backgroundTint="@color/colorPrimaryDark">
<LinearLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:orientation="vertical"
android:padding="12dp">
<TextView
android:id="@+id/message_text_view"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
Palette
Component Tree
Attributes
item_message_sent.xml
This is a sent message
Spicely
app
src
main
res
layout
item_message_sent.xml
1:1
CRLF
UTF-8
4 spaces


Android
item_selling_time.xml
product_card.xml
progress_dialog.xml
buyer_bottom_navigation.xml
menu/buyer_bottom_navigation.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
farmer_bottom_navigation.xm
mipmap
ic_launcher (6)
ic_launcher_foreground (5)
ic_launcher_round (6)
<?xml version="1.0" encoding="utf-8"?>
<menu
xmlns:android="http://schemas.android.com/apk/res/android"
android:id="@+id/bottom_navigation"
>
<item
android:id="@+id/home"
android:title="Home"
android:icon="@drawable/home"
/>
<item
android:id="@+id/marketPlace"
android:title="MarketPlace"
android:icon="@drawable/market"
/>
<item
android:id="@+id/Prediction"
android:title="Prediction"
android:icon="@drawable/predic"
/>
<item
android:id="@+id/message"
android:title="Messaging"
android:icon="@drawable/message"
/>
Spicely
Home
MarketPlace
Prediction
Messaging
Profile
Palette
Component Tree
Attributes
Spicely › app › src › main › res › menu › buyer_bottom_navigation.xml
1:1 CRLF UTF-8 4 spaces


Android
product_card.xml
progress_dialog.xml
buyer_bottom_navigation.xml
farmer_bottom_navigation.xml
menu/farmer_bottom_navigation.xml
activity_on_board_3.xml
activity_price_prediction.xml
activity_product_detail.xml
activity_profile_settings.xml
activity_register.xml
activity_splash_screen.xml
buyer_card.xml
buyer_product_card.xml
buyer_seller_card.xml
dialog_add_product.xml
item_buyer_product.xml
item_conversation.xml
item_favorite_farmer.xml
item_message_received.xml
item_message_sent.xml
item_selling_time.xml
product_card.xml
progress_dialog.xml
menu
buyer_bottom_navigation.xm
farmer_bottom_navigation.xm
mipmap
ic_launcher (6)
ic_launcher_foreground (5)
ic_launcher_round (6)
<?xml version="1.0" encoding="utf-8"?>
<menu
xmlns:android="http://schemas.android.com/apk/res/android"
android:id="@+id/bottom_navigation"
>
<item
android:id="@+id/home"
android:title="Home"
android:icon="@drawable/home"
/>
<item
android:id="@+id/trends"
android:title="Trends"
android:icon="@drawable/trends"
/>
<item
android:id="@+id/buyers"
android:title="Buyers"
android:icon="@drawable/buyer"
/>
<item
android:id="@+id/message"
android:title="Messaging"
android:icon="@drawable/message"
/>
Spicely
Home
Trends
Buyers
Messaging
Profile
Palette
Component Tree
Attributes
Spicely › app › src › main › res › menu › farmer_bottom_navigation.xml
1:1 CRLF UTF-8 4 spaces

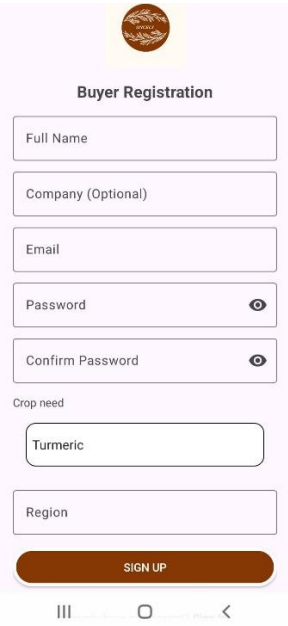

SPICELY
Empowering Farmers Everywhere
Our platform connects spice farmers directly to global markets, ensuring fair prices and better opportunities.
SKIP
NEXT

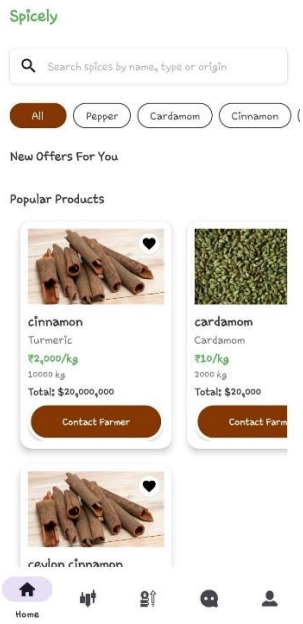

SPICELY
Direct Selling, More Profit
Cut out the middleman and sell your products directly to buyers. Increase your profit margins and gain more control over your business.
SKIP
NEXT

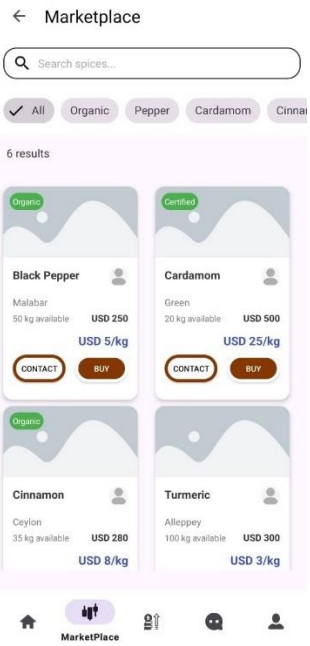

SPICELY
Your Privacy, Our Priority
We are committed to protecting your personal data. Our platform uses advanced security measures to ensure your information stays private.
GET STARTED

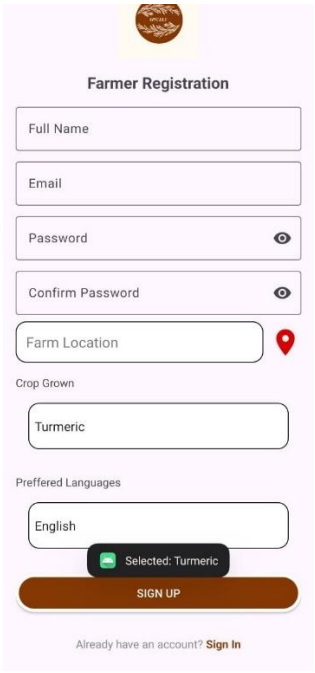

Choose Your Role
Please select your role to begin signing up. Choose 'Farmer' if you are a producer, or 'Buyer' if you are looking to purchase spices.
I'M A FARMER
I'M A BUYER

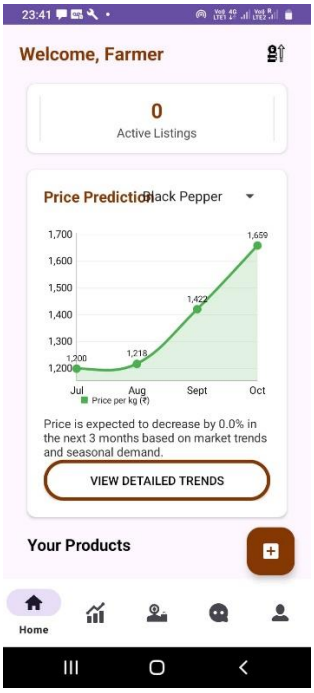

Sign In
Enter your email
Enter your password
Forgot Password?
SIGN IN
Don't have an account? Sign Up

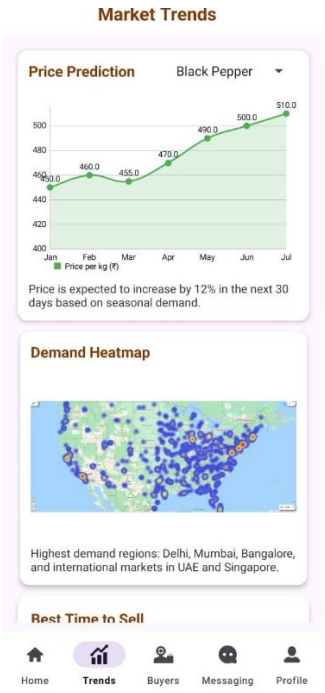

Buyer Registration
Full Name
Company (Optional)
Email
Password
Confirm Password
Crop need
Turmeric
Region
SIGN UP

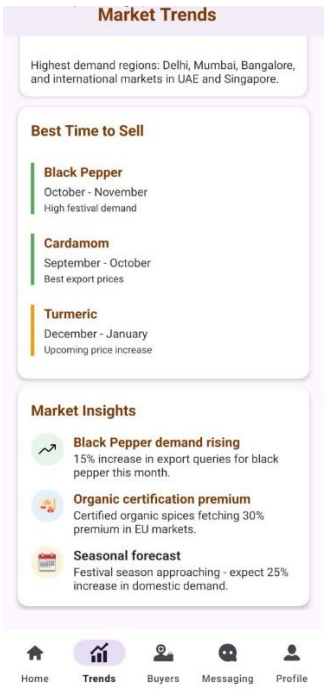

Spicely
Search spices by name, type or origin
All
Pepper
Cardamom
Cinnamon
New Offers For You
Popular Products
cinnamon
Turmeric
₹2,000/kg
10000 kg
Total: $20,000,000
Contact Farmer
cardamom
Cardamom
₹10/kg
2000 kg
Total: $20,000
Contact Farm
Home


Marketplace
Search spices...
All
Organic
Pepper
Cardamom
6 results
Organic
Black Pepper
Malabar
50 kg available
USD 250
USD 5/kg
CONTACT
BUY
Certified
Cardamom
Green
20 kg available
USD 500
USD 25/kg
CONTACT
BUY
Organic
Cinnamon
Ceylon
35 kg available
USD 280
USD 8/kg
Turmeric
Alleppey
100 kg available
USD 300
USD 3/kg
MarketPlace


Farmer Registration
Full Name
Email
Password
Confirm Password
Farm Location
Crop Grown
Turmeric
Preffered Languages
English
Selected: Turmeric
SIGN UP
Already have an account? Sign In

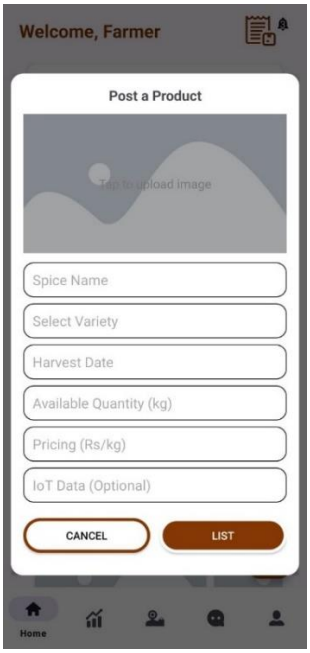

23:41
Welcome, Farmer
0
Active Listings
Price Prediction
Black Pepper
1,700
1,600
1,500
1,400
1,300
1,200
1,200
1,218
1,422
1,659
Jul
Aug
Sept
Oct
Price per kg (₹)
Price is expected to decrease by 0.0% in the next 3 months based on market trends and seasonal demand.
VIEW DETAILED TRENDS
Your Products
Home

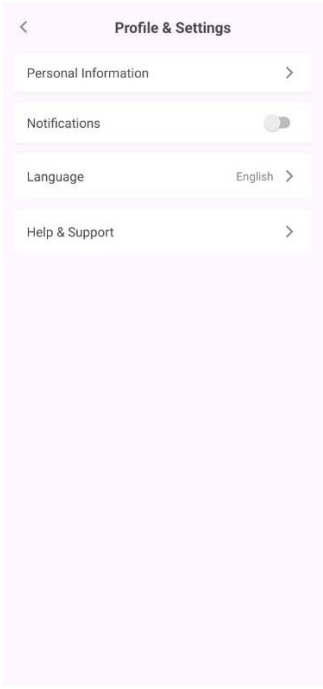

Market Trends
Price Prediction
Black Pepper
500
480
460
440
420
400
450.0
460.0
455.0
470.0
490.0
500.0
510.0
Jan
Feb
Mar
Apr
May
Jun
Jul
Price per kg (₹)
Price is expected to increase by 12% in the next 30 days based on seasonal demand.
Demand Heatmap
Highest demand regions: Delhi, Mumbai, Bangalore, and international markets in UAE and Singapore.
Best Time to Sell
Home
Trends
Buyers
Messaging
Profile

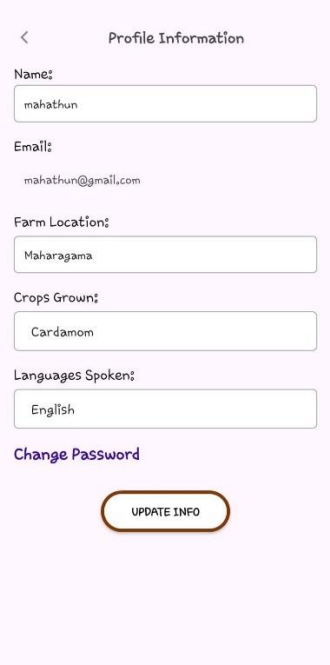

Market Trends
Highest demand regions: Delhi, Mumbai, Bangalore, and international markets in UAE and Singapore.
Best Time to Sell
Black Pepper
October - November
High festival demand
Cardamom
September - October
Best export prices
Turmeric
December - January
Upcoming price increase
Market Insights
Black Pepper demand rising
15% increase in export queries for black pepper this month.
Organic certification premium
Certified organic spices fetching 30% premium in EU markets.
Seasonal forecast
Festival season approaching - expect 25% increase in domestic demand.
Home
Trends
Buyers
Messaging
Profile

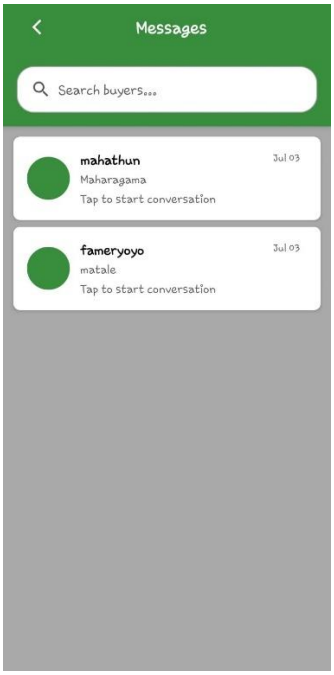

Potential Buyers
search by using buyer name/ spice/ location
Best Matches for Your Products
buytest pvt
india
Interested in: Turmeric
Contact Buyer
buyerSL
india
Interested in: Turmeric
Contact Buyer
All Buyers
buytest pvt
india
Interested in: Turmeric
buyerSL
india
Interested in: Turmeric
Home
Trends
Buyers
Messaging
Profile

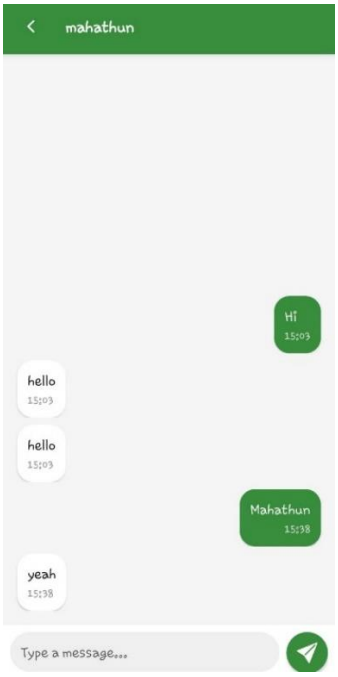

Spice Price Prediction
Prediction Result
Predicted Price: -- USD per kg
Input Parameters
Spice Type
Turmeric
Region
Kerala
Demand Index (0-100)
0-100
Supply Index (0-100)
0-100
Rainfall (mm)
Enter rainfall
Temperature (°C)
Enter temperatur
Month (1-12)
1-12
Year
2025
Predict Price

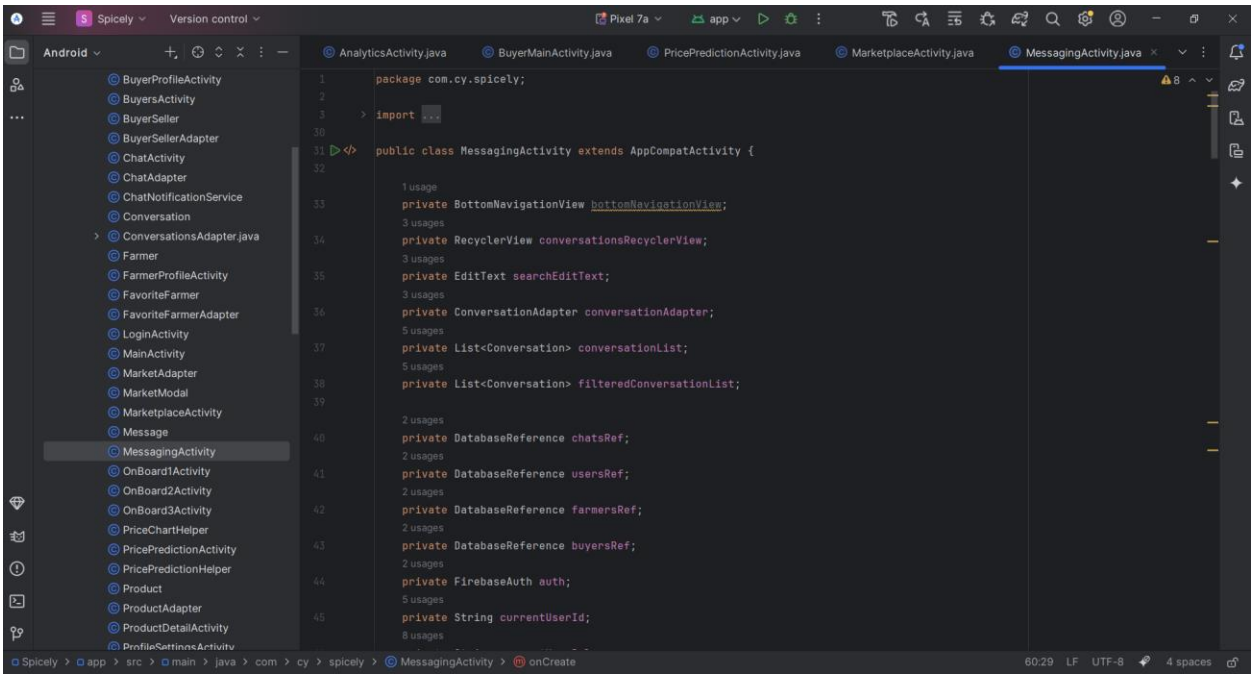

Welcome, Farmer
Post a Product
Tap to upload image
Spice Name
Select Variety
Harvest Date
Available Quantity (kg)
Pricing (Rs/kg)
IoT Data (Optional)
CANCEL
LIST
Home

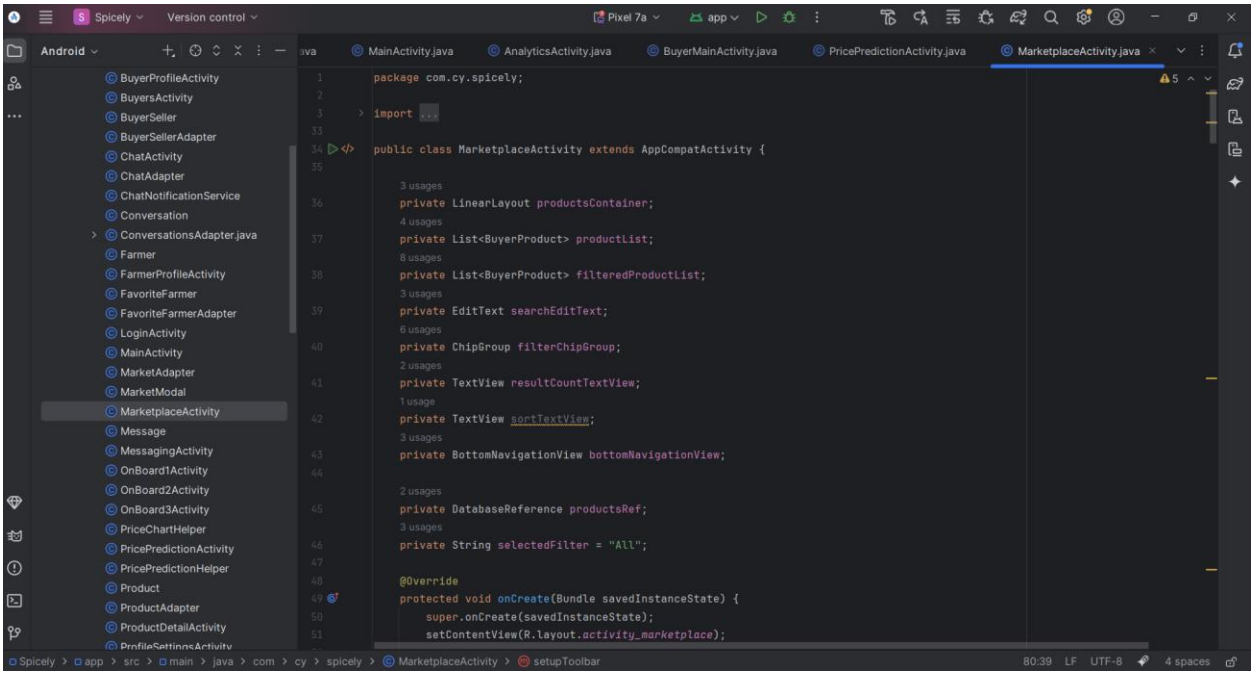

Profile & Settings
Personal Information
Notifications
Language
English
Help & Support

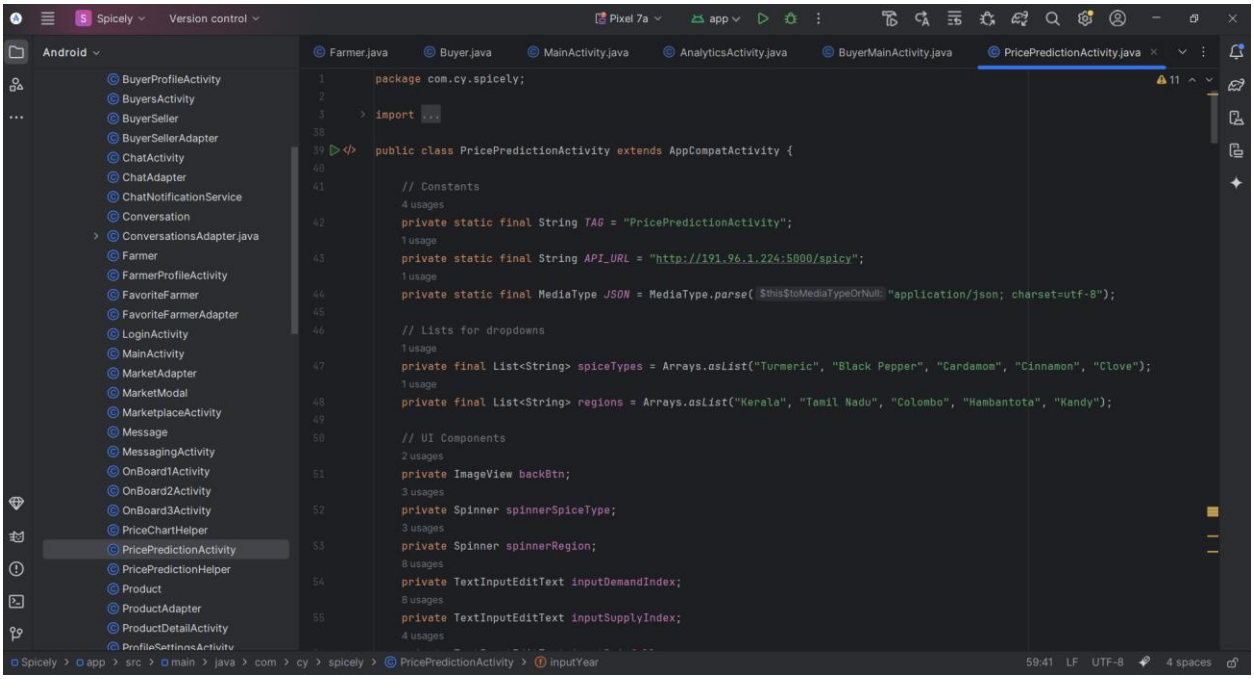

Profile Information
Name:
mahathun
Email:
mahathun@gmail.com
Farm Location:
Maharagama
Crops Grown:
Cardamom
Languages Spoken:
English
Change Password
UPDATE INFO

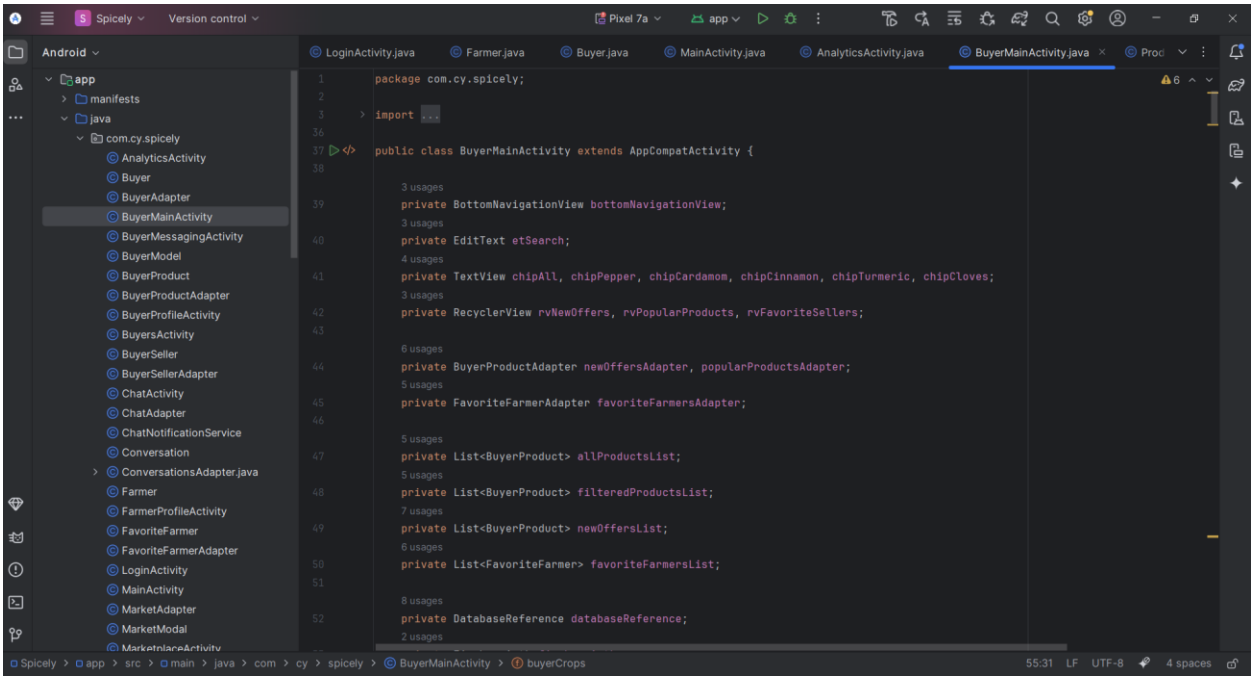

Messages
Search buyers...
mahathun
Jul 03
Maharagama
Tap to start conversation
fameryoyo
Jul 03
matale
Tap to start conversation

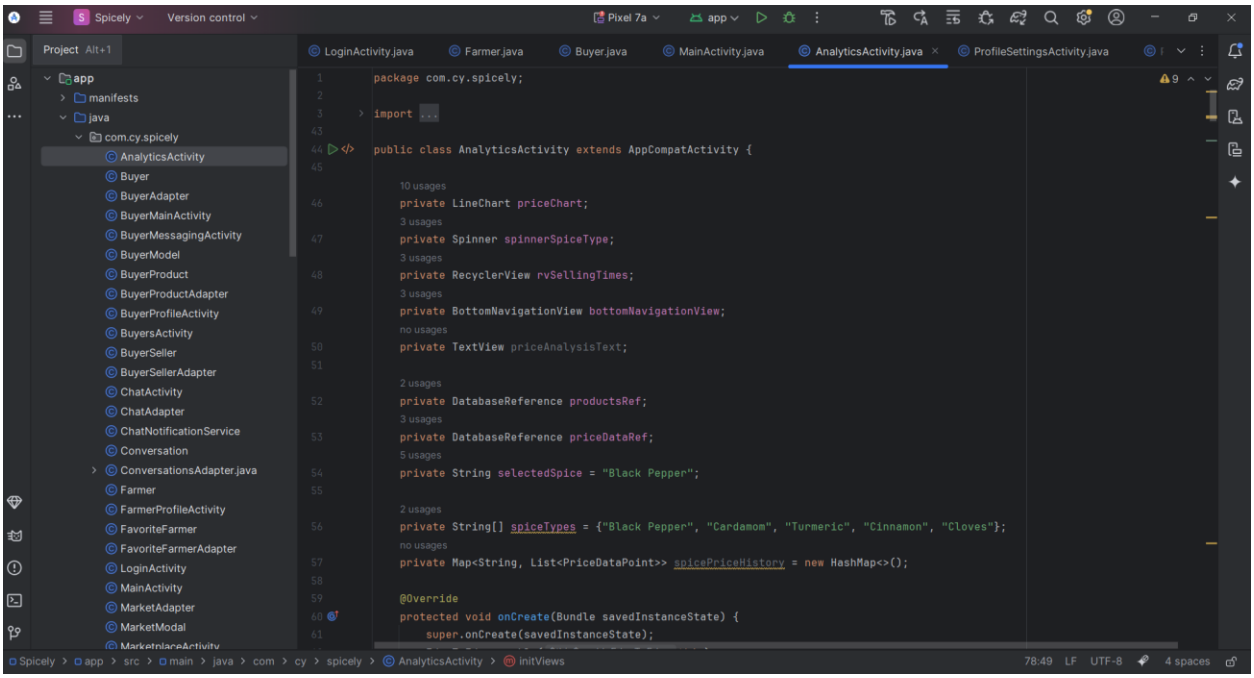

mahathun
Hi
15:03
hello
15:03
hello
15:03
Mahathun
15:38
yeah
15:38
Type a message...

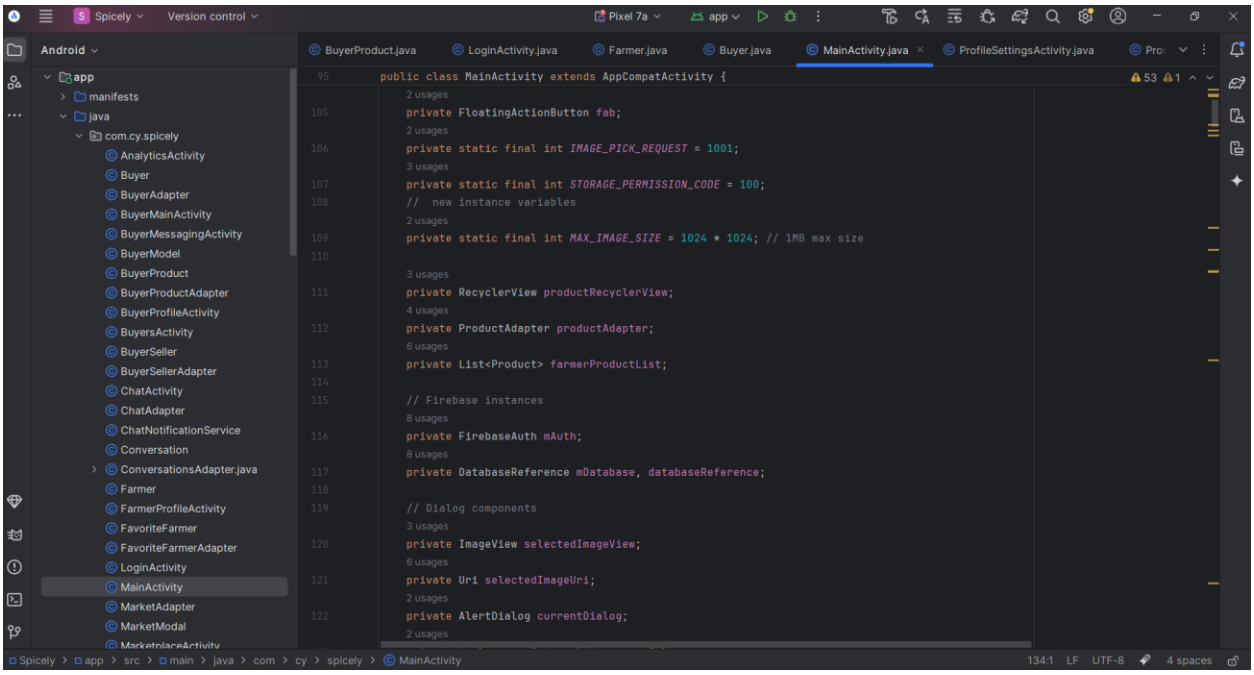

Spicely
Version control
Pixel 7a
app
Android
AnalyticsActivity.java
BuyerMainActivity.java
PricePredictionActivity.java
MarketplaceActivity.java
MessagingActivity.java
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
MarketplaceActivity
Message
MessagingActivity
OnBoard1Activity
OnBoard2Activity
OnBoard3Activity
PriceChartHelper
PricePredictionActivity
PricePredictionHelper
Product
ProductAdapter
ProductDetailActivity
package com.cy.spicely;
import ...
public class MessagingActivity extends AppCompatActivity {
1 usage
private BottomNavigationView bottomNavigationView;
3 usages
private RecyclerView conversationsRecyclerView;
3 usages
private EditText searchEditText;
3 usages
private ConversationAdapter conversationAdapter;
5 usages
private List<Conversation> conversationList;
5 usages
private List<Conversation> filteredConversationList;
2 usages
private DatabaseReference chatsRef;
2 usages
private DatabaseReference usersRef;
2 usages
private DatabaseReference farmersRef;
2 usages
private DatabaseReference buyersRef;
2 usages
private FirebaseAuth auth;
5 usages
private String currentUserId;
8 usages
Spicely > app > src > main > java > com > cy > spicely > MessagingActivity > onCreate
60:29
LF
UTF-8
4 spaces

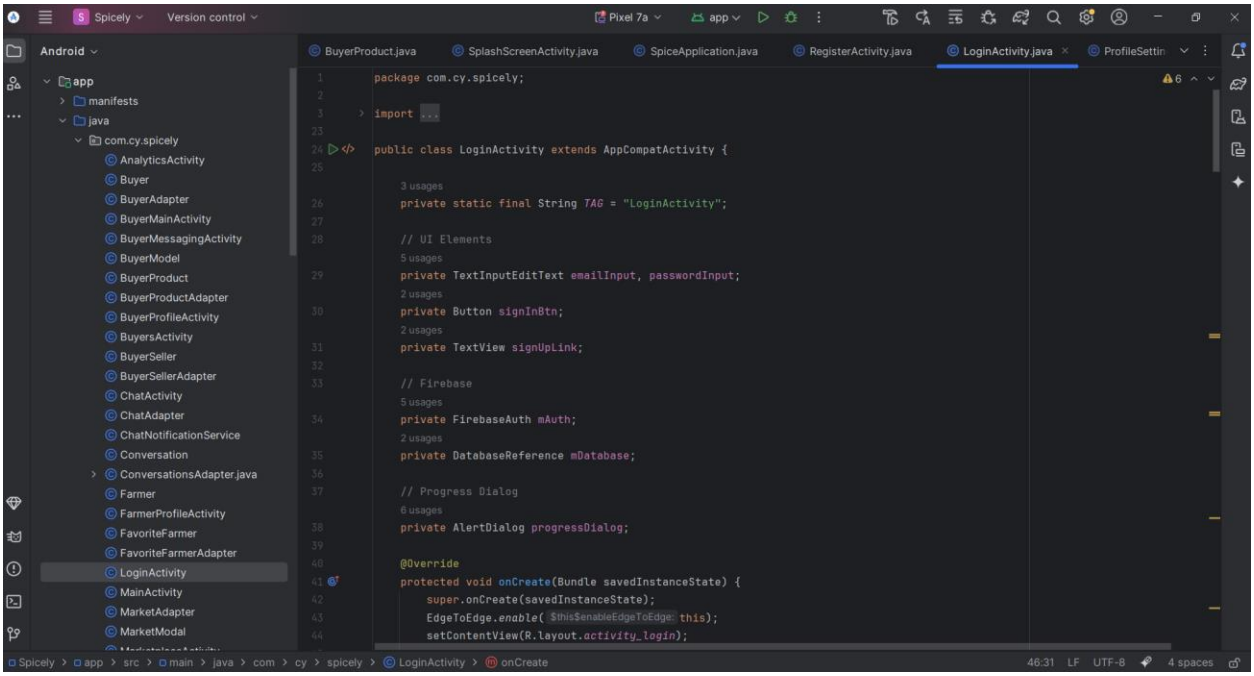

Spicely
Version control
Pixel 7a
app
Android
MainActivity.java
AnalyticsActivity.java
BuyerMainActivity.java
PricePredictionActivity.java
MarketplaceActivity.java
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
MarketplaceActivity
Message
MessagingActivity
OnBoard1Activity
OnBoard2Activity
OnBoard3Activity
PriceChartHelper
PricePredictionActivity
PricePredictionHelper
Product
ProductAdapter
ProductDetailActivity
package com.cy.spicely;
import ...
public class MarketplaceActivity extends AppCompatActivity {
3 usages
private LinearLayout productsContainer;
4 usages
private List<BuyerProduct> productList;
8 usages
private List<BuyerProduct> filteredProductList;
3 usages
private EditText searchEditText;
6 usages
private ChipGroup filterChipGroup;
2 usages
private TextView resultCountTextView;
1 usage
private TextView sortTextView;
3 usages
private BottomNavigationView bottomNavigationView;
2 usages
private DatabaseReference productsRef;
3 usages
private String selectedFilter = "All";
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_marketplace);
Spicely › app › src › main › java › com › cy › spicely › MarketplaceActivity › setupToolbar
80:39
LF
UTF-8
4 spaces

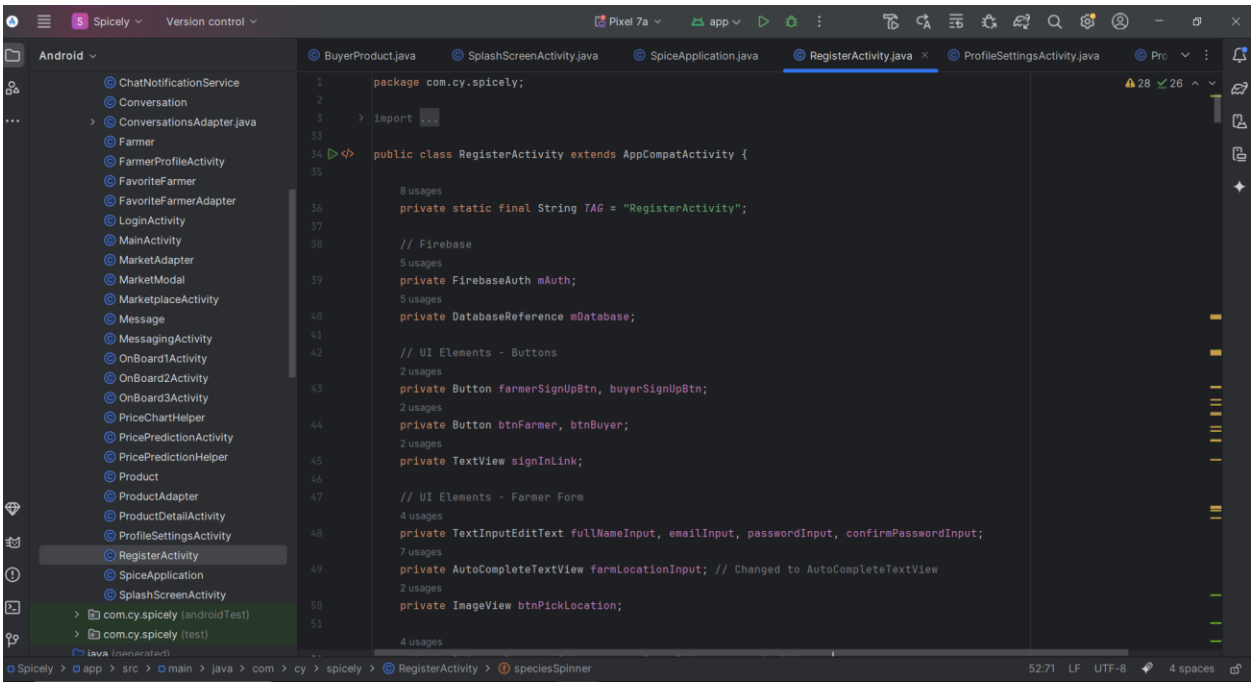

Spicely
Version control
Pixel 7a
app
Android
Farmer.java
Buyer.java
MainActivity.java
AnalyticsActivity.java
BuyerMainActivity.java
PricePredictionActivity.java
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
MarketplaceActivity
Message
MessagingActivity
OnBoard1Activity
OnBoard2Activity
OnBoard3Activity
PriceChartHelper
PricePredictionActivity
PricePredictionHelper
Product
ProductAdapter
ProductDetailActivity
package com.cy.spicely;
import ...
public class PricePredictionActivity extends AppCompatActivity {
// Constants
4 usages
private static final String TAG = "PricePredictionActivity";
1 usage
private static final String API_URL = "http://191.96.1.224:5000/spicy";
1 usage
private static final MediaType JSON = MediaType.parse($this$toMediaTypeOrNull: "application/json; charset=utf-8");
// Lists for dropdowns
1 usage
private final List<String> spiceTypes = Arrays.asList("Turmeric", "Black Pepper", "Cardamom", "Cinnamon", "Clove");
1 usage
private final List<String> regions = Arrays.asList("Kerala", "Tamil Nadu", "Colombo", "Hambantota", "Kandy");
// UI Components
2 usages
private ImageView backBtn;
3 usages
private Spinner spinnerSpiceType;
3 usages
private Spinner spinnerRegion;
8 usages
private TextInputEditText inputDemandIndex;
8 usages
private TextInputEditText inputSupplyIndex;
4 usages
Spicely › app › src › main › java › com › cy › spicely › PricePredictionActivity › inputYear
59:41
LF
UTF-8
4 spaces


Spicely
Version control
Pixel 7a
app
Android
LoginActivity.java
Farmer.java
Buyer.java
MainActivity.java
AnalyticsActivity.java
BuyerMainActivity.java
app
manifests
java
com.cy.spicely
AnalyticsActivity
Buyer
BuyerAdapter
BuyerMainActivity
BuyerMessagingActivity
BuyerModel
BuyerProduct
BuyerProductAdapter
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
package com.cy.spicely;
import ...
public class BuyerMainActivity extends AppCompatActivity {
3 usages
private BottomNavigationView bottomNavigationView;
3 usages
private EditText etSearch;
4 usages
private TextView chipAll, chipPepper, chipCardamom, chipCinnamon, chipTurmeric, chipCloves;
3 usages
private RecyclerView rvNewOffers, rvPopularProducts, rvFavoriteSellers;
6 usages
private BuyerProductAdapter newOffersAdapter, popularProductsAdapter;
5 usages
private FavoriteFarmerAdapter favoriteFarmersAdapter;
5 usages
private List<BuyerProduct> allProductsList;
5 usages
private List<BuyerProduct> filteredProductsList;
7 usages
private List<BuyerProduct> newOffersList;
6 usages
private List<FavoriteFarmer> favoriteFarmersList;
8 usages
private DatabaseReference databaseReference;
2 usages
Spicely › app › src › main › java › com › cy › spicely › BuyerMainActivity › buyerCrops
55:31 LF UTF-8 4 spaces


Spicely
Version control
Pixel 7a
app
Project Alt+1
LoginActivity.java
Farmer.java
Buyer.java
MainActivity.java
AnalyticsActivity.java
ProfileSettingsActivity.java
app
manifests
java
com.cy.spicely
AnalyticsActivity
Buyer
BuyerAdapter
BuyerMainActivity
BuyerMessagingActivity
BuyerModel
BuyerProduct
BuyerProductAdapter
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
package com.cy.spicely;
import ...
public class AnalyticsActivity extends AppCompatActivity {
10 usages
private LineChart priceChart;
3 usages
private Spinner spinnerSpiceType;
3 usages
private RecyclerView rvSellingTimes;
3 usages
private BottomNavigationView bottomNavigationView;
no usages
private TextView priceAnalysisText;
2 usages
private DatabaseReference productsRef;
3 usages
private DatabaseReference priceDataRef;
5 usages
private String selectedSpice = "Black Pepper";
2 usages
private String[] spiceTypes = {"Black Pepper", "Cardamom", "Turmeric", "Cinnamon", "Cloves"};
no usages
private Map<String, List<PriceDataPoint>> spicePriceHistory = new HashMap<>();
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
Spicely › app › src › main › java › com › cy › spicely › AnalyticsActivity › initViews
78:49 LF UTF-8 4 spaces


Spicely
Version control
Pixel 7a
app
Android
BuyerProduct.java
LoginActivity.java
Farmer.java
Buyer.java
MainActivity.java
ProfileSettingsActivity.java
app
manifests
java
com.cy.spicely
AnalyticsActivity
Buyer
BuyerAdapter
BuyerMainActivity
BuyerMessagingActivity
BuyerModel
BuyerProduct
BuyerProductAdapter
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
public class MainActivity extends AppCompatActivity {
2 usages
private FloatingActionButton fab;
2 usages
private static final int IMAGE_PICK_REQUEST = 1001;
3 usages
private static final int STORAGE_PERMISSION_CODE = 100;
// new instance variables
2 usages
private static final int MAX_IMAGE_SIZE = 1024 * 1024; // 1MB max size
3 usages
private RecyclerView productRecyclerView;
4 usages
private ProductAdapter productAdapter;
6 usages
private List<Product> farmerProductList;
// Firebase instances
8 usages
private FirebaseAuth mAuth;
8 usages
private DatabaseReference mDatabase, databaseReference;
// Dialog components
3 usages
private ImageView selectedImageView;
6 usages
private Uri selectedImageUri;
2 usages
private AlertDialog currentDialog;
2 usages
Spicely › app › src › main › java › com › cy › spicely › MainActivity
134:1 LF UTF-8 4 spaces


Spicely
Version control
Pixel 7a
app
Android
BuyerProduct.java
SplashScreenActivity.java
SpiceApplication.java
RegisterActivity.java
LoginActivity.java
ProfileSettin
app
manifests
java
com.cy.spicely
AnalyticsActivity
Buyer
BuyerAdapter
BuyerMainActivity
BuyerMessagingActivity
BuyerModel
BuyerProduct
BuyerProductAdapter
BuyerProfileActivity
BuyersActivity
BuyerSeller
BuyerSellerAdapter
ChatActivity
ChatAdapter
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
package com.cy.spicely;
import ...
public class LoginActivity extends AppCompatActivity {
3 usages
private static final String TAG = "LoginActivity";
// UI Elements
5 usages
private TextInputEditText emailInput, passwordInput;
2 usages
private Button signInBtn;
2 usages
private TextView signUpLink;
// Firebase
5 usages
private FirebaseAuth mAuth;
2 usages
private DatabaseReference mDatabase;
// Progress Dialog
6 usages
private AlertDialog progressDialog;
@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
EdgeToEdge.enable($this$enableEdgeToEdge: this);
setContentView(R.layout.activity_login);
Spicely › app › src › main › java › com › cy › spicely › LoginActivity › onCreate
46:31 LF UTF-8 4 spaces


Spicely
Version control
Pixel 7a
app
Android
BuyerProduct.java
SplashScreenActivity.java
SpiceApplication.java
RegisterActivity.java
ProfileSettingsActivity.java
ChatNotificationService
Conversation
ConversationsAdapter.java
Farmer
FarmerProfileActivity
FavoriteFarmer
FavoriteFarmerAdapter
LoginActivity
MainActivity
MarketAdapter
MarketModal
MarketplaceActivity
Message
MessagingActivity
OnBoard1Activity
OnBoard2Activity
OnBoard3Activity
PriceChartHelper
PricePredictionActivity
PricePredictionHelper
Product
ProductAdapter
ProductDetailActivity
ProfileSettingsActivity
RegisterActivity
SpiceApplication
SplashScreenActivity
com.cy.spicely (androidTest)
com.cy.spicely (test)
1 package com.cy.spicely;
3 import ...
34 public class RegisterActivity extends AppCompatActivity {
8 usages
36 private static final String TAG = "RegisterActivity";
38 // Firebase
5 usages
39 private FirebaseAuth mAuth;
5 usages
40 private DatabaseReference mDatabase;
42 // UI Elements - Buttons
2 usages
43 private Button farmerSignUpBtn, buyerSignUpBtn;
2 usages
44 private Button btnFarmer, btnBuyer;
2 usages
45 private TextView signInLink;
47 // UI Elements - Farmer Form
4 usages
48 private TextInputEditText fullNameInput, emailInput, passwordInput, confirmPasswordInput;
7 usages
49 private AutoCompleteTextView farmLocationInput; // Changed to AutoCompleteTextView
2 usages
50 private ImageView btnPickLocation;
4 usages
28
26
Spicely › app › src › main › java › com › cy › spicely › RegisterActivity › speciesSpinner
52:71
LF
UTF-8
4 spaces